Course Structure

R22

Mechanical Engineering

For

BTech 4-Year Degree Course

(Applicable for the students admitted into E1 from the Academic Year 2022-23)

(I – IV Years Syllabus)

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES

Basar, Nirmal, Telangana – 504107

Course Structure of Credits

S.No	Course Category	Course Type	Standard Credits as per AICTE	Department Credits
1	BSC	Basic Science Course	25	20.5
2	ESC	Engineering Science Course	24	22.5
3	HSMC	Humanities and Social Sciences including Management Course	12	10
4	PCC	Program Core Course	48	70
5	PEC	Program Elective Course	18	15
6	OEC	Open Elective Course	18	9
7	MC	Mandatory Course	0	0
8	SIP	Seminar + Internship + Project Work + Comprehensive Viva	15	13
Total			**160**	**160**

FIRST YEAR (E1) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	MA1101	Linear Algebra and Calculus	BSC	3	1	0	4	4
2	PH1101	Engineering Physics	BSC	3	0	0	3	3
3	EE1103	Basic Electrical and Electronics Engineering	ESC	3	0	0	3	3
4	HS1101	English	HSMC	2	0	0	2	2
5	PH1701	Engineering Physics Lab	BSC	0	0	3	3	1.5
6	EE1701	Basic Electrical and Electronics Engineering Lab	ESC	0	0	2	2	1
7	HS1701	English Lab	HSMC	0	0	2	2	1
8	ME1701	Design Thinking	ESC	0	0	2	2	1
9	ME1702	Engineering Drawing and computer graphics	ESC	0	1	4	5	3
10	BS1101	Environmental science	MC	3	0	0	3	0
Total								**19.5**

FIRST YEAR (E1) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	MA1201	Engineering Mathematics II	BSC	3	1	0	4	4
2	CY1201	Engineering Chemistry	BSC	3	0	0	3	3
3	CS1201	Programming for Problem Solving	ESC	3	0	0	3	3
4	ME1201	Engineering Mechanics	ESC	3	1	0	4	4
5	ME1202	Introduction to Manufacturing processes	PCC	2	0	0	2	2
6	CS1801	Programming for Problem Solving Lab	ESC	0	0	3	3	1.5
7	ME1801	Fuels and Lubricants Lab	BSC	0	0	2	2	1
8	ME1802	Engineering workshop	ESC	0	1	2	3	2
9	BM1201	Constitution of India	MC	2	0	0	2	0
Total								**20.5**

SECOND YEAR (E2) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	MA2101	Engineering Mathematics III	BSC	3	1	0	4	4
2	ME2101	Kinematics of Machinery	PCC	3	0	0	3	3
3	ME2102	Strength of Materials	PCC	3	1	0	4	4
4	ME2103	Thermodynamics	PCC	3	0	0	3	3
5	ME2104	Materials Engineering	PCC	3	0	0	3	3
6	ME2701	Strength of Materials Lab	PCC	0	0	2	2	1
7	ME2702	Materials Engineering Lab	PCC	0	0	2	2	1
8	ME2703	Computational mathematics lab	ESC	0	0	2	2	1
9	HS210X	Start up Entrepreneurship	MC	0	0	2	2	0
Total								**20**

SECOND YEAR (E2) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	ME2201	Fluid Mechanics and Hydraulic Machines	PCC	3	1	0	4	4
2	ME2202	Manufacturing Technology I	PCC	3	1	0	4	4
3	ME2203	Dynamics of Machinery	PCC	3	1	0	4	4
4	EC2201	Electronics ICs application	ESC	3	0	0	3	3
5	ME221X	Professional Elective-1	PEC	3	0	0	3	3
6	ME2801	Fluid Mechanics & Hydraulic Machinery Lab	PCC	0	0	2	2	1
7	ME2802	Theory of Machines Lab	PCC	0	0	2	2	1
8	ME2803	Manufacturing Technology I Lab	PCC	0	0	2	2	1
9	HS2201	Soft skills	HSMC	0	0	2	2	1
10	HS2202	Essence of Indian Traditional Knowledge	MC	2	0	0	2	0
Total								**22**

THIRD YEAR (E3) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	ME3101	Heat Transfer	PCC	3	0	0	3	3
2	HS3101	Operations research	HSMC	3	0	0	3	3
3	ME3102	Design of Machine Members	PCC	3	0	0	3	3
4	ME3103	IC Engine and Hybrid Vehicle	PCC	3	0	0	3	3
5	ME3104	Manufacturing Technology II	PCC	3	0	0	3	3
6	XX33xx	Open Elective – 1	OEC	3	0	0	3	3
7	ME3701	Heat Transfer lab	PCC	0	0	2	2	1
8	ME3702	Manufacturing Technology II lab	PCC	0	0	2	2	1
9	ME3703	Computer Aided Machine Drawing Practice	PCC	0	0	3	3	1.5
Total								**21.5**

THIRD YEAR (E3) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	ME3201	Power plant engineering	PCC	3	0	0	3	3
2	ME3202	Design of Transmission Elements	PCC	3	0	0	3	3
3	ME3203	Computer Aided Engineering	PCC	3	0	0	3	3
4	ME3204	Instrumentation and Control system	PCC	3	0	0	3	3
5	ME324x	Professional Elective-2	PEC	3	0	0	3	3
6	BM3201	Managerial Economics and Financial Analysis	HSMC	3	0	0	3	3
7	ME3801	Thermal Engineering lab	PCC	0	0	2	2	1
8	ME3802	Computer Aided Engineering Lab	PCC	0	0	3	3	1.5
9	ME3803	Instrumentation and Control system lab	PCC	0	0	2	2	1
10	ME3900	Theme based project		0	0	2	0	1
Total								**22.5**

FOURTH YEAR (E4) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	ME4101	Automation in Manufacturing	PCC	3	0	0	3	3
2	ME4102	Refrigeration and Air Conditioning	PCC	3	0	0	3	3
3	ME415X	Professional Elective – 3	PEC	3	0	0	3	3
4	ME415X	Professional Elective – 4	PEC	3	0	0	3	3
5	ME415X	Professional Elective – 5	PEC	3	0	0	3	3
6	ME4701	Automation in Manufacturing Lab	PCC	0	0	2	2	1
7	ME4000	Comprehensive Viva		0	0	0	0	1
8	ME4XXX	Internship		0	0	0	0	1
9	ME4901	Project – I						4
Total								**22**

FOURTH YEAR (E4) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	XX44xx	Open Elective – 2	OEC	3	0	0	3	3
2	XX44xx	Open Elective – 3	OEC	3	0	0	3	3
3	ME4902	Project – II	PROJ	0	0	12	12	6
Total								**12**

Program Elective Courses (PECs)

Sl. No.	Course Code	Course Title	Course Category
1	ME2211	Composite materials	PEC-I
2	ME2212	Nanotechnology	PEC-I
3	ME2213	Biomaterials	PEC-I
4	ME3121	Robotics	PEC-II
5	ME3122	Nontraditional Manufacturing Processes	PEC-II
6	ME3123	Green Engineering	PEC-II
7	ME3231	Automotive systems	PEC-III
8	ME3232	Production, Planning and Control	PEC-III
9	ME3233	Computational Fluid Dynamics	PEC-III
10	ME3234	Internet of things	PEC-III
11	ME3235	Mechanical Vibrations	PEC-III
12	ME4141	AI and ML for Mechanical Systems	PEC-IV
13	ME4142	Biomechanical Engineering	PEC-IV
14	ME4143	Turbomachinery	PEC-IV
15	ME4144	Additive Manufacturing Technology	PEC-IV
16	ME4145	Systems Engineering	PEC-IV
17	ME4151	Laser applications in Manufacturing systems	PEC-V
18	ME4152	Solar Technology	PEC-V
19	ME4153	Non Destructive Technology	PEC-V
20	ME4154	Finite Element Methods	PEC-V
21	ME4155	Industrial Engineering	PEC-V

Open Elective Courses (OECs)

Sl. No.	Course Code	Course Title	Course Category
1	ME3301	Smart Materials	OEC-I
2	ME3302	Finite element analysis	OEC-I
3	ME3303	Advanced Fluid Mechanics	OEC-I
4	ME4401	3D Printing Technology	OEC-II
5	ME4402	Alternate sources of Energy	OEC II
6	ME4403	Computer aided design	OEC II
7	ME4411	Nano Science and Technology	OEC III
8	ME4412	Welding Technology	OEC III
9	ME4413	Industrial Robotics	OEC III

FIRST YEAR (E1) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	MA1101	Engineering Mathematics I	BSC	3	1	0	4	4
2	PH1101	Engineering Physics	BSC	3	0	0	3	3
3	EE1101	Basic Electrical and Electronics Engineering	ESC	3	0	0	3	3
4	HS1101	English	HSMC	2	0	0	2	2
5	PH1701	Engineering Physics Lab	BSC	0	0	3	3	1.5
6	EE1701	Basic Electrical and Electronics Engineering Lab	ESC	0	0	2	2	1
7	HS1701	English Lab	HSMC	0	0	2	2	1
8	ME1701	Design Thinking	ESC	0	0	2	2	1
9	ME1702	Engineering Drawing and computer graphics	ESC	0	1	4	5	3
10	BS1101	Environmental science	MC	3	0	0	3	0
Total								**19.5**

Category: **Basic Science Course** Subject Code**: MA1101**

Engineering Mathematics I

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 1 - 0 - 4**

Course outcomes: After learning the concepts of this paper the student must be able to

1. Write the matrix representation of set of linear equations and to analyze the solution of the system of equations.
2. Find the Eigen values and Eigen vectors.
3. Reduce the quadratic form to canonical form using orthogonal transformations.
4. Analyze the nature of sequence and series.
5. Solve the applications on the mean value theorems.
6. Evaluate the improper integrals using Beta and Gamma functions.
7. Find the extreme values of functions of two variables with/without constraints.

UNIT-I

Matrix Theory: Types of Matrices, Symmetric, Hermitian, Skew-Symmetry, Skew-Hermitian, Orthogonal matrices, Unitary matrices; Elementary row and column operations on a matrix, Rank of a matrix by Echelon form and Normal form, Inverse of a Non-singular matrix by Gauss-Jordan method; Consistency and solutions of system of linear equations using elementary operations, Gauss elimination method; Gauss Seidel Iteration method.

UNIT-II

Eigen values and Eigen vectors: Linear Transformation and Orthogonal Transformation; Characteristic roots and vectors of a matrix; Diagonalization of a matrix; Cayley-Hamilton theorem(without proof) ; finding inverse and power of a matrix by Caylay-Hamilton Theorem; Quadratic forms and Nature of the Quadratic forms; Reduction of quadratic form to canonical form by Orthogonal transformation.

UNIT-III

Sequences & Series: Definition of a sequence, limit; Convergent, Divergent and Oscillatory sequences. Convergent, Divergent and Oscillatory Series; Series of positive terms; Comparison test, p-test, D-Alembert's ratio test; Raabe's test; Cauchy's Integral test; Cauchy's root test; Logarithmic test. Alternating series; Leibnitz test; Alternating Convergent series; Absolute and conditionally convergence.

UNIT-IV

Calculus: Mean value theorems: Roll's theorem, Lagrange's Mean value theorem, Cauchy's Mean value theorem; Taylor's and Macaurin's series with remainders, Expansions; Applications of definite integrals to evaluate surface area and volumes of revolutions of curves (Only in Cartesian coordinates): Definition of Improper Integrals and their convergence, Beta and Gamma functions and their applications.

UNIT-V

Multivariable Calculus (Partial Differentiation and applications): Definitions of Limits and continuity. Partial Differentiation; Euler's theorem; Total Derivative; Jacobian; Functional dependence and independence; Maxima and minima of functions of several variables (two and three variables) using Lagrange Multipliers.

Maxima and minima of functions of several variables (two and three variables) using Lagrange Multipliers.

TEXTBOOKS:

1. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, 8th Edition,
2. R.K.Jain and S.R.K.Iyengar Advanced Engineering Mathematics, Narosa Publications House.2008
3. B.S. Grewal, Higher Engineering Mathematics, Khanna Publications, 2009.

REFERENCES:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson,Reprint,2002.
2. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
3. N.P. bail and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint,2008

WEB REFERENCES:

1. https://nptel.ac.in/courses/111108157
2. https://nptel.ac.in/courses/111107112
3. https://archive.nptel.ac.in/courses/111/106/111106146/

Category: **Basic Science Course** Subject Code**: PH1101**

ENGINEERING PHYSICS

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Unit I: Vectors and Mathematical Physics (6)

Gradient, Divergence, Curl and its applications, Line, surface and volume integrals, Stokes and Gauss theorem: Applications, Curvilinear Coordinates: Polar, Cylindrical and spherical co-ordinates, Problems.

Unit II: Electrodynamics (12)

Electrodynamics before Maxwell, Fixing of Ampere's Law, Maxwell Equations in Matter, Boundary Conditions, Continuity Equation, Poynting Theorem, Wave equation for E and B, Monochromatic Plane Waves, Energy and Momentum in EM Waves. Propagation in Linear Media, Reflection and Transmission at Normal Incidence. EM Waves in Conductors, Reflection at Conducting Surface.

Unit III: Quantum Mechanics (8)

Introduction to Quantum Mechanics, De-Broglie’s waves and uncertainty principle, Wave Function and its Significance, Time dependant and time independent Schrodinger wave equations, Particle in a box - Problems.

Unit IV: Electron Structure of Solids (10)

Introduction to Crystallography, Bravais Lattices and crystal systems, Atomic Packing, Atomic Radii, Crystal Structures (SC, BCC and FCC), Miller Indices, Classical Free electron Theory, Kronig Penny model (E vs K), Band theory of solids.

Unit V: Semiconductor Physics (8)

Intrinsic and extrinsic semiconductors, Fermi level and carrier- concentration, Effect of temperature on Fermi level. Mobility of charge carriers and effect of temperature on mobility, Hall Effect, Energy band gap determination of semiconductors by four probe method, Direct and Indirect Band gap semiconductors.

Text Books:

1. Singh, M. Engg Physics: McGraw-Hill Education (India) Pvt Limited.
2. Griffiths, David J.. Introduction to Electrodynamics. United Kingdom: Pearson Education, 2014.
3. Aruldhas, G.. Quantum Mechanics. India: Phi Learning, 2008.
4. Kittel, Charles., McEuen, Paul. Introduction to solid state physics. United Kingdom: Wiley, 2005.

Category: Engineering Science Course **Subject Code: EE1101**

BASIC ELECTRICAL AND ELETRONICS ENGINEERING

Internals: 40Marks **L-T-P-C**
Externals: 60Marks **3-0-0-3**

Course Objectives:
This course introduces the concept of
1. Electrical DC and AC circuits, basic law's of electricity and methods to solve the electrical networks
2. Construction operational features of energy conversion devices i.e. transformers , DC motors and induction motors.
3. Basics of electronics, semiconductor devices and their characteristics and operational features.

Course Outcomes:
At the end of the course the student will be able to:
1. Understand the basic concept of electrical circuits under DC and AC excitation and solve basic electrical circuit problems
2. Understand basic concept and performance of transformers and motors used as various industrial drives.

UNIT- I DC CIRCUIT ANALYSIS

Electrical circuit elements: R-L-C Parameters, V–I relationship for Passive elements, Diode, Voltage and Current Independent and Dependent Sources
Circuit analysis: Kirchoff's Laws, Network reduction techniques – series, parallel, series parallel, star–to-delta, delta-to-star transformation, Source Transformation, Mesh Analysis and Nodal Analysis
Network Theorems - Thevenin's, Norton's, Maximum Power Transfer, Superposition
Step response of RL, RC and RLC circuits

UNIT- II AC CIRCUIT ANALYSIS

Single Phase AC Circuits - R.M.S. and Average values, Form Factor, steady state analysis of
series, Parallel and Series parallel Combinations of R, L and C with Sinusoidal excitation, concept of reactance, Impedance, Susceptance and Admittance – phase and phase difference, Concept of Power Factor, j-notation, complex and Polar forms of representation.
Resonance – Series resonance and Parallel resonance circuits

UNIT- III THREE PHASE AC CIRCUITS

Three phase ac circuits -Three phase EMF generation, delta and Y connections, line and phase quantities, solution of three phase circuits, balanced supply voltage and balanced load, phasor diagram, measurement of power in three phase circuits

UNIT- IV BASIC ELECTRONICS

Introduction to electronics and electronic systems, Diode and Rectifier circuits (Half and Full wave), BJT, Transistor biasing. Small signal transistor amplifiers (CE), Operational amplifiers and their basic application, Introduction to digital circuits

UNIT- V ELECTRICAL MACHINES (12 Hrs)

Transformers: Construction, EMF equation, ratings, phasor diagram on no load and full load, equivalent circuit, regulation and efficiency calculations, open and short circuit test , applications

DC machines: Construction, EMF and Torque equations, Characteristics of DC generators and motors, applications

Induction motors: The revolving magnetic field, principle of orientation, ratings, equivalent circuit, Torque-speed characteristics, applications

Text Books:

1. Hughes, Edward. Electrical Technology. United Kingdom: Longman, 1977.
2. SMITH, Stanley Parker. Norman N. Parker. Problems in Electrical Engineering Sixth Edition. United Kingdom: Constable & Company, 1954.
3. Boylestad, R.L., and L. Nashelsky. Electronic Devices and Circuit Theory. Pearson Prentice Hall, 2006
4. Millman, Jacob, and Christos C. jt auth Halkias. Millman's Electronic Devices and Circuits. India: Tata McGraw-Hill, 2007.
5. Hayt, William H.., Kemmerly, Jack Ellsworth., Durbin, Steven. Engineering Circuit Analysis. United Kingdom: McGraw-Hill, 2001.
6. Kothari, D.P., and I.J. Nagrath. Electric Machines. Tata McGraw-Hill, 2004.

Reference Books:

1. Kishore, K.L. (2008). Electronic Devices and Circuits: BS Publications.
2. Maini, A.K., & Agrawal, V. (2009). Electronic Devices and Circuits: Wiley India Pvt. Limited.
3. Jagan, N.C., &Lakshminarayana, C. (2005). Network Theory: Anshan.
4. SUDHAKAR. NETWORK THEORY -: Tata McGraw-Hill Education.
5. P. S. Bimbhra, G.C.G. Electrical Machines-I: KHANNA PUBLISHING HOUSE.

Web References:

1. https://nptel.ac.in/courses/108108076
2. https://nptel.ac.in/courses/108105053

Category: **HSM Course** Subject Code**: HS1101**

English

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **2 - 0 - 0 - 2**

Course Objectives: The course will help to

a. Improve the language proficiency of students in English with an emphasis on Vocabulary, Grammar, Reading and Writing skills.

b. Equip students to study academic subjects more effectively and critically using the theoretical and practical components of English syllabus.

c. Develop study skills and communication skills in formal and informal situations.

Course Outcomes: Students should be able to

1. Use English Language effectively in spoken and written forms.
2. Comprehend the given texts and respond appropriately.
3. Communicate confidently in various contexts and different cultures.
4. Acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.

UNIT –I

'The Raman Effect' from the prescribed textbook 'English for Engineers' published by Cambridge University Press.

Vocabulary Building: The Concept of Word Formation --The Use of Prefixes and Suffixes. **Grammar**: Identifying Common Errors in Writing with Reference to Articles and Prepositions. **Reading**: Reading and Its Importance- Techniques for Effective Reading. **Basic Writing Skills**: Sentence Structures -Use of Phrases and Clauses in Sentences Importance of Proper Punctuation- Techniques for writing precisely – Paragraph writing – Types, Structures and Features of a Paragraph - Creating Coherence-Organizing Principles of Paragraphs in Documents.

UNIT –II

'Ancient Architecture in India' from the prescribed textbook 'English for Engineers' published by Cambridge University Press.

Vocabulary: Synonyms and Antonyms. **Grammar:** Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement. **Reading:** Improving Comprehension Skills – Techniques for Good Comprehension. **Writing:** Format of a Formal Letter-Writing Formal Letters E.g.., Letter of Complaint, Letter of Requisition, Job Application with Resume.

UNIT –III

‘Blue Jeans’ from the prescribed textbook ‘English for Engineers’ published by Cambridge University Press.
Vocabulary: Acquaintance with Prefixes and Suffixes from Foreign Languages in English to form Derivatives-Words from Foreign Languages and their Use in English. **Grammar:** Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses.
Reading: Sub-skills of Reading- Skimming and Scanning. **Writing:** Nature and Style of Sensible Writing- Defining- Describing Objects, Places and Events – Classifying- Providing Examples or Evidence

UNIT –IV

‘What Should You Be Eating’ from the prescribed textbook ‘English for Engineers’ published by Cambridge University Press.

Vocabulary: Standard Abbreviations in English. **Grammar:** Redundancies and Clichés in Oral and Written Communication. **Reading:** Comprehension- Intensive Reading and Extensive Reading. **Writing:** Writing Practices--Writing Introduction and Conclusion - Essay Writing-Précis Writing.

UNIT –V

‘How a Chinese Billionaire Built Her Fortune’ from the prescribed textbook ‘English for Engineers’ published by Cambridge University Press.

Vocabulary: Technical Vocabulary and their usage. **Grammar:** Common Errors in English. **Reading**: Reading Comprehension-Exercises for Practice. **Writing**: Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report.

Textbook:

1. Sudarshana, N.P. and Savitha, C. (2018). English for Engineers. Cambridge University Press.

References:

1. Swan, M. (2016). Practical English Usage. Oxford University Press.
2. Kumar, S and Lata, P.(2018). Communication Skills. Oxford University Press.
3. Wood, F.T. (2007).Remedial English Grammar. Macmillan.
4. Zinsser, William. (2001). On Writing Well. Harper Resource Book.
5. Hamp-Lyons, L. (2006).Study Writing. Cambridge University Press.
6. Exercises in Spoken English. Parts I –III. CIEFL, Hyderabad. Oxford University Press

Category: **Basic Science Course** Subject Code**: PH1701**

Engineering Physics laboratory

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 3 - 1.5**

List of experiments:

1. Photoelectric effect

2. Hall effect

3. Ultrasonic Interferometer

4. Melde's Experiment

5. Four probe Method

6. Frank hertz Experiment

7. Seebeck and Peltier effect

8. Solar cell

9. Couple pendulum

10. Dipersive power of prism

11. Diffraction Grating

12.Fly wheel

Category: **Engineering Science Course** Subject Code**: EE1701**

Basic Electrical and Electronics Engineering laboratory

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 2 - 1**

COURSE OBJECTIVE:

To provide practical exposure to

➢ To expose the students to the concepts of electrical and electronics circuits and their applications

➢ To expose the students to the operation of dc machines and transformer and give them experimental skills.

COURSE OUTCOMES:

Student will be able to

- Understand principles of measuring instruments of voltage, current and power
- Analyze the characteristics of semiconductor devices and understand their applications
- Analyze the characteristics and evaluate performance of DC machines and transformers

LIST OF EXPERIMENTS:

1. Introduction to Lab:

(a) Basic safety precautions. Introduction and use of measuring instruments – voltmeter, ammeter, multi-meter, oscilloscope. Real-life resistors, capacitors and inductors.

(b) Demonstration of cut-out sections of machines: dc machine (commutator-brush arrangement), induction machine (squirrel cage rotor), synchronous machine (field winging - slip ring arrangement) and single-phase induction machine**.**

2. Measuring the steady-state and transient time-response of R-L, R-C, and R-L-C circuits to a step change in voltage (transient may be observed on a storage oscilloscope). Sinusoidal steady state response of R-L, and R-C circuits – impedance calculation and verification. Observation of phase differences between current and voltage. Resonance in R-L-C circuits.

3. Transformers: Observation of the no-load current waveform on an oscilloscope (non sinusoidal wave-shape due to B-H curve nonlinearity should be shown along with a discussion about harmonics). Loading of a transformer: measurement of primary and secondary voltages and currents, and powe**r**

4. Open circuit & short circuit test on single phase transformer.
5. Verification of KCL&KVL.
6. Characteristic of the lamps (Tungsten, Fluorescent and Compact Fluorescent Lamps)
7. Verification of Network Theorems.
8. V-I characteristics of Diodes and BJT
9. Half-wave and full-wave rectifiers, rectification with capacitive filters, zener diode
10. Studies on logic gates

Category: **HSM Course** Subject Code**: HS1701**

English Laboratory

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0 - 0 - 2 - 1**

Course Objectives:

- To facilitate computer-assisted multi-media instruction enabling individualized and independent language learning.
- To sensitize students to the nuances of English speech sounds, word accent, intonation and rhythm.
- To bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking
- To improve the fluency of students in spoken English and neutralize their mother tongue influence
- To train students to use language appropriately for public speaking and interviews

Learning Outcomes: Students will be able to attain

- Better understanding of nuances of English language through audio- visual experience and group activities
- Neutralization of accent for intelligibility
- Speaking skills with clarity and confidence which in turn enhances their employability skills

UNIT – I

Understand: Listening Skill- Its importance – Purpose- Process- Types- Barriers of Listening -Communication at Work Place- Spoken vs. Written language.

Practice: Introduction to Phonetics – Speech Sounds – Vowels and Consonants -Ice-Breaking Activity and JAM Session- Situational Dialogues – Greetings – Taking Leave – Introducing Oneself and Others.

UNIT – II

Understand: Structure of Syllables – Word Stress and Rhythm– Weak Forms and Strong Forms in Context- Features of Good Conversation – Non-verbal Communication.

Practice: Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms in Context-Situational Dialogues – Role-Play- Expressions in Various Situations –Making Requests and Seeking Permissions - Telephone Etiquette.

UNIT – III

Understand: Intonation-Errors in Pronunciation-the Influence of Mother Tongue (MTI)- How to make Formal Presentations.

Practice: Common Indian Variants in Pronunciation – Differences in British and American Pronunciation- Formal Presentations.

UNIT – IV

Understand: Listening for General Details-Public Speaking – Exposure to Structured Talks.
Practice: Listening Comprehension Tests- Making a Short Speech – Extempore

UNIT – V

Understand: Listening for Specific Details- Interview Skills.
Practice: Listening Comprehension Tests- Mock Interviews.

Suggested References:

1. Clarity English Success - Software
2. Connected Speech- Software
3. Issues in English 2- Software
4. http://www.clarityenglish.com/program/practicalwriting/
5. http://www.clarityenglish.com/program/roadtoielts/
6. http://www.clarityenglish.com/program/clearpronunciation1/
7. http://www.clarityenglish.com/program/resultsmanager/

Category: **Engineering Science Course** Subject Code**: ME1701**

DESIGN THINKING

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 2 - 1**

COURSE OUTCOMES:

Upon Completion of the course, the students should be able to:

- Define key concepts of design thinking
- Practice design thinking in all stages of problem solving
- Apply design thinking approach to real world problems

UNIT I: INTRODUCTION

Why Design? - Four Questions, Ten Tools - Principles of Design Thinking - The process of Design Thinking – How to plan a Design Thinking project.

UNIT II: UNDERSTAND, OBSERVE AND DEFINE THE PROBLEM

Search field determination - Problem clarification - Understanding of the problem – Problem analysis- Reformulation of the problem - Observation Phase - Empathetic design - Tips for observing -Methods for Empathetic Design - Point-of-View Phase - Characterization of the target group - Description of customer needs.

UNIT III: IDEATION AND PROTOTYPING

Ideate Phase - The creative process and creative principles - Creativity techniques - Evaluation of ideas - Prototype Phase - Lean Startup Method for Prototype Development - Visualization and presentation techniques.

UNIT IV: TESTING AND IMPLEMENTATION

Test Phase - Tips for interviews - Tips for surveys - Kano Model - Desirability Testing - How to conduct workshops - Requirements for the space - Material requirements - Agility for Design Thinking.

UNIT V: FUTURE

Design Thinking meets the corporation – The New Social Contract – Design Activism – Designing tomorrow.

TEXT BOOKS:

1. Christian Mueller-Roterberg, Handbook of Design Thinking - Tips & Tools for how to design thinking. [Unit 1, 2, 3, 4]
2. Designing for Growth: a design thinking tool kit for managers By Jeanne Liedtka and Tim Ogilvie. [Unit 1]
3. Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation by Tim Brown. [Unit 5]

REFERENCES:

1. Johnny Schneider, "Understanding Design Thinking, Lean and Agile", O'Reilly Media, 2017.
2. Roger Martin, "The Design of Business: Why Design Thinking is the Next Competitive Advantage", Harvard Business Press , 2009.
3. Hasso Plattner, Christoph Meinel and Larry Leifer (eds), "Design Thinking: Understand – Improve – Apply", Springer, 2011

WEB REFERENCES:

- http://ajjuliani.com/design-thinking-activities/
- https://venturewell.org/class-exercises
- https://themechanicalengineering.com/
- https://mechanicalengineering.blog/
- https://engineerine.com/
- https://www.mechanicalbooster.com/

Category: **Engineering Science Course** Subject Code**: ME1702**

Engineering Drawing and computer graphics

Internals: 50 Marks **L - T - P - C**

Externals: 50 Marks **1 - 0 - 4 - 3**

Course Objectives:

- To introduce the students to the "Universal Language of Engineers" for effective communication through drawing.
- To understand the basic concepts of drawing through modern techniques.
- To impart knowledge about standard principles of projection of objects.
- To provide the visual aspects of Engineering drawing using AutoCAD.

Course Outcomes: At the end of the course, the student will be able to

- Use Engineering principles and techniques to understand and interpret engineering drawings.
- Understand the concepts of AutoCAD.
- Draw orthographic projections of lines, planes and solids using AutoCAD.
- Use the techniques, skills and modern engineering tools necessary for engineering practices.

UNIT-1:

Introduction to Engineering Drawing: Principles of Engineering Graphics and their significance, usage of Drawing instruments, lettering, types of lines and Dimensioning.
Over view of AutoCAD: Theory of CAD software (The Menu System, Tool Bars, Drawing area, Dialogue boxes, Shortcut Menu, the command lines, Select and erase objects, Introduction to layers etc.), Drawing simple figures- lines, planes, solids.

UNIT-II

Geometrical constructions: Construction of regular polygons.
Conic sections: Construction of Ellipse, Parabola, Hyperbola (General method only), Cycloid, Epicycloid, Hypocycloid and Involutes.
Scales: Construction of Plain, Diagonal and Vernier scales.

UNIT-III

Orthographic projections: Principles of Orthographic Projections
Projections of Points: Projections of Points placed in different quadrants
Projection of lines: lines parallel and inclined to both the planes (Determination of true lengths and true inclinations and traces)
Projection of planes: Planes inclined to both the reference planes

UNIT-IV

Projection of Solids: Projection of solids whose axis is parallel to one of the reference planes and inclined to the other plane, axis inclined to both the planes
Projection of sectioned solids: Sectioning of simple solids like prism, pyramid, cylinder and cone in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section.

UNIT-V

Development of surfaces: .Development of surfaces of Right Regular Solids - Prism, Pyramid, Cylinder and Cone

Isometric Projections: Principles of Isometric projection – Isometric Scale, Isometric Views of planes and simple solids

Perspective projections: Basic concepts of perspective views.

Text/Reference Books:

1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing House,2014
2. Gopalakrishna K.R., "Engineering Drawing" (Vol. I&II combined), Subhas Stores, Bangalore, 2007.
3. Shah, M.B. &Rana B.C, Engineering Drawing and Computer Graphics, Pearson Education,2008
4. Venugopal K. and Prabhu Raja V., "Engineering Graphics", New Age publications
5. Agrawal B. & Agrawal C. M., Engineering Graphics, TMH Publication 2012,
6. Narayana, K.L. & P Kannaiah, Text book on Engineering Drawing, Scitech Publishers, 2008
7. (Corresponding set of) CAD Software Theory and User Manuals

WEB REFERENCES

8. https://nptel.ac.in/courses/112103019
9. https://www.youtube.com/@TIKLESACADEMY

COURSE ASSESSMENT METHODS		
Assessment Method	**Description**	**Weight**
Assignment & Continuous monitoring	Assignments & Evaluation of drawings	25%
Mid Term	2nd mid examinations will be conducted and best of 2 will be considered.	25%
End Term	Students will be evaluated based on the understanding of the principles, skills and practices of the course	50%

Category: **Mandatory Course** Subject Code**: BS1001**

ENVIRONMENTAL SCIENCE

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 0**

Learning Objectives:

1. Stimulate interest in the environment and endeavours to generate awareness about environmental concerns among students.
2. Develop an understanding of how natural resources and the environment affect quality of life and the quest for sustainable development.
3. Develop knowledge and understanding of environmental issues and principle and apply their knowledge to mitigate the environmental problems.
4. Understand and resolve some of today's most challenging scientific and policy issues—including global climate change, pollution, biodiversity conservation, sustainability, environmental pollution and toxic waste disposal, disease control, disaster management, socio-environmental issues and balancing resource use and preservation.
5. Design and evaluate strategies, technologies, and methods for sustainable management of environmental systems and for the remediation or restoration of degraded environments.
6. Recognizes the global changes and responses for attaining a more sustainable environment.

LEARNING OUTCOMES:

The Environmental Science minor supplements other majors to facilitate students' understanding of complex environmental issues from a problem-oriented, interdisciplinary perspective. Students:

- ❖ Understand core concepts and methods from ecological and physical sciences and their application in environmental problem-solving.
- ❖ Appreciate key concepts from economic, political, and social analysis as they pertain to the design and evaluation of environmental policies and institutions.
- ❖ Appreciate the ethical, cross-cultural, and historical context of environmental issues and the links between human and natural systems.
- ❖ Appreciate that one can apply systems concepts and methodologies to analyze and understand interactions between social and environmental processes.
- ❖ Reflect critically about their roles and identities as citizens, consumers and environmental actors in a complex.

UNIT 1: MULTIDISCIPLINARY NATURE OF ENVIRONMENTAL STUDIES

Definition, scope and importance, need for public awareness.

UNIT 2: NATURAL RESOURCES:

Renewable and non-renewable resources : Natural resources and associated problems.

a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people.

b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems.

c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.

d) Food resources: World food problems, changes caused by agriculture and over-grazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies.

e) Energy resources: Growing energy needs, renewable and non renewable energy sources, use of alternate energy sources.

f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification.

- .Role of an individual in conservation of natural resources.
- Equitable use of resources for sustainable lifestyles.

UNIT 3: ECOSYSTEMS & BIODIVERSITY

Concept of an ecosystem. Structure and function of an ecosystem. Producers, consumers and decomposers. Energy flow in the ecosystem. Ecological succession. Food chains, food webs and ecological pyramids.

Introduction, types, characteristic features, structure and function of the following ecosystems:-

a. Forest ecosystem, b. Grassland ecosystem, c. Desert ecosystem, d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries).

b. Biodiversity- Definition : genetic, species and ecosystem diversity. Biogeographical classification of India Value of biodiversity : consumptive use, productive use, social, ethical, aesthetic and option values.

c. Biodiversity at global, National and local levels. Inida as a mega-diversity nation Hot-sports of biodiversity.

d. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT 4: ENVIRONMENTAL POLLUTION

Definition, Cause, effects and control measures of :- Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, Nuclear hazards

- Solid waste Management: Causes, effect s and control measures of urban and industrial wastes.
- Role of an individual in prevention of pollution
- Pollution case studies.
- Disaster management: floods, earthquake, cyclone and landslides.
- Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies.
- Environment Protection Act., Air (Prevention and Control of Pollution) Act. Water Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act .

UNIT 5 : SOCIAL ISSUES & THE ENVIRONMENT

Human Rights. Value Education. HIV/AIDS. Women and Child Welfare. Role of Information Technology in Environment and human health.

Field work : Visit to a local area to document t environmental assets river/ forest/grassland/hill/mountain Visit to a local polluted site-Urban/Rural/Industrial/Agricultural . Study of common plants, insects, birds. Study of simple ecosystems-pond, river, hill slopes, etc.

REFERENCES :

1. Agrawal, K. C.. Fundamentals Of Environmental Biology. India: Nidhi Publishers, 2001.
2. Bharucha, Erach. The Biodiversity of India. India: Mapin Pub., 2002.
3. Reynolds, Joseph., Theodore, Louis., Santoleri, Joseph J.., Reynolds, Joseph P.. Introduction to Hazardous Waste Incineration. United Kingdom: Wiley, 2000.
4. Clark, R. B.. Marine Pollution. United Kingdom: Oxford University Press, 2001.
5. Environmental Encyclopedia. United States: Gale/Thomson-Gale, 2003

FIRST YEAR (E1) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	MA1201	Engineering Mathematics II	BSC	3	1	0	4	4
2	CY1201	Engineering Chemistry	BSC	3	0	0	3	3
3	CS1201	Programming for Problem Solving	ESC	3	0	0	3	3
4	ME1201	Engineering Mechanics	ESC	3	1	0	4	4
5	ME1202	Introduction to Manufacturing processes	PCC	2	0	0	2	2
6	CS1801	Programming for Problem Solving Lab	ESC	0	0	3	3	1.5
7	ME1801	Fuels and Lubricants Lab	BSC	0	0	2	2	1
8	ME1802	Engineering workshop	ESC	0	1	2	3	2
9	BM1201	Constitution of India	MC	2	0	0	2	0
Total								**20.5**

Category: **Basic Science Course** Subject Code**: MA1201**

Engineering Mathematics II

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 1 - 0 - 4**

Course Outcomes:

At the end of the course student will be able to

- Solve first order linear differential equations and special non linear first order equations like Bernouli , Riccati & Clairaut's equations
- Compute double integrals over rectangles and type I and II" regions in the plane
- Explain the concept of a vector field and make sketches of simple vector fields in the plane.
- Explain concept of a conservative vector field, state and apply theorems that give necessary and sufficient conditions for when a vector field is conservative, and describe applications to physics.
- Recognize the statements of Stokes' Theorem and the Divergence Theorem and understand how they are generalizations of the Fundamental Theorem of Calculus.
- Able to solve the problems in diverse fields in engineering science using numerical methods.

UNIT-I

Ordinary Differential Equations of first order: Exact first order differential equation, finding integrating factors, linear differential equations, Bernoulli's , Riccati , Clairaut's differential equations, finding orthogonal trajectory of family of curves, Newton's Law of Cooling, Law of Natural growth or decay.

UNIT-II

Ordinary Differential Equations of higher order: Linear dependence and independence of functions, Wronskian of n- functions to determine Linear Independence and dependence of functions, Solutions of Second and higher order differential equations (homogeneous & non-homogeneous) with constant coefficients, Method of variation of parameters, Euler-Cauchy equation.

UNIT-III

Integral Calculus: Convergence of improper integrals, tests of convergence, Beta and Gamma functions - elementary properties, differentiation under integral sign, differentiation of integrals with variable limits - Leibnitz rule. Rectification, double and triple integrals, computations of surface and volumes, change of variables in double integrals - Jacobians of transformations, integrals dependent on parameters – applications.

UNIT-IV

Laplace Transform (LT): Improper integrals; Beta and Gamma functions and their properties.Definition and existence of LT, LT of elementary functions, First and second shifting properties,Change of scale property, LT of , LT of , LT of derivatives of $f(t)$, LT of integral of $f(t)$, Evaluation of improper integrals using LT, LT of periodic and step functions,

Inverse LT: Definition and its properties, Convolution theorem (statement only) and its application to the evaluation of inverse LT, Solution of linear ODE with constant coefficients (initial value problem) using LT

UNIT-V

Vector Differentiation: Vector point functions and scalar point functions. Gradient, Divergence and Curl. Directional derivatives, Tangent plane and normal line. Vector Identities. Scalar potential functions. Solenoidal and Irrotational vectors.

Vector Integration: Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications. Numerical Methods: Introduction and motivation about numerical methods, True value, approximate value, error, error percentage, algebraic equations, transcendental equations, Newton-Raphson method, Bisection method.

Text Books:

1. Jain, Rajinder Kumar, and Satteluri RK Iyengar. Advanced engineering mathematics. Alpha Science Int'l Ltd., 2007.

References Books

1. Kreyszig, E. (2017). Advanced Engineering Mathematics: Wiley.
2. M.D., Raisinghania. Ordinary and Partial Differential Equations, 20th Edition: S. Chand Publishing.

Category: Basic Science Course **Subject Code:** CY1201

Engineering Chemistry

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

Unit I: Electrochemistry

Introduction to electrochemistry: Galvanic cell (Daniel cell), Nernst equation. Types of electrodes: metal-metal ion electrodes, metal-insoluble salt-anion electrodes, calomel electrode, gas-ion electrodes, hydrogen and chlorine electrodes, oxidation-reduction electrodes (quinhydrone electrode), amalgam electrodes and ion exchange electrode (glass electrode). EMF and applications of EMF: determination of pH of the solution, potentiometric titrations, Classification of commercial cells - primary cells (dry cell) and secondary cells (Lithium ion battery, Pb-acid storage battery). Fuel cells: H2-O2 fuel cell.

UNIT - II: Corrosion and water treatment.

Dry and wet corrosion and their mechanisms. Pilling - Bedworth Rule. Types of Corrosion: galvanic corrosion, concentration cell corrosion, pitting corrosion and stress corrosion. Factors influencing the rate of corrosion: Temperature, pH and dissolved oxygen. Corrosion Prevention methods: Cathodic protection – Sacrificial Anodic method and Impressed current method. Metallic coatings: galvanization and tinning methods. Water: Hardness of water, Degrees of hardness. Calculation of hardness by EDTA method. Disadvantages of hard water in boilers: priming, foaming, scales, sludges and caustic embrittlement. Treatment of boiler feed water: Zeolite process, Ion exchange process.

UNIT - III: Energy sources

Introduction. Definition and classification of fuels. Calorific value of a fuel, Characteristics of a good fuel. Coal, types of Coal. Analysis of Coal: Proximate and Ultimate analysis. Bomb Calorimeter and Junker's gas Calorimeter. Problems on calculation of calorific value. Liquid fuels Introduction .Synthetic Petrol: Fisher Tropsch process. Introduction to Bio-fuels: Bio-diesel, Biogas

Unit IV: Chemical kinetics

Introduction to rate of reaction and rate constant determination. Factors influencing rate of reaction. Complex reactions: definition and classification of complex reactions, definition of reversible reactions with examples, rate law derivation for reversible reactions. Consecutive reactions: definition, rate law derivation and examples of consecutive reactions. Parallel reactions: definition, rate law derivation and examples of parallel reactions. Steady-state approximation: introduction, kinetic rate law derivation by applying steady state approximation in case of the oxidation of NO and pyrolysis of methane.

UNIT - V: Nanochemistry

Introduction to nanomaterials, classification: Carbon based nanomaterials, metallic nanoparticles, metal oxide nanoparticles. Properties at nanoscale. Synthetic approaches: Top-Down (Photolithography, ball milling) and Bottom-Up (Sol-gel, Hydrothermal). Brief overview on characterization of nanomaterials: X-ray, SEM and TEM. Applications of nanomaterials

Reference Books

1. Jain P C. Engineering Chemistry. N.P.: Dhanpat Rai, 1997.
2. Chemistry for Engineers. India: Laxmi Publications Pvt Limited, 2008.
3. Srivastava, H. C.. Engineering Chemistry. India: Pragati Prakashan, 2011.
4. Agarwal, Shikha. Engineering Chemistry: Fundamentals and Applications. India: Cambridge University Press, 2019.

Category: **Engineering Science Course** Subject Code**: CS1201**

Programming for Problem Solving

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT-I: Introduction to Programming & Arithmetic expressions and precedence, Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.). **Idea of Algorithm:** steps to solve logical and numerical problems. **Representation of Algorithm:** Flowchart/Pseudocode with examples. From algorithms to programs; source code, variables (with data types) variables and memory locations, Syntax and Logical Errors in compilation, object and executable code. Arithmetic expressions and precedence.

UNIT-II: Conditional Branching , Loops & Arrays, Writing and evaluation of conditionals and consequent branching, Iteration and loops, Arrays (1-D, 2-D), Character arrays and Strings.

UNIT-III: Function & Basic Algorithms, Functions (including using built in libraries), Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference, Searching, Basic Sorting Algorithms (Bubble, Insertion and Selection), Finding roots of equations, notion of order of complexity through example programs (no formal definition required)

UNIT-IV: Recursion & Structure, Recursion, as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc. Quick sort or Merge sort, Structures, Defining structures and Array of Structures

UNIT-V: Pointers & File handling, Idea of pointers, Defining pointers, Use of Pointers in self-referential structures, notion of linked list (no implementation) File handling (only if time is available, otherwise should be done as part of the lab)

Text Books

1. Gottfried, Byron S.. Schaum's Outline of Programming with C. United States: McGraw-Hill Education, 1996.
2. Balagurusamy. (2008). Programming in ANSI C: Tata McGraw-Hill.

Reference Books

1. Kernighan, B.W., & Ritchie, D.M. (1988). The C Programming Language: 2nd Edition: Prentice Hall.

Category: Professional core course **Subject Code: ME1201**

Engineering Mechanics

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 1 - 0 - 4**

Course Outcomes:

CO1: To understand representation of force, moments for drawing free-body diagrams and analyze friction-based systems in static condition

CO2: To locate the centroid of an area and calculate the moment of inertia of a section. CO3: Apply of conservation of momentum & energy principle for particle dynamics and rigid body kinetics

CO4: Understand and apply the concept of virtual work, rigid body dynamics and systems under vibration

Unit I: Introduction to Engineering Mechanics

Force Systems Basic concepts, Particle equilibrium in 2-D & 3-D; Rigid Body equilibrium; System of Forces, Coplanar Concurrent Forces, Components in Space – Resultant- Moment of Forces and its Application; Couples and Resultant of Force System, Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems and Spatial Systems; Vector Mechanics- dot product, cross product, Problems

Unit II: Structural analysis and friction

Equilibrium in three dimensions; Method of Sections; Method of Joints; How to determine if a member is in tension or compression; Simple Trusses; Zero force members; Beams & types of beams; Frames & Machines, Problems

Types of friction, Limiting friction, Laws of Friction, Static and Dynamic Friction; Motion of Bodies, wedge friction, screw jack & differential screw jack, Problems

Unit III: Properties of surfaces and solids

Distributed Force: Centroid and Centre of Gravity; Centroids of a triangle, circular sector, quadrilateral, etc., Centroid of simple figures from first principle, centroid of composite sections; Centre of Gravity and its implications, Problems

Area moment of inertia- Definition, Moment of inertia of plane sections from first principles, Theorems of moment of inertia, Moment of inertia of standard sections and composite sections; Mass moment inertia of circular plate, Cylinder, Cone, Sphere, Hook, Problems

Unit IV: Virtual Work and Energy Method:

Virtual displacements, principle of virtual work for particle and ideal system of rigid bodies, degrees of freedom. Active force diagram, systems with friction, mechanical efficiency. Conservative forces and potential energy (elastic and gravitational), energy equation for equilibrium. Applications of energy method for equilibrium. Stability of equilibrium, Problems

Unit V: Dynamics

Rectilinear motion; Plane curvilinear motion (rectangular, path, and polar coordinates). 3-D curvilinear motion; Relative and constrained motion; Newton's 2nd law (rectangular, path, and polar coordinates). Work-kinetic energy, power, potential energy. Impulse-momentum (linear, angular); Impact (Direct and oblique), Problems

Basic terms, general principles in dynamics; Types of motion, Instantaneous centre of rotation in plane motion and simple problems; D'Alembert's principle and its applications in plane motion and connected bodies;

Text books:

1. Irving H. Shames (2006), Engineering Mechanics, 4th Edition, Prentice Hall
2. F. P. Beer and E. R. Johnston (2011), Vector Mechanics for Engineers, Vol I - Statics, Vol II, – Dynamics, 9th Ed, Tata McGraw Hill
3. R.C. Hibbler (2006), Engineering Mechanics: Principles of Statics and Dynamics, Pearson Press.
4. Andy Ruina and Rudra Pratap (2011), Introduction to Statics and Dynamics, Oxford University Press
5. Shanes and Rao (2006), Engineering Mechanics, Pearson Education,
6. Hibler and Gupta (2010), Engineering Mechanics (Statics, Dynamics) by Pearson Education

Reference books:

1. Reddy Vijaykumar K. and K. Suresh Kumar (2010), Singer's Engineering Mechanics
2. Bansal R.K. (2010), A Text Book of Engineering Mechanics, Laxmi Publications
3. Khurmi R.S. (2010), Engineering Mechanics, S. Chand & Co.
4. Tayal A.K. (2010), Engineering Mechanics, Umesh Publications

Web References

1. https://nptel.ac.in/courses/112106286
2. https://www.digimat.in/nptel/courses/video/112103108/L01.html
3. https://unacademy.com/course/engineering-mechanics-gate-mechanical/J692OFFK

Category: **Engineering Science Course** Subject Code**: ME1202**

Introduction to Manufacturing processes

Internals (Theory) : 40 Marks **L - T - P - C**
Externals (Practical) : 60 Marks **2 - 0 - 0 - 2**

(i) THEORY

Course Objectives:

- To understand the basic manufacturing process of producing a component by casting, forming plastic molding, joining processes, machining of a component either by conventional or by unconventional processes.
- To understand the advanced manufacturing process of additive manufacturing process.

Course Outcome:

- Students will gain knowledge of the different manufacturing processes which are commonly employed in the industry, to fabricate components using different materials.

Module – 1: ***Metal Casting*****:** Introduction, Tools, Types of Patterns, Pattern Materials, Types of casting – Sand, Die and other casting processes and Applications

Module – 2: ***Metal Forming*****:** Introduction, Classification, Types of Bulk and sheet metal forming and Applications.

Module – 3: ***Powder Metallurgy*****:** Introduction, Powder production methods, Compaction, Sintering, Secondary operations and Applications.

Module – 4: ***Joining:*** Types of Joining, Introduction to Welding, Brazing and soldering, Arc, Solid state welding processes.

Module – 5: ***Conventional Machining processes*****:** Introduction to machining operations; Lathe operations, Drilling, Milling and Grinding.

Module – 6: ***Unconventional Machining processes***.

Module – 7: ***CNC Machining and Additive manufacturing***

Text Books:

1. Elements Of Workshop Technology Volume –1&2. India, Indian Book Distributing Company Calcutta, 2010.
2. Kalpakjian, Serope. Manufacturing Engineering and Technology. India, Pearson Education, 2001..

Reference Books

1. Gowri P. Hariharan and A. Suresh Babu,"Manufacturing Technology – I" Pearson Education, 2008.
2. Roy A. Lindberg, "Processes and Materials of Manufacture", 4th edition, Prentice Hall India, 1998.
3. Rao P.N., "Manufacturing Technology", Vol. I and Vol. II, Tata McGrawHill House, 2017.

Web References

1. https://onlinecourses.nptel.ac.in/noc20_me67/preview
2. https://nptel.ac.in/courses/112107144

Category: **Engineering Science Course** Subject Code**: CS1801**

Programming for Problem Solving Laboratory

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0 - 0 - 3 - 1.5**

List of Experiments:

1. Familiarization with programming environment
2. Simple computational problems using arithmetic expressions
3. Problems involving if-then-else structures
4. Iterative problems e.g., sum of series
5. 1D Array manipulation
6. Matrix problems, String operations
7. Simple functions
8. Recursive functions
9. Pointers and structures
10. File operations

Category: **Basic Science Course** Subject Code**: ME1801**

Fuels and Lubricants Laboratory

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 2 - 1**

List of Experiments:

1. Determination Of Flash Point And Fire Point Of Liquid Fuels/Lubricants Using Ables Apparatus.
2. Determination Of Flash Point And Fire Point Of Liquid Fuels/Lubricants Using Pesky Martens Test
3. Carbon Residue Test: Liquid Fuels.
4. Determination Of Viscosity Of Liquid Lubricants And Fuels Using Saybolt Viscometer
5. Determination Of Viscosity Of Liquid Lubricants And Fuels Using red wood viscometer-I & II.
6. Determination Of Viscosity Of Liquid Lubricants And Fuels Using engler viscometer
7. Determination of calorific value of gaseous fuels using Junkers gas calorimeter.
8. Determination of calorific value of solid/liquid fuels using bomb calorimeter
9. Drop point and penetration apparatus for grease
10. ASTM distillation test apparatus
11. Cloud and pour point apparatus

Category: **Engineering Science Course** Subject Code**: ME1802**

Engineering Workshop

Internals: 40 Marks **L - T - P - C**

Externals : 60 Marks **0 - 1 - 2 - 2**

Course Outcomes: Upon completion of this laboratory course

- Students will be able to fabricate components with their own hands.

List of Experiments:

1. **Fitting –** To produce a Step Fit on the given workpiece.

   - To produce a V Fit on the given workpiece.

2. **Carpentry** – To produce a Half lap joint on the given wooden work part.

   - To produce a Dove tail joint on the given wooden work part.

3. **House Wiring** – To perform and understand the Series and Parallel wiring connections.

   - To perform and understand Staircase and Godown wiring connections.

4**. Tin Smithy** – To produce a Tray from the given sheet metal.

   - To produce a Cylinder from the given sheet metal.

5. **Welding** – To practice formation of a Bead on the given workpiece.

   - To perform a Butt and a Lap joint on the given workpiece.

6**. Foundry** – To prepare a Mold cavity using a Single piece pattern.

   - To prepare a Mold cavity using a Split piece pattern.

7. **Machining** – To perform a Plain turning operation, Facing operation on the given workpiece.

   - To perform a Step and a Taper turning operation on the given workpiece.

8. **Plastic molding** – Demonstration

9. **WIRE EDM, CNC, 3D Printer** - Demonstration

SECOND YEAR (E2) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	MA2101	Engineering Mathematics III	BSC	3	1	0	4	4
2	ME2101	Kinematics of Machinery	PCC	3	0	0	3	3
3	ME2102	Strength of Materials	PCC	3	1	0	4	4
4	ME2103	Thermodynamics	PCC	3	0	0	3	3
5	ME2204	Materials Engineering	PCC	3	0	0	3	3
6	ME2701	Strength of Materials Lab	PCC	0	0	2	2	1
7	ME2702	Materials Engineering Lab	PCC	0	0	2	2	1
8	ME2703	Computational mathematics lab	ESC	0	0	2	2	1
9	HS210X	Start up Entrepreneurship	MC	0	0	2	2	0
Total								**20**

Category: **Basic Science Course** Subject Code**: MA2101**

Engineering Mathematics III

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 1 - 0 - 4**

UNIT-I: Complex Differentiation

Limit, Continuity and Differentiation of Complex functions, Analyticity, Cauchy-Riemann equations (without proof), finding harmonic conjugate, elementary analytic functions (exponential, trigonometric, logarithm) and their properties, Conformal mappings, Mobius transformations.

UNIT-II: Complex Integration

Line integral, Cauchy's theorem, Cauchy's Integral formula, Zeros of analytic functions, Singularities, Taylor's series, Laurent's series, Residues, Cauchy Residue theorem (All theorems without Proof)..

Unit-III:

Numerical Methods

Introduction to error analysis, Calculus of finite difference. Interpolation: Newton forward and backward interpolation, Lagrange's interpolation, Newton's divided difference interpolation formula.

Numerical integration: Trapezoidal rule, Simpson's 1/3 rule, Weddle's rule. Numericalsolution of ordinary differential equation: Euler method, Modified Euler method, Fourth order Runge-Kutta method

UNIT-IV:

Probability

Random Variables and Probability Distributions: Concept of a Random Variable, Discrete Probability Distributions, Continuous Probability Distributions, Statistical Independence, Definition of Cumulative Distribution function and its properties for Discrete and Continuous distributions.

Mathematical Expectation: Mean of a Random Variable, Variance and Covariance of Random Variables, Means and Variances of Linear Combinations of Random Variables, Chebyshev's Theorem.

Discrete Distributions:Binomial, Poisson, Negative-Binomialdistributions, Geometric Distribution.

UNIT-V:

Continuous Distributions: Uniform Distribution, Normal Distribution, Exponential Distribution.

Fundamental Sampling Distributions: Random Sampling, Some Important Statistics, SamplingDistributions, Sampling Distribution of Means and the Central Limit Theorem, Sampling Distribution of S2, t –Distribution, F-Distribution.

Textbooks/References:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9^{th} Edition, John Wiley & Sons, 2006.
2. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2010.
3. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003.
4. S. Ross, A First Course in Probability, 6th Ed., Pearson Education India, 2002.
5. Advanced Engineering Mathematics (3rd Edition) by R. K. Jain and S. R. K. Iyengar, Narosa Publishing House, New Delhi
6. Dr. M.D. Raisinghania,Ordinary andPartialdifferentialequations,S.CHAND,17^{th} Edition,2014

WEB REFERENCES

1. https://archive.nptel.ac.in/courses/111/105/111105090/

Category: **Professional Core Course** Subject Code**: ME2101**

Kinematics of Machinery

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

Course Outcome

- Upon successful completion of this course, the student will be able to
- Calculate the degrees of freedom for a given mechanism, perform number synthesis
- Perform velocity and acceleration analysis of simple mechanisms, determine Power transmitted in belt drives
- Generate cam profile for a specified motion of follower.
- Determine number of teeth and basic dimensions for a gear pair avoiding interference
- Determine number of teeth for gears of a gear train with a predefined speed ratio.

UNIT I

Terminology: Definitions of link, pair, chain and mechanism, degrees of freedom, Isomers,Inversion, Type, Number and Dimensional Synthesis

Basic Laws: Kutzbach and Grubbler's criterion. Grashof's Law

Simple mechanisms: four bar mechanism, single and double slider crank mechanisms. Ackerman and Davis steering gear mechanism, Hooke's Joint,Geneva mechanism.

Straight line mechanisms: Pantograph Peaucellier, Hart, Scott-Russel, Watt and Tchebicheff mechanisms

UNIT II

Velocity and acceleration analysis of planar mechanisms: Velocities in mechanisms by instantaneous center method, Instantaneous Centre, Space Centrode and Body Centrode, Kennedy Theorem. velocity and acceleration of mechanisms by using relative velocity method including Coriolis component of acceleration, Klien's construction

UNIT III

Belt, Rope and chain drives: Types of belt drives, Action of Belts on pulleys, Velocity ratio, Slip, material for belts & ropes, crowning of pulleys, Types of pulleys, Law of belting, Length of belt in case of open belt drive and crossed-belt drive, Ratio of friction tensions, power transmitted, Centrifugal effect on belts, Maximum power transmitted by a belt, initial tension, Creep, Types of Chains, chain length, Angular speed ratio**.**

UNIT IV

Cam and follower: Types of Cams and followers, motion of the follower, follower displacement diagram, Cam profile for specified follower motion and Cams with specified contours.

UNIT V

Gears: Theory of Gearing, Terminology and Definitions, Law of Gearing, Tooth profiles, Path of contact and Arc of contact. Interference, methods of avoiding interference. Contact Ratio. Introduction to Helical, Bevel and worm gears.

Gear Trains: Simple, Compound, Reverted and Epicyclic gear trains. Differential of an Automobile.

TEXT BOOKS:

1. Theory of Machines, S.S Rattan, 4th Edition, 2015, Tata Mc-Graw Hill.
2. Theory of Mechanisms and machines, Amitabha Ghosh & A. K. Malik,3rd Edition, 2008, East West Press private limited.
3. Mechanism and machine theory, Ashok G. Ambekar, 1st Edition, 2007, Prentice Hall India

REFERENCES:

1. Theory of machines, R.S. Khurmi& J. K. Gupta,14thRevised edition, S Chand,2005
2. Theory of Machines, Thomas Bevan, 3rd Edition, 2005, CBS Publishers and Distributors
3. Kinematics and Dynamics of machinery, Robert. Norton, 2009, Tata Mc- Grawhill
4. Theory of Machines and Mechanisms, Shigley J.E and Uicker J.J, 3rd Edition, 2009, Oxford university press
5. Mechanisms and Machine Theory, J.S. Rao and R.V. Dukkipati, 1992, Wiley Eastern Limited

WEB REFERENCES

1. https://archive.nptel.ac.in/courses/112/104/112104121/
2. https://onlinecourses.nptel.ac.in/noc20_me21/preview

Category: **Professional Core Course** Subject Code:**ME2102**

Strength of Materials

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 1 - 0 - 4**

COURSE OUTCOMES:

- Able to understand the basic concept of stress and strains and deformation for basic geometries subjected to axial loading and thermal effect.
- Able to find the maximum shear force and maximum bending moment by drawing the shear force and bending moment diagrams for different types of beams with different lateral loading condition.
- Able to find the strength of the various cross sectional beams such as rectangular, hollow circular, circular T, I sections etc.
- Able to calculate the deflections of the beam using different methods under different boundary and loading conditions
- Able to find the shear strength of the solid and hollow shafts which are subjected to torsional loading in power transmission.
- Able to learn about the strain energy concept.

Unit – I

Simple stresses and strains: Types of stresses and strains. Hooks's Law, Stress- Strain curve for ductile materials, moduli of elasticity.Poisson's ratio, linear strain, volumetric strain, relations between elastic constants. Bars of varying sections, bars of uniform strength, compound bars and temperature stresses, change in length.

Complex Stresses: Stresses on oblique planes, principle stresses and principle planes. Mohr circle of stress.

Unit-II

Shear Force and Bending Moment: Relation between intensity of loading. Shear force and bending moment, shear force and bending moment diagrams for cantilever and simply supported beams for point loads, uniformly distributed loads, uniformly varying loads and couples.

Unit-III

Theory of simple bending: Assumptions derivation of basic equation: $M/I = f/y = E/R$, Modulus of section, Moment of resistance, determination of flexural stresses.

Direct and Bending Stresses: Basic concepts, core of sections for rectangular, solid and hollow circular and I sections.

Unit-IV

Distribution of shear stress: Equation of shear stress, distribution across rectangular, circular, T and I sections.

Deflections: Deflections of cantilever and simply supported beams for point loads and uniformly distributed loads by double integration and Macaualay's method.

Strain Energy: Strain energy in bars due to gradually applied loads, sudden loads, impact loads and shock loads.

Unit-V

Torsion-Theory of pure torsion- derivation of basic equation **$T/J = y/R = G\Theta/L$** and hollow circular shafts, power transmission, combined bending and torsion.

Columns – Euler column theory, Expression for crippling load, Limitations of Euler's Formula

Basics of thin cylinder, thick cylinder and theories of failure

Suggested Readings:

TEXTBOOKS:

1. S.Ramamrutam, R.Narayanan," Strength of materials", DhanpatRai Publications, 14th edition
2. James M. Gere And Barry Goodier ,"Mechanics of materials", CENGAGE Learning Custom Publishing, 8^{th}edition.
3. S.S. Rattan, "Strength of materials", Tata McGraw-Hill Publications, 2nd edition.
4. Dr. R K Bansal, "A text book of strength of materials", Lakshmi Publications, 3rd edition
5. R.K. Rajput, Strength of Materials, S. Chand & Co., 2003.
6. R. Subramanian Strength Of Materials 3/E , Published by Oxford University Press.

REFERENCES:

1. B.C. Punmia, Strength of Materials and Theory of Structures, Laxmi Publishers, Delhi, 2000.
2. Timoshenko, Elements Of Strength Of Materials 5ed Paperback – 2003
3. Hibbeler, R. C. Mechanics of Materials. 6th ed. East Rutherford, NJ: Pearson Prentice Hall, 2004.
4. N. M. Belyaev., Strength of Materials , Mir Publishers, Moscow, fourteenth edition

WEB REFERENCES

1. https://nptel.ac.in/courses/112107146
2. https://nptel.ac.in/courses/112106141
3. https://onlinecourses.nptel.ac.in/noc19_ce18/preview
4. https://www.youtube.com/watch?v=8CP714_wKVk

Category: **Professional Core Course** Subject Code**: ME2103**

Thermodynamics

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OUTCOMES:

- Able to understand the thermodynamic properties, process, cycle, equilibrium and concepts of systems, surroundings and universe andthe energy transfer in the form of work and heat.
- The students will be able to apply energy balance to systemsand control volumes, in situations involving heat and work interactions.
- Able to understand the Carnot cycle and major difference in the working principles of heat engine, heat pump and refrigerator to calculate the maximum efficiency / COP of the cycle. Also student will learn the irreversibity processes, entropy change and maximum available energyby a process.
- Able to understand the concept of phase change of a pure substance and graphical representation of a pure substance on p-v, p-T, T-v, h-s and T-s diagrams, the usage of Steam tables and Mollier diagrams to solve problems. And the students will also learn to derive the thermodynamics relations involving entropy, enthalpy and internal energy.
- The students will be able to understand the working principles of air standard cycles such as Otto, Diesel and dual cycles.

UNIT I

Fundamentals - System & Control volume; Property, State & Process; Exact & Inexact differentials; Work - Thermodynamic definition of work; examples; Displacement work; Path dependence of displacement work and illustrations for simple processes; electrical, magnetic, gravitational, spring and shaft work. Temperature, Definition of thermal equilibrium and Zeroth law; Temperature scales; Various Thermometers- Definition of heat; Examples of heat/work interaction in systems.

UNIT II

First Law for Cyclic & Non-cyclic processes; Concept of total energy (E), Various modes of energy, Internal energy and Enthalpy. First Law for Flow Processes - Derivation of general energy equation for a control volume; Steady state steady flow processes including throttling; Examples of steady flow devices; Unsteady processes; Examples of unsteady processes. First law applications for system and control volume.

UNIT III

Second law - Definitions of direct and reverse heat engines; Definitions of thermal efficiency and COP; Kelvin-Planck and Clausius statements; Definition of reversible process; Internal and external irreversibility; Carnot cycle; Absolute temperature scale.

Clausius inequality; Definition of entropy (S), Evaluation of S for ideal gases undergoing various processes; - Principle of increase of entropy; Illustration of processes in T-s coordinates; Definition of Isentropic efficiency for compressors, turbines and nozzles- Irreversibility and Availability, Availability function for systems and Control volumes undergoing different processes, Lost work. Second law analysis for a control volume. Exergy balance equation and Exergy analysis.

UNIT IV

Thermodynamic properties relations involving Entropy, Enthalpy and Internal Energy.TdS equations – difference in heat capacities and ratio of heat capacities. Maxwell's relations, Joule-Kelvin Expansion.Clausius-Clayperon equation.

Definition of Pure substance, Ideal Gases and ideal gas mixtures, Real gases and real gas mixtures, Compressibility charts- Properties of two phase systems - Const. temperature and Const. pressure heating of water; Definitions of saturated states; P-v-T surface; Use of steam Tables, Saturation tables; Superheated tables; Identification of states &determination of properties. Mollier's chart.

UNIT V

Air Standard Cycles - nomenclature for reciprocating piston cylinder engine. Thermodynamics analysis of Air standard cyclessuch as Otto, Diesel and Dual cycles. Comparison of Otto, Diesel and dual cycles based on same compression ratio or same maximum pressure and temperature. Atkinson cycle and Lenoir cycle.

Text Books:

1. Nag P.K, "*Engineering Thermodynamics'*: Tata McGraw Hill Publishing, 8th Edn, 3rd Reprint 2010.

2.Yunus A Cengel and Michael A Boles, "*Thermodynamics-An Engineering Approach*", Tata Mc Graw Hill Publishing Company Ltd. ,6thEdn., Fifth Reprint, 2009.

3. Nag P.K, "*Basic & Applied Thermodynamics'*: Tata McGraw Hill Publishing, 8th Reprint 2006.

4. Richard E.Sonntag, C.Borgnakke, G.J Van Wylen, "Fundamentals of Thermodynamics': John Wiley & Sons, 7th Edn., 2009.

5. Rajput R K, " *Engineering Thermodynamics*" Laxmi Publications, 4th Edition, 2010

REFERENCES :

1. Sonntag, Richard E.., Borgnakke, Claus., Van Wylen, Gordon J.. Fundamentals of Thermodynamics. Singapore: Wiley, 1998.
2. Çengel, Yunus A.., Boles, Michael A.. Thermodynamics: An Engineering Approach. India: McGraw-Hill, 2011.
3. Holman, Jack Phillip. Thermodynamics J.P. Holman. United States: Mcgraw-Hill, 1974.
4. Rao, Y. V. C. An introduction to thermodynamics. India: Universities Press, 2004.
5. Dugan, R. E, Jones, James Beverly. Engineering Thermodynamics. United Kingdom: Prentice Hall, 1996.

Web References:

1. https://nptel.ac.in/courses/112105123
2. https://nptel.ac.in/courses/112104113
3. https://nptel.ac.in/courses/127106135

Category: **Professional Core Course** Subject Code**: ME2104**

Materials Engineering

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OUTCOMES:

- Students will get knowledge on bonds, crystallization of metals and effect of grain boundaries on the properties
- Students will be able to carry out the different mechanical testing methods to determine the mechanical properties of the materials.
- Students will be able to construct the equilibrium diagrams by experimental methods and knowing all types of equilibrium diagrams
- Students will be able to learn the structure and properties of all cast irons, steels and Non-ferrous metal alloys

Unit I:

Introduction: Classification of materials, properties of materials, advanced material, future and modern materials. Atomic structure, inter atomic bonding and structure of crystalline solids, Influence on properties of materials. Crystal structures, crystallography, planes and directions.

Imperfections in solids: Solidification process and Imperfections point, line, surface and volume defects, characteristics of dislocations, interactions between dislocations.

Unit II:

Deformation behaviors of materials: Elastic deformation, plastic deformation, and time dependent deformation processes, failure of materials, Fracture, fatigue and creep concepts and their significance.

Mechanical Properties of material and testing: Stress vs Strain graph, Tension test, Compression Test, Brinells, Vickers, Rockwell hardness test and micro hardness testing. Impact testing, creep test, fatigue test and fracture of materials and testing.

Methods of Melting: Crucible melting and cupola operation, steel making processes.

Unit III:

Phase Diagrams: Gibbs phase rule, cooling curves for pure metals and alloy, construction of phase diagrams, Equilibrium of phase diagrams (isomorphous, eutectic, partial eutectic and layered system), lever and tie line rule, phase transformation, iron-iron carbide phase diagram, different phases and applications in iron carbon system.

Unit IV:

Heat treatment and TTT curves: Transformation rate effects and TTT and CCT diagrams, microstructure and property changes in iron-carbon diagrams. Heat treatment of steel, Annealing, Normalizing, Hardening, Tempering, Austempering and Martempering of steels. Surface hardening of steels. Carburizing, Nitriding, Cyaniding, Flame and induction hardening methods.

Unit V:

Cast Irons and Steels: Structure and properties of White Cast iron, Malleable Cast iron, grey cast iron, Spheriodal graphite cast iron, Alloy cast irons. Classification of steels, structure and properties of plain carbon steels, Low alloy steels, tool and die steels.

Text Books :

1. Avner, Sidney H.. Introduction to Physical Metallurgy. Germany: McGraw-Hill,
2. Fulay, Pradeep P.., Askeland, Donald R.. Essentials of Materials Science & Engineering - SI Version. United States: Cengage Learning, 2009.

References :

1. Dr. VD Kodgire and Dr. S V Kodgire. Material Science and Metallurgy for Engineers
2. Askeland, D.R., Fulay, P.P., & Wright, W.J. (2011). The Science and Engineering of Materials, SI Edition: Cengage Learning.
3. Rethwisch, David G.., Callister, William D.. Materials Science and Engineering. India: Wiley, 2014.
4. Raghavan, V.. Materials Science And Engineering: A First Course. India: Phi Learning, 2015.
5. Flinn, Richard Aloysius., Trojan, Paul K.. Engineering Materials and Their Applications. United States: Houghton Mifflin, 1986.
6. Engineering Materials and Metallurgy. India: S. Chand Limited, 2006.

Web References:

1. https://onlinecourses.nptel.ac.in/noc20_me78/preview

Category: **Professional Core Course** Subject Code**: ME2701**

Strength of Materials Laboratory

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0 - 0 - 2 - 1**

Course Outcomes: After the completion of the course student should be able to

- Clearly understands the concepts of deciding the shape or type of specimen for assessing different strengths against various straining actions.
- Design the specimens for assessing a particular property of the material with the available machines.
- Decide the suitability of the material to the particular situation e.g., dynamic loads, vibrations, impacts, fatigue, etc.,
- Design the experiments making use of various techniques of load measuring or deformation measuring instruments.
- Understand about the different destructive and Non-destructive Testing

List of the Experiments

1. Uni-axial Tension test on a specimen of Ductile Material.
2. Torsion test on a specimen of ductile material
3. BrinellHardness test
4. Rockwell Hardness test
5. Compression tests on helical spring.
6. Compression test on Structural materials
7. Impact test-Izod Tests.
8. Impact test -Charpy Impact
9. Direct Shear test.
10. Verification of Maxwell's Reciprocal theorem on beams.
11. Bending test on Cantilever beam of steel or timber.
12. Bending test on simply supported beam.
13. Simple bending of sheet
14. Composite Material testing
15. Non-destructive Testing

Category: **Professional Core Course** Subject Code**: ME2702**

Materials Engineering Laboratory

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0 - 0 - 2 - 1**

Objectives:

- To familiarize the procedure for specimen preparation
- To prepare different metal specimen for identification
- To study the microstructure of metals and alloys
- To understand the heat treatment procedures
- To study the microstructure after heat treatment

List of Experiments:

Study of: Metallurgical Microscope, Iron-Iron Carbide diagram, Procedure for specimen preparation

1. Metallographic Study of Low / Medium carbon steel

2. Metallographic Study of Eutectoid steel

3. Metallographic Study of Hyper Eutectoid steel

4. Metallographic Study of Nodular cast iron

5. Metallographic Study of Grey cast iron

6. Metallographic Study of White cast iron

7. Metallographic Study of Brass and Bronze

8. Metallographic Study of microstructure after hardening, normalizing and annealing of steel specimen.

9. Quantitative analysis of Grain Size, ASTM Grain number

10. Quantitative analysis of Volume fraction

Category: **Professional Core Course** Subject Code**: ME2703**

Computational Mathematics laboratory

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 2 - 1**

Objectives:

- Solve the algebraic and transcendental equations within given range range using MAT LAB programs
- Utilize MAT LAB programs for verifying properties of limits, derivatives of a function
- Interpret rank, Eigen values and vectors with matrix transformations.
- Utilize MAT LAB programs for solving differential equations and multiple integrals
- Make use of MAT LAB programs for interpolating values of differential equations
- Use MAT LAB programs for vector operations on vector field.

List of Experiments:

1. BASIC FEATURES: Features and uses, Local environment setup.
2. ALGEBRA: Solving basic algebraic equations, Solving system of equations, Two dimensional plots
3. CALCULUS: Calculating limits, Solving differential equations, Finding definite integral.
4. MATRICES: Addition, subtraction and multiplication of matrices, Transpose of a matrix, Inverse of a matrix.
5. SYSTEM OF LINEAR EQUATIONS: Rank of a matrix, Gauss Jordan method, LU decomposition method.
6. LINEAR TRANSFORMATION: Characteristic equation, Eigen values, Eigen vectors.
7. DIFFERENTIATION AND INTEGRATION: Higher order differential equations, Double integrals, Triple integrals.
8. INTERPOLATION AND CURVE FITTING: Lagrange polynomial, Straight line fit, Polynomial curve fit
9. ROOT FINDING: Bisection method, Regula false method, Newton Raphson method.
10. NUMERICAL DIFFERENTION AND INTEGRATION: Trapezoidal, Simpson’s method, Euler method, Runge Kutta method
11. 3D PLOTTING: Line plotting, Surface plotting, Volume plotting.
12. VECTOR CALCULUS: Gradient, Divergent, Curl.

Reference Books:

1. Cleve Moler, “Numerical Computing with MATLAB”, SIAM, Philadelphia, 2nd Edition, 2008.
2. Dean G. Duffy, “Advanced Engineering Mathematics with MATLAB”, CRC Press, Taylor & Francis Group, 6th Edition, 2015.

Category: Mandatory Course **Subject Code: HS210X**

Start Up Entrepreneurship

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 2 - 0**

Course Outcomes:

1. Understand the behavioural aspects of entrepreneurs and time management
2. Creative thinking and transform ideas into reality
3. Importance of innovation in new business opportunities
4. Create a complete business plan and workout the budget plan.
5. write a project proposal with budget statement

UNIT I: Creativity & Discovery: Definition of Creativity, self test creativity, discovery and delivery skills, The imagination threshold, Building creativity ladder, Collection of wild ideas, Bench marking the ideas, Innovative to borrow or adopt, choosing the best of many ideas, management of tradeoff between discovery and delivery, Sharpening observation skills, reinventing self, Inspire and aspire through success stories

UNIT II: From Idea to Startup : Introduction to think ahead backward, Validation of ideas using cost and strategy, visualizing the business through value profile, activity mapping, Risks as opportunities, building your own road map

UNIT III: Innovation career lessons : Growing & Sharing Knowledge, The Role of Failure In Achieving Success, Creating vision, Strategy, Action & Resistance: Differentiated Market Transforming Strategy; Dare to Take Action; Fighting Resistance; All About the startup Ecosystem; Building a Team; Keeping it Simple and Working Hard.

UNIT IV: Action driven business plan: Creating a completed non-business plan (a series of actions each of which moves your idea toward implementation), including a list of the activities to be undertaken, with degrees of importance (scale of 1 to 3, where 1 is _most important‘). A revision of the original product or service idea, in light of information gathered in the process, beginning to design the business or organization that will successfully implement your creative idea. Preparing an activity map.

UNIT V: Startup financing cycle: Preparing an initial cash flow statement, showing money flowing out (operations; capital) and flowing in. Estimate your capital needs realistically. Prepare a bootstrapping option (self financing). Prepare a risk map. Prepare a business plan comprising five sections: The Need; The Product; Unique Features; The Market; Future Developments. Include a Gantt chart (project plan – detailed activities and starting and ending dates); and a project budget.

Textbooks:

1. Vasant Desai, ―Dynamics of Entrepreneurial Development and Management‖, Himalaya Publishing House, 1997.
2. Prasanna Chandra, ―Project – Planning , Analysis, Selection, Implementation and Review‖, TataMcGraw-Hill Publishing Company Ltd., 1995.
3. B. Badhai, ―Entrepreneurship for Engineers‖, Dhanpath Rai & Co., Delhi, 2001.
4. Stephen R. Covey and A. Roger Merrill, ―First Things First‖, Simon and Schuster, 2002.
5. Robert D. Hisrich and Michael P.Peters, ― Entrepreneurship‖, Tata McGRaw Hill Edition, 2002.

Web resources:

1. **Startup India Learning Program:** **https://www.startupindia.gov.in/content/sih/en/learning-and-development_v2.html**
2. **Entrepreneurship Essentials:** **https://onlinecourses.nptel.ac.in/noc22_ge03/preview**
3. **Entrepreneurship:** **https://onlinecourses.nptel.ac.in/noc21_mg70/preview**

SECOND YEAR (E2) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	ME2201	Fluid Mechanics and Hydraulic Machines	PCC	3	1	0	4	4
2	ME2202	Manufacturing Technology I	PCC	3	1	0	4	4
3	ME2203	Dynamics of Machinery	PCC	3	1	0	4	4
4	EC2201	Electronics ICs application	ESC	3	0	0	3	3
5	ME221X	Professional Elective-1	PEC	3	0	0	3	3
6	ME2801	Fluid Mechanics & Hydraulic Machinery Lab	PCC	0	0	2	2	1
7	ME2802	Theory of Machines Lab	PCC	0	0	2	2	1
8	ME2803	Manufacturing Technology I Lab	PCC	0	0	2	2	1
9	HS220X	Soft skills	HSMC	0	0	2	2	1
Total								**22**

Category: Professional Core Course **Subject Code:** ME2201

Fluid Mechanics and Hydraulic Machines

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks 3 - 1 - 0 - 4

COURSE OUTCOMES:

- Able to know the fluid properties and their engineering significance.
- Able to determine the pressure at a point and identify the variation of pressure in a fluid.
- Able to understand the basic concepts of fluid motion.
- Able to analyze different flow characteristics of laminar and turbulent flows
- Able to understand the boundary layer and its significance along with the various concepts of boundary layer like its growth, thickness and separation.
- Able to understand the concept of flow around the submerged objects
- Able to know the characteristics of compressible fluids flow and Mach number and its significance

UNIT-I

Properties of Fluids: Introduction, definition of fluid, Units of measurement, Fluid Properties- mass density, specific weight, specific gravity, Viscosity, Newton's law of viscosity – Newtonian and non Newtonian fluids. Classification of fluids- Ideal and real.

Fluid Statics: Fluid pressure at a point, variation of Pressure in a fluid, measurement of Pressure-simple manometers, differential manometers.
Fluid Kinematics: Fundamentals of fluid flow –types of fluid flow, description of flow pattern, basic principles of fluid flow, continuity equation, acceleration of a fluid particles.

UNIT-II

Fluid dynamics: Introduction, forces acting on a fluid in motion, Euler's equation of motion, Bernoulli's equation, application of Bernoulli's equation – venturimeter, pilot tube. Impulse momentum equation, application of impulse momentum equation – Forces on a pipe bend.

Flow through pipes: Introduction, two types of flow – laminar and turbulent – Reynold's experiment. Laws of fluid friction, Darcy- Weisbach equation. Steady laminar flow- circular pipes – Hagen-Poiscuille's law. Hydrodynamically smooth and rough boundaries and it's criteria and resistance to flow of fluid in smooth and rough boundaries – variation of friction factor.

UNIT-III

Boundary layer theory: Introduction, thickness of boundary layer, boundary layer along a flat thin plate and its characteristics. Laminar and turbulent boundary layer, laminar sub layer, separation of boundary layer and its control.
Fluid flow around submerged objects: Drag and lift – Introduction, types of drag, drag on a flat plate. Development of lift on immersed bodies – lift of an airfoil

UNIT-IV
Flow of compressible fluids: Introduction, concepts of compressible flow, continuity and energy equation, propagation of elastic waves due to compression of fluid, velocity of sound, Mach number and its significance, propagation of elastic waves due to disturbance of fluid stagnation properties, area velocity relationship for compressible flows.

UNIT-V
Pumps : Euler's equation – theory of Rotodynamic machines – various efficiencies – velocity components at entry and exit of the rotor, velocity triangles – Centrifugal pumps, working principle, work done by the impeller, performance curves – Cavitation in pumps
Turbines : Classification of water turbines, heads and efficiencies, velocity triangles- Axial, radial and mixed flow turbines- Pelton wheel, Francis turbine and Kaplan turbines, working principles – draft tube- Specific speed, unit quantities

Reference books

1. K.Subramanya,Theory and Applications of fluid Mechanics, Tata McGraw-Hill Publishing Company Ltd., New Delhi,1993.
2. Vijay Gupta and Santhosh K. Gupta, Fluid Mechanics and its applications, wiley Eastern Ltd.,New Delhi,1984.
3. K.L.Kumar, Engineering Fluid Mechanics, Eurasia Publishing House PVT Ltd, New delhi,2009.
4. P.N.MOdi,andS.M.Seth., Hydralics and Fluid Mechanics, Standard Book House, 1995.
5. Fluid Mechanics & Hydraulic Machines, S.C. Gupta, Pearson Publishers.

Web References:

1. https://nptel.ac.in/courses/112105206
2. https://nptel.ac.in/courses/112104117
3. https://www.youtube.com/watch?v=h7jW4yAHUl0&list=PL9RcWoqXmzaJipa5bafQ3fgHEMC5iPNn9

Category: Professional Core Course **Subject Code:** ME2202

Manufacturing Technology I

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 1 - 0 - 4**

COURSE OUTCOMES:

- Able to understand the elements of casting, construction of patterns and gating systems, moulds, methods of moulding, moulding machines and solidification of castings of various metals.
- Able to understand the different types of welding processes, welds and weld joints, their characteristics, cutting of ferrous and non-ferrous metals by various methods.
- Able to understand the basic concept on one, two and three dimensional stress analysis, theory of plasticity; strain hardening, hot and cold working process.
- Able to understand the principles of Extrusion, rolling, forging processes, wire drawing and sheet metal processes, their applications and defects.
- Able to understand the limits, fits and tolerance. Indian standard system, international standard organization system
- Able to know the principles of working of the most commonly used instruments for measuring linear and angular distances
- Able to study the different types of comparators, optical measuring instruments, flatness measurement methods and measuring methods of surface roughness

UNIT – I

Casting: Introduction, Pattern allowances and their Construction. Principles of Gating, Gating ratio and design of Gating systems, time of filling the cavity. Moulds: definition, mould materials, types of moulds, moulding methods, moulding machines, tests. Solidification of casting – Concept – Solidification of pure metal and alloys, short & long freezing range alloys.

Risers – Types, function and design, casting design considerations, Design of feeding systems i.e., sprue, runner, gate and riser, moulding flasks. Problems, Casting inspection and defects

UNIT – II

Welding : Classification of welding process, power characteristics, types of welds and welded joints and their characteristics, design of welded joints, Thermit welding and Plasma (Air and water) welding. Defects, causes and remedies. Problems

UNIT – III

Forming: Hot working, cold working, strain hardening, recovery, recrystallisation and grain growth, Comparison of properties of Cold and Hot worked parts, **Rolling:** Theory of rolling, Mechanics of rolling. **Extrusion:** Basic extrusion process and its characteristics, Analysis of wire drawing and extrusion. **Forging:** Principles of forging – Tools and dies, Analysis of Forging, **Deep Drawing:** Analysis of deep drawing, tests for measuring of formability.

UNIT-IV

Introduction, Accuracy and precision, Limits, Fits and Tolerances, ISO system. Types of interchangeability. Slip gauges and end bars. Height gauges, Abbe's rule, Types of micrometers. Tomlinson gauges, sine bar, autocollimator, calibration of precision polygons and circular scales. Dial indicator, Sigma mechanical comparator. Free flow and back pressure type Pneumatic comparators. Contact & non-contact tooling, Applications of single and multijet gauge heads; computation and match gauging.

UNIT-V

Taylor's principles for plain limit gauges. Usage and limitations of Ring and Snap gauges. Indicating type limit gauges. Position and receiver gauges, principles of thread gauging. Gauge materials and steps in gauge manufacture. Surface roughness characteristics and its measurement. Tool maker's microscope, Floating carriage diameter measuring machine and coordinate measuring machine. Measurement of straightness and flatness using autocollimator. Roundness measurement with intrinsic datum (V-block, Bench centers) and extrinsic datum (TALYROND).

TEXT BOOKS:

1. Rao, P.N. (2013). Manufacturing Technology: Metal Cutting and Machine Tools: McGraw Hill Education (India).
2. Kalpakjian, Serope. Manufacturing Engineering and Technology. India: Pearson Education, 2001.

REFERENCES:

1. R. k Jain,Production Technology. India: Khanna Pub., 1989.
2. Lindberg, Roy A.. Processes and Materials of Manufacture. India: Allyn and Bacon, 1990.
3. Heine, R.W., Loper, C.R., & Rosenthal, P.C. (1976). Principles of Metal Casting: McGraw-Hill Education.
4. Parmar, R. S.. Welding Processes and Technology. India: Khanna Publishers, 2001.
5. Rajput, RK. (2007). A Textbook of Heat and Mass Transfer: S. Chand Publishing.
6. Noorani, Rafiq. Rapid prototyping : principles and applications. United Kingdom: Wiley, 2006.
7. Jain VK (2002) Advanced Machining Processes. Allied, New Delhi.
8. Narayana, K. L.. Production Technology. India: I.K. International Publishing House Pvt. Limited, 2010.

Web References:

1. https://archive.nptel.ac.in/courses/112/107/112107219/
2. https://onlinecourses.nptel.ac.in/noc22_me28/preview

Category: **Professional Core Course** Subject Code**: ME2203**

Dynamics of Machinery

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 1 - 0 - 4**

COURSE OBJECTIVES:

- To find static and dynamic forces on planar mechanisms.
- To know the causes and effects of unbalanced forces in machine members.
- To determine natural frequencies of undamped, damped and forced vibrating systems of one, two and multi degree freedom systems.

COURSE OUTCOMES:

- Able to analyze the planar mechanism by performing static and dynamics force analysis.
- Able to apply gyroscopic principles on Aero plane, ship, four wheel and two wheel vehicles
- Able to understand the basic concepts of friction in inclined plane, in screw and nuts, pivots and collars with uniform pressure and uniform wear
- Able to understand how to draw turning moment diagram and can design a flywheel for IC engine
- Able to understand the basics concepts of governors and forces acting on various governors and able to solve numerical problems on different governors
- Able to balance rotating and reciprocating mass in various planes and able to understand balancing of V- engine and multi cylinder engines
- Able to perform analysis of the response of one degree of freedom systems with free and forced vibrations and can evaluate the critical speed of the shaft and can understand torsional vibrations
- Able to understand two and three rotor systems and can solve simple vibration calculations of rotor systems.

UNIT – I

Static and Dynamic force analysis: Analysis of four bar and slider crank mechanism, Inertia Forces of a Reciprocating Engine Mechanism

Flywheel: Turning moment diagram for steam engine, I.C. engine and multi cylinder engine. Crank effort - coefficient of Fluctuation of energy, coefficient of Fluctuation of speed – Fly wheels and their design.

UNIT –II

Gyroscope: effect of precession motion on the stability of moving vehicles such as motor car, motor cycle, aero planes and ships.

Governors: Watt, Porter and Proell governors. Spring loaded governors – Hartnell and Hartung governors with auxiliary springs. Sensitiveness, isochronism and hunting –effort and power of a governor.

UNIT – III

Balancing: Balancing of rotating masses - single and multiple – single and different planes.

Balancing of Reciprocating Masses: Primary and Secondary balancing of reciprocating masses. Analytical and graphical methods. Unbalanced forces and couples – V, multi cylinder, in -line and radial engines for primary and secondary balancing, locomotive balancing – Hammer blow, Swaying couple, variation of tractive force.

UNIT – IV

Brakes*:* block brakes, band brakes, band and block brakes, internal expanding brake.

Dynamometers: Introduction, types - prony, rope brake, epi-cyclic, Bevis Gibson and belt transmission dynamometers.

UNIT – V

Vibrations: Introduction, types of vibrations, free longitudinal vibrations, damped vibrations, logarithmic decrement, forced vibrations, vibrations isolation and transmissibility, transverse vibrations, whirling of shafts, critical speeds.

TEXT BOOKS:

1. Rattan, S.S. (2014). Theory of Machines: McGraw-Hill Education (India) Private.
2. Khurmi, R. S.., Gupta, J. K.. Theory of Machines. India: Eurasia Publishing House, 2005.

REFERENCES:

1. Rao, J. S.., Dukkipati, Rao V.. Mechanism and Machine Theory. India: New Age International (P) Limited, 2007.
2. Ballaney, P. L.. Theory Of Machines And Mechanisms. India: Khanna, 2003.
3. Bevan, Thomas. Theory of Machines, 3/e. India: Pearson Education, 2010.
4. Lal, Jagdish. Theory of Mechanisms & Machines. India: Metropolitan Book Company, 1985.

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1. https://archive.nptel.ac.in/courses/112/106/112106270/
2. https://www.digimat.in/nptel/courses/video/112105268/L01.html

Category: **Engineering Science Course** Subject Code**: EC2201**

Electronics IC Applications

Internals: 40 Marks	**L - T - P - C**
Externals: 60 Marks	**3 - 0 - 0 - 3**

UNIT - I

Integrated Circuits: Classification, chip size and circuit complexity, basic information of Op-amp, ideal and practical Op-amp, internal circuits, Op-amp DC and AC characteristics, 741 op-amp and its features, modes of operation-inverting, non-inverting, differential.

Op-Amp Applications: Basic applications of Op-amp- instrumentation amplifier, V to I and I to Vconverters, Differentiators and Integrators, Comparators, Schmitt Trigger.

UNIT - II

Active Filters & Oscillators: Introduction, 1st order LPF, HPF filters. Band pass, Band reject and all pass filters. Oscillator types and principle of operation – RC phase shift and, Wien bridge oscillators.

waveform generators – triangular, saw tooth, square wave generator

UNIT - III

Timers: Introduction to 555 timer, functional diagram, monostable and astable operations and applications.

D-A and A-D Converters: Introduction, basic DAC techniques, weighted resistor DAC, R-2R ladder DAC, inverted R-2R DAC, Different types of ADCs - parallel comparator type ADC, counter type ADC, successive approximation ADC and dual slope ADC. DAC and ADC specifications.

UNIT - IV

Number systems: Digital number systems; codes; Boolean algebra and logic gates: Karnaugh Map

Combinatorial circuits: Introduction, Adders and subtractor, Code converters, encoder and decoder; multiplexer, demultiplexer,

UNIT – V

Latches and Flip-flops, registers, counters

Memories: ROM architecture, types & applications, RAM architecture, Static & Dynamic RAMs.

TEXT BOOKS:

1. Jain, Shail., Roy Choudhury, D.. Linear Integrated Circuits. United Kingdom: New Age Science Limited, 2011.
2. Gayakwad, Ramakant A.. Op-amps and Linear Integrated Circuit Technology. United Kingdom: Prentice-Hall, 1983.
3. Jain, R. P.. Modern Digital Electronics. India: McGraw-Hill Education, 1997.

REFERENCE BOOKS:

1. Driscoll, Frederick F.., Coughlin, Robert F.. Operational Amplifiers & Linear Integrated Circuits. United Kingdom: Prentice Hall, 1998.
2. Dailey, Denton J.. Operational Amplifiers and Linear Integrated Circuits: Theory and Applications. United Kingdom: McGraw-Hill, 1989.
3. Franco, Sergio. Design with operational amplifiers and analog integrated circuits. Boston: WCB/McGraw-Hill, 1998.
4. Digital Fundamentals. India: Pearson Education, 2005.

Web References:

1. https://nptel.ac.in/courses/108108111

Category: Professional Core Course **Subject Code: ME2801**

Fluid Mechanics & Hydraulic Machinery Laboratory

Internals: 40 Marks
Externals: 60 Marks

L - T - P - C
0 - 0 - 2 - 1

Objectives:

- To understand the principles and performance characteristics of flow devices
- To know about the measurement of the fluid properties

Course Outcomes:

The students who have undergone the course will be able to measure various properties of fluids and characterize the performance of fluid machinery

List of the Experiments:

1. Measurement of Coefficient of Discharge of given Orifice meter
2. Measurement of Coefficient of Discharge of given Venturi meter
3. Measurement of minor losses in a given pipe
4. Determination of the performance characteristics of a single stage centrifugal pump
5. Determine the surface profile of free vortex comparison with their theoretical values.
6. Measurement of velocity of flowing fluid using pitot tube
7. Determination of the performance characteristics of Pelton Wheel
8. To demonstrate the phenomenon of pipe surge resulting from a change in velocity of water flowing through a pipe
9. Determine the centre of pressure for curved surface
10. Determine Reynold's number for Reynold's apparatus
11. Determine the metacentric height for ship model
12. Observe the stream lines for stream line apparatus

Category: **Professional Core Course** Subject Code**: ME2802**

Theory of Machines Laboratory

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0 - 0 - 2 - 1**

List of Experiments:

1 Study various types of kinematics links, pairs & mechanisms.
2 Working models of various commonly used mechanisms and its inversions
3 Working models of various types of gears trains- Simple, Compound, Reverted and Epi- cyclic trains
4 To find experimentally the Gyroscopic couple on Motorized Gyroscope and compare with applied couple
5 To evaluate the performance on gravity controlled governors
6 To evaluate the performance on spring controlled governors
7 To find out experimentally the coriolli's component of acceleration and compare with theoretical values
8 To perform the experiment of Balancing of rotating parts and find the unbalanced couple and forces
9 To calculate the torque on a Planet Carrier and torque on internal gear using epicyclic gear train and holding torque apparatus
10 Working model of a various types of gear box i) Sliding mesh ii) Constant mesh iii) Synchromesh gear box
11 Rope Brake Dynamometer
12 Study on various types of Brakes
13 Study on working of rack and pinion steering gear mechanism
14 Study on working of differential
15 To study various types of cam and follower arrangements and To find out jump phenomenon of cams and followers with the help of test kit
16 To plot the follower displacement vs. angle of cam rotation curves for different cam follower pairs.

Category: **Professional Core Course** Subject Code**: ME2803**

Manufacturing Technology I Laboratory

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 2 - 1**

List of Experiments:

1. Cold Metal Transfer Welding
2. Resistance welding.
3. Plastic Extrusion. Sine Bar and Slip gauges.
4. Combination Die(Hydraulic Press)
5. Study and Applications of Profile Projector
6. Study and Applications of Tool Maker's Microscope

*Usage of Micrometers, Bevel Protractor height gauges, slip gauges, sine bar is performed wherever relevant

Category: HSM Course **Subject Code: HS2201**

Soft Skills

Externals: 60 Marks **L-T-P-C***
Internals: 40 Marks **2-0-0-1**

Course Outcomes

1. communicate effectively in formal and informal situations
2. understand the structure and mechanics of writing resumes, reports, documents and e-mails
3. present effectively in academic and professional contexts
4. develop communication in writing for a variety of purposes
5. identify areas of evaluation in Group Discussions conducted by organizations as part of the selection procedure
6. overcome stage fear and tackle questions

UNIT-I

Activities on Fundamentals of Inter-personal Communication
Starting a conversation - responding appropriately and relevantly - using the right body language-Role Play in different situations & Discourse Skills using visuals.

UNIT-II

Activities on Reading Comprehension
General Vs Local comprehension- reading for facts- guessing meanings from context- scanning- skimming- inferring meaning- critical reading – surfing Internet

UNIT-III

Activities on Writing Skills
Structure and presentation of different types of writing- Resume writing/ e-correspondence/ Technical report writing- planning for writing - improving one's writing.

UNIT-IV

Activities on Presentation Skills
Oral presentations (individual and group) through JAM sessions/seminars/PPTs and written presentations

UNIT-V

Activities on Group Discussion, Debate and Interview Skills - Dynamics of group discussion- intervention- summarizing-modulation of voice-body language-relevance-fluency and organization of ideas and rubrics for evaluation- Concept and process-pre-interview planning-opening strategies-answering strategies- interview through tele-conference & video-conferencing - Mock Interviews.

Text Books

1. Wentz, Frederick H.. Soft Skills Training: A Workbook to Develop Skills for Employment. United States: Create Space Independent Publishing Platform, 2012.
2. Maxwell, John C.. Everyone Communicates, Few Connect: What the Most Effective People Do Differently. United Kingdom: Thomas Nelson, 2010.
3. Lowndes, Leil. How to Talk to Anyone: 92 Little Tricks for Big Success in Relationships. United Kingdom: Thorsons, 1999.
4. Maxwell, John C.. Teamwork 101: What Every Leader Needs to Know. United States: HarperCollins Leadership, 2009.
5. Ryan, Mary Jane. Adaptability: How to Survive Change You Didn't Ask for. United States: Broadway Books, 2009.
6. Miller, R. Conflict Communication: A New Paradigm in Conscious Communication. YMAA Publication Center, 2016.

THIRD YEAR (E3) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	ME3101	Heat Transfer	PCC	3	0	0	3	3
2	HS3101	Operations research	HSMC	3	0	0	3	3
3	ME3102	Design of Machine Members	PCC	3	0	0	3	3
4	ME3103	IC Engine and Hybrid Vehicle	PCC	3	0	0	3	3
5	ME3104	Manufacturing Technology II	PCC	3	0	0	3	3
6	XX43xx	Open Elective – 1	OEC	3	0	0	3	3
7	ME3701	Heat Transfer lab	PCC	0	0	2	2	1
8	ME3702	Manufacturing Technology II lab	PCC	0	0	2	2	1
9	ME3703	Computer Aided Machine Drawing Practice	PCC	0	0	3	3	1.5
10	ME3XXX	Internship						1
Total								**22.5**

Category: **Professional Core Course** Subject Code**: ME3101**

Heat Transfer

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OBJECTIVES:

- To understand the basic concepts of heat transfer.
- To study the concepts of conduction, convection, radiation and heat exchangers applicable for commercial and industrial use.
- To study and solve problems on different modes of heat transfer which are related to thermal power plants, refrigeration and air conditioning.

COURSE OUTCOMES:

- Able to grasp the concept of steady state conduction. Student can learn representing conduction equation in various forms
- Able to understand the concept of extended surfaces and its applications. Also, will aware transient heat conduction and how it varies with respect to time.
- Expected to develop the ability to formulate practical conduction heat transfer problems by transforming the physical system into a Mathematical model and selecting an appropriate solution technique and evaluating the significance of results
- Able to formulate practical forced and natural convection heat transfer problems by transforming the physical system into a mathematical model.
- Able to calculate heat transfer in condensation and boiling systems, turbulent and laminar film condensation.

UNIT – I:

Introduction: Modes and mechanisms of heat transfer – Basic laws of heat transfer –General applications of heat transfer.

Conduction Heat Transfer: Fourier rate equation – General heat conduction equation in Cartesian, Cylindrical and Spherical coordinates.

UNIT – II:

Simplification and forms of the field equation – steady, unsteady and periodic heat transfer – boundary and Initial conditions.

One Dimensional Steady State Heat Conduction: in Homogeneous slabs, hollow cylinders and spheres – overall heat transfer coefficient – electrical analogy – Critical radius/thickness of insulation-with Variable Thermal conductivity –with internal heat sources or Heat generation. Extended surface (fins) Heat Transfer – Long Fin, Fin with insulated tip and Short Fin, Application to errors in Temperature measurement.

One Dimensional Transient Heat Conduction: in Systems with negligible internal resistance – Significance of Biot and Fourier Numbers - Chart solutions of transient conduction systems- Problems on semi-infinite body.

UNIT – III:

Convective Heat Transfer: Dimensional analysis–Buckingham π Theorem and its application for developing semi – empirical non- dimensional correlations for convective heat transfer – Significance of non-dimensional numbers – Concepts of Continuity, Momentum and Energy Equations.

Forced convection: External Flows: Concepts of hydrodynamic and thermal boundary layer and use of empirical correlations for convective heat transfer for flow over-Flat plates, Cylinders and spheres..

Internal Flows: Division of internal flow through Concepts of Hydrodynamic and Thermal Entry Lengths – Use of empirical relations for convective heat transfer in Horizontal Pipe Flow, annular flow.

Free Convection: Development of Hydrodynamic and thermal boundary layer along a vertical plate – Use of empirical relations for convective heat transfer on plates and cylinders in horizontal and vertical orientation.

UNIT IV:

Heat Transfer with Phase Change: Boiling: Pool boiling – Regimes, determination of heat transfer coefficient in Nucleate boiling, Critical Heat flux and Film boiling.

Condensation: Film wise and drop wise condensation –Nusselt's Theory of Condensation on a vertical plate - Film condensation on vertical and horizontal cylinders using empirical correlations.

Heat Exchangers:

Classification of heat exchangers – overall heat transfer Coefficient and fouling factor – Concepts of LMTD and NTU methods - Problems using LMTD and NTU methods.

UNIT V:

Radiation Heat Transfer

Emission characteristics and laws of black-body radiation – Irradiation – total and monochromatic quantities– laws of Planck, Wien, Kirchoff, Lambert, Stefan and Boltzmann– heat exchange between two black bodies – concepts of shape factor – Emissivity – heat exchange between gray bodies – radiation shields– electrical analogy for radiation networks.

Mass Transfer

Basic Concepts – Diffusion Mass Transfer – Fick's Law of Diffusion – Steady state Molecular Diffusion – Convective Mass Transfer – Momentum, Heat and Mass Transfer Analogy – Convective Mass Transfer Correlations

TEXT BOOKS:

1. Sachdeva, R.C. (2017). Fundamentals of Engineering Heat and Mass Transfer: New Academic Science.

2. Thirumaleshwar, M. (2009). Fundamentals of Heat and Mass Transfer: Pearson Education.

REFERENCE BOOKS:

1. Nag, P.K. (2002). Heat Transfer: Tata Mcgraw-Hill Publishing Company Limited.

2. Holman, J.P. (2010). Heat Transfer: Tenth Edition: McGraw-Hill Education.

3. Cengel. (2011). Heat and Mass Transfer: McGraw-Hill Education.

4. Rajput, RK. A Textbook of Heat and Mass Transfer: S. Chand Publishing.– S.Chand& Company Ltd.

5. Kothandaraman, C.P. (2006). Fundamentals of Heat and Mass Transfer: New Age International.

WEB REFERENCES:

- IIT video lecturers (NPTEL)
- http://www.wisc-online.com/Objects/ViewObject.aspx?ID=SCE304
- http://web.cecs.pdx.edu/~gerry/heatAnimations/sphereTransient/#TOC
- http://rpaulsingh.com/animated%20figures/animationlisttopic.htm
- http://www.slideshare.net/meenng/transfer-of-heat
- http://www.phy.cuhk.edu.hk/contextual/heat/hea/heatp01_e.html

Category: **HSM Course** Subject Code**: HS3101**

Operations Research

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT - I

Development — Definition— Characteristics and Phases — Types of models — Operations Research models — applications.

Allocation: Linear Programming Problem Formulation — Graphical solution — Simplex method — Artificial variables techniques: Two—phase method, Big M method.

UNIT - II

Transportation Problem — Formulation — Optimal solution, unbalanced transportation problem — Degeneracy. Assignment problem — Formulation — Optimal solution – Variants of Assignment Problem- Traveling Salesman problem.

UNIT - III

Sequencing — Introduction — Flow —Shop sequencing — n jobs through two machines — n jobs through three machines — Job shop sequencing — two jobs through m machines.

Replacement: Introduction — Replacement of items that deteriorate with time — when money value is not counted and counted — Replacement of items that fail completely- Group Replacement.

UNIT - IV

Theory of Games: Introduction —Terminology— Solution of games with saddle points and without saddle points- 2 x 2 games — dominance principle — m x 2 & 2 x n games - graphical method.

Inventory: Introduction — Single item, Deterministic models — Purchase inventory models with one price break and multiple price breaks —Stochastic models — demand may be discrete variable or continuous variable — Single Period model and no setup cost.

UNIT - V

Waiting Lines: Introduction —Terminology-Single Channel — Poisson arrivals and Exponential Service times — with infinite population and finite population models— Multichannel — Poisson arrivals and exponential service times with infinite population.

Dynamic Programming: Introduction — Terminology- Bellman's Principle of Optimality — Applications of dynamic programming- shortest path problem — linear programming problem.

Simulation: Introduction, Definition, types of simulation models, Steps involved in the simulation process- Advantages and disadvantages- applications of simulation to queuing and inventory.

TEXT BOOKS

1. Sharma, J K. (2009). Operations Research:Theory and Applications: Macmillan Publishers India Limited.
2. Hillier, F.S., & Lieberman, G.J. (2001). Introduction to Operations Research: McGraw-Hill.

Reference Books

1. Taha, HamdyA.. Operations Research: An Introduction. United Kingdom: Prentice Hall, 2011.
2. Raju, N. V. S.. Operations Research: Operations Research: Theory and Practice. United States: Taylor & Francis Group, 2019.
3. Natarajan, A. M.., Balasubramani, P.. Operations Research. India: Pearson Education, 2006.
4. Wagner, Harvey M.. Principles of Operations Research: With Applications to Managerial Decisions. India: Prentice-Hall, 1975..
5. Durga Prasad, M. V. (2012). Operations research. New Delhi: Cengage Learning.

Web References:

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Category: **Professional Core Course** Subject Code**: ME3102**

Design of Machine Members

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0- 0 - 3**

Course Outcomes:

1. Apply the knowledge of stress analysis, theories of failure, manufacturing and material science, and ergonomics principles in design of machine elements.
2. Analyze the stress and strain on mechanical components under different loadings; and understand, identify and quantify failure modes for mechanical parts.
3. Design various machine elements such as temporary and permanent fasteners, Mechanical Springs, pressure vessels and IC engine parts such as piston, connecting rod and crank.
4. Able to make proper assumptions, perform correct analysis and finally decide the size of machine elements while giving due consideration to material, manufacturing method and cost of the element.
5. Approach design problem successfully, and able to take decisions when no unique solution exists

Unit-I Introduction

Materials used in machine design and their specifications to Indian standards. Important mechanical properties of materials used in design. Codes and standards used in design. Reliability, Principles of good Ergonomic Design, Manufacturing considerations. Preferred numbers. Analysis of Stress and Strain : Definition of stress and strain, Types of loading, Direct normal stress, bending stress, Torisonal stress, crushing and bearing stresses, Biaxial stress and Triaxial stress. Theories of elastic failure, Stress concentration factor, factor of safety, Design of components for static loads, Introduction to thermal stresses.

Unit-II Design against fluctuating load

Importance of fatigue in design, Fluctuating stresses, fatigue strength and endurance limit. Factors affecting fatigue strength. S-N Diagram, Soderberg and Modified Goodman's diagrams for fatigue design. Cumulative fatigue, Miner's rule, Design of components for fatigue. Design of components for impact loading.

Unit-III Design of permanent and temporary joints

riveted and welded joints under direct and eccentric loading Design of cotter and knuckle joints,. Design of bolts and nuts, locking devices, bolt of uniform strength, design of gasket joints, design of power screws and screw jack.

Unit-IV Design of springs and pressure vessels

Mechanical springs: Introduction. Different types of springs. Materials used for springs

Helical Springs: Whal factor, calculation of stress, Deflection and energy stored in spring. Design for static and fluctuating loads. Leaf Springs: Stress and Deflection. Nipping of Leaf springs. Design for static and fluctuating loads. Thick and thin cylinders.

Unit-V Design of I.C. Engine parts

Introduction. Materials used. Design of piston, connecting rod and crank for I.C. Engines.

Text Books

1. V.B. Bhandari, Machine Design, Tata Mc Graw Hill Publication, 1991.
2. J.E. Shigley, C.R. Mischne, Mechanical Engineering Design, Tata Mc Graw Hill Publications, 2003.
3. Robert L. Norton, Machine Design: An Integrated Approach,2/e Pearson Education, 2000

REFERENCES:

1. Robert C. Juvinall, Fundamentals of Machine Component Design, John Wiley & Sons, 2005
2. M.F. Spotts, Design of Machine Elements, Prentice Hall of India, 1964.
3. P. Kannaiah, Machine Design, 2nd Edition, Scitech Publications. 2012,

Web References:

1. https://archive.nptel.ac.in/courses/112/105/112105125/
2. http://www.nptelvideos.com/course.php?id=791

Category: **Professional Core Course** Subject Code**: ME3103**

IC Engines & Hybrid Vehicle

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 –3**

Course outcomes

- To understand the operating principles of cycles involved in I.C.Engines.
- To understand the working principles, combustion process and performance characteristics of S.I. and C.I. engines.
- To understand the basics of electric and hybrid electric vehicles, their architecture, technologies and fundamentals
- To understand the plug – in hybrid electric vehicle architecture, design and component sizing in hybrid electric vehicles.

Unit I: Introduction: Introduction, energy conversion, basics engine components and nomenclature, working principle of engines, air standard cycles, fuel-air cycles and actual cycles. Alternate Fuels: Alcohol -Hydrogen -Natural Gas and Liquefied Petroleum Gas – Biodiesel-Biogas-Properties -Merits and Demerits of fuels - Emissions and Control.

Unit II:

Spark Ignition Engines: Air-fuel mixture requirements - Carburetors –Properties of Gasoline fuel -Injection systems -Monopoint and Multipoint injection –Gasoline Direct Injection – Ignition Systems-Stages of combustion -Normal and Abnormal combustion-Factors affecting on knock -Combustion Chambers.

Unit III:

Compression Ignition Engines: States of combustion in C.I. Engine -–Combustion chambers - Properties of Diesel fuel -Fuel spray behavior -spray structure -spray penetration and evaporation –Air motion - Normal and Abnormal combustion-Factors affecting on knock ;Turbocharging.

Performance parameters and their characteristics: Introduction, engine power calculation methods, efficiencies, characteristics curves, heat balance sheet.

Unit IV:

Hybrid Vehicles: Introduction: Sustainable Transportation, A Brief History of HEVs, Why EVs Emerged and Failed, Architectures of HEVs, Interdisciplinary Nature of HEVs, State of the Art of HEVs, Challenges and Key Technology of HEVs.Hybridization of the Automobile: Vehicle Basics, Basics of the EV, Basics of the HEV,Basics of Plug-In Hybrid Electric Vehicle (PHEV), Basics of Fuel Cell Vehicles (FCVs).

Unit V:

EV Fundamentals: Introduction, Vehicle Model, Vehicle Performance, EV PowertrainComponent Sizing, Series Hybrid Vehicle, Parallel Hybrid Vehicle, Wheel SlipDynamics.

Plug-in Hybrid Electric Vehicles: Introduction to PHEVs, PHEV Architectures,Equivalent Electric Range of Blended PHEVs, Fuel Economy of PHEVs, PowerManagement of PHEVs, Component Sizing of EREVs, Component Sizing of BlendedPHEVs, Vehicle-to-Grid Technology.

Text Books

1. Ganesan, V.. IC Engines. United Kingdom: Tata McGraw-Hill, 2007.
2. Ehsani, Mehrdad., Gao, Yimin., Longo, Stefano., Ebrahimi, Kambiz. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles. United States: CRC Press, 2018.
3. R.B.Mathur and R.P.Sharma, (2010), Internal Combustion Engines., Dhanpat Rai & Sons

References

1. Khajepour, Amir., Goodarzi, Avesta., Fallah, M. Saber. Electric and Hybrid Vehicles: Technologies, Modeling and Control - A Mechatronic Approach. United Kingdom: Wiley, 2014.
2. Colin R.Feriguson, and Allan.T.Kirkpatrik, (2000), I.C.engines Applied Thermosciences
3. John B. Heywood, (2000), Internal Combustion Engine Fundamentals, McGraw Hill.
4. Rowland S.Benson and N.D.Whitehouse, (2000) Internal combustion Engines, Vol. I and II, Pergamon Press.

WEB REFERENCES:

1. https://onlinecourses.nptel.ac.in/noc22_me65/preview
2. https://archive.nptel.ac.in/courses/108/103/108103009/
3. https://onlinecourses.nptel.ac.in/noc23_ee01/preview

Category: **Professional Core Course** Subject Code**: ME3104**

Manufacturing Technology II

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OUTCOMES:

- Able to understand the basic concepts of the philosophy of metal cutting and the mechanism of chip formation. Student will understand the effect of various cutting parameters on cutting forces
- Able to know the various concepts about tool wear, tool life, cutting fluid etc.,
- Able to understand the basic concepts of the philosophy of metal cutting and the mechanism of chip formation.
- Able to understand about machine tools such as Lathe, Drilling, Milling etc.,
- Able to calculate the machining time
- Able to understand the principles of design of Jigs and fixtures and their uses

UNIT-I

Mechanics of Machining: Single point cutting tool-types of reference systems–ASA,ORS and NRS systems and their Inter-relationships. Mechanism of chip formation, shear plane model. types of chips, effect of cutting parameters Forces in chip formation-Cutting force analysis- Ernst and Merchant analysis-theory of Lee and Shaffer. Effect of various cutting parameters on cutting forces, Theory of strain and strain rate in metal cutting and Energy considerations.

UNIT-II

Tool Wear Life and Machinability: Different causes-various forms of tool wear-measurement of tool wear. Tool life. Machinability-criterion for machinability-influence of variables affecting machinability. Measurement of Cutting Forces and Temperatures Tool Materials: Various tool materials, their properties and general guidelines for selection. Cutting Fluids: Functions, properties, types and selection. Economics of Metal Cutting: Various types of costs and their estimation. Determination of cutting speed for maximum production rate and minimum cost criteria.

UNIT – III

Introduction to machine tools: Lathe: Description, types, operations, accessories, attachments and machine time calculations. Introduction to Capstan and Turret Lathe and Automatic Machine. Drilling: Description, types of drilling machines, drilling operations, machine time Calculations

UNIT-IV

Milling: Description, types of milling machines, Mounting of milling cutters, types of milling operations, machining time calculation, types of indexing methods. Gear cutting process, Gear Milling , Thread Milling, Shaping, Planning and Slotting: Description, types of machines and operations, tool setting and quick return mechanisms. Machining time calculations.

UNIT V

Grinding machine –Theory of grinding – classification of grinding machine – cylindrical and surface grinding machine – Tool and cutter grinding machine – special types of grinding machines – Grinding wheel, Different types of abrasives – bonds, specification and selection of a grinding wheel

Principles of design of Jigs and fixtures and uses. Classification of Jigs & Fixtures – Principles of location and clamping – Types of clamping & work holding devices.

Text Books :

1. R.K. Jain, *Engineering Metrology*, Khanna Publications, 2008.

2. I.C. Gupta, *A Text Book of Engineering Metrology*, Dhanpat Rai & Sons, 1984.

3. Rao, P.N. (2013). Manufacturing Technology: Metal Cutting and Machine Tools: McGraw Hill Education (India).

4. Hajra Choudhury, A. K.., Hajra Choudhury, S. K.. Elements of Workshop Technology. Vol. I: Manufacturing Processes. India: Media Promoters & Publishers,

5. Ghosh, Amitabha., Mallik, Asok Kumar. Manufacturing Science. United Kingdom: Ellis Horwood, 1986.

References:

1. Dotson, Connie L. Fundamentals of Dimensional Metrology. United States: Cengage Learning, 2015.

2. Machine Tools – C.Elanchezhian and M. Vijayan /Anuradha Agencies Publishers.

3.Schmid, Steven R.., Kalpakjian, Serope. Manufacturing Engineering and Technology. Singapore: Pearson, 2014.

4. P.C.Sharma, Production Engineering, Dhanpat Rai & Sons, New Delhi.

Web References:

1. https://onlinecourses.nptel.ac.in/noc21_me04/preview

Category: **Professional Core Course** Subject Code**: ME3701**

Heat Transfer Laboratory

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0 - 0 - 2 - 1**

Course Outcomes:

Students undergoing this course are able to

- Design of experiments to study thermal power cycles and other thermal systems including compressors, turbines and combustion systems.

LIST OF EXPERIMENTS

1. Evaluation of thermal conductivity using lagged pipe apparatus.
2. Determination of thermal conductivity using guarded plate apparatus.
3. Evaluation of Stefan Boltzmann Constant.
4. Determination of radiation from a grey body.
5. Determination of heat transfer co-efficient using pin-fin apparatus.
6. Evaluation of COP of refrigerant
7. Experiment on parallel flow heat exchanger
8. Experiment on counter flow heat exchanger
9. Determination of convective heat transfer coefficient during natural convection.
10. Determination of convective heat transfer coefficient during forced convection.
11. Study of air-conditioning test rig
12. Study of air blower
13. Study of air compressor

Category: **Professional Core Course** Subject Code**: ME3702**

Manufacturing Technology II Laboratory

Internals: 40 Marks	**L - T - P - C**
Externals: 60 Marks	**0 - 0 - 2 - 1**

List of experiments:

I. Casting:

1. Pattern design and preparation
2. Moulding properties like permeability, Green hardness, Dry tensile & compression strength, Green tensile & compression strength, Moisture measurement.
3. Riser design & sieve analysis.

II. Forming

1. Simulation of Extrusion & deep drawing process

III. Welding:

1. Arc welding characteristics
2. Characteristics of MIG welding
3. Demo of TIG and Resistance spot welding

IV. Machining:

1. Milling of Spur gear. Usage of Profile projector, Tool Maker's Microscope etc.
2. Effect of process parameter and machining on shear angle in orthogonal cutting on chip formation in turning. Usage of Tool cutter grinder
3. Effect of process parameters in turning on cutting forces & temperatures
4. Grinding of single point cutting tool and to perform surface roughness measurement
5. Lathe Threading Profile calibration using Profile projector, Tool Maker's Microscope etc.

Category: **Professional Core Course** Subject Code**: ME3702**

Computer Aided Machine Drawing Practice

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **0 - 0 - 3 - 1.5**

Objectives:

- To understand format of drawing sheet, angle of projections and practice of simple machine elements
- To practice free hand sketching of machine elements
- To understand assembly drawings of typical machine parts such as Connecting rod, Eccentric, Cross head, Machine vice, Screw jack, Non-return valves, Safety valves, Bearings, Tail stock etc.

I. Machine Drawing Conventions:

Need for drawing conventions – introduction to IS conventions

a) Conventional representation of materials, common machine elements and parts such as screws, nuts, bolts, keys, gears, webs, ribs.

b) Types of sections – selection of section planes and drawing of sections and auxiliary sectional views. Parts not usually sectioned.

c) Methods of dimensioning, general rules for sizes and placement of dimensions for holes, centers, curved and tapered features.

d) Title boxes, their size, location and details – common abbreviations & their liberal usage

e) Types of Drawings – working drawings for machine parts.

II. Drawing of Machine Elements and simple parts

Selection of Views, additional views for the following machine elements and parts with every drawing proportions.

a) Popular forms of Screw threads, bolts, nuts, stud bolts, tap bolts, set screws.

b) Rivetted joints for plates

c) Shaft coupling, spigot and socket pipe joint.

d) Journal, pivot and collar and foot step bearings.

III. Assembly Drawings:

Drawings of assembled views for the part drawings of the following using conventions and easy drawing proportions.

a) Engine parts – stuffing boxes, cross heads, Eccentrics, Petrol Engine connecting rod, piston assembly.

b) Other machine parts – Screws jacks, Machine Vices Plummer block, Tailstock.

c) Valves : Steam stop valve, spring loaded safety valve, feed check valve and air cock.

NOTE : First angle projection to be adopted. The student should be able to provide working drawings of actual parts.

TEXT BOOKS :

1. Singh, Ajeet. Machine Drawing. India: Tata McGraw Hill Education, 2012.
2. Narayana, K.L. Machine Drawing. New Age International Pvt, 2016.

REFERENCES :

1. Gill, P. S.., Hira, D. S.. Text Book of Mechanical Engineering Drawing. United States: n.p., 1970.
2. Luzadder, Warren j. Fundamentals of Engineering Drawing, by Warren J. Luzadder. N.p.Prentice-hall, 1965.
3. Rajput, R. K. A Textbook of Manufacturing Technology: Manufacturing Processes. India: Laxmi, 2007.
4. TJohn, K. C.TextbookOf Machine Drawing. India: Phi Learning, 2009.

THIRD YEAR (E3) – SEMESTER – II

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	ME3201	Power plant engineering	PCC	3	0	0	3	3
2	ME3202	Design of Transmission Elements	PCC	3	0	0	3	3
3	ME3203	Computer Aided Engineering	PCC	3	0	0	3	3
4	ME323x	Instrumentation and Control system	PCC	3	0	0	3	3
5	ME324x	Professional Elective-2	PEC	3	0	0	3	3
6	BM320X	Managerial Economics and Financial Analysis	HSMC	3	0	0	3	3
7	ME3801	Thermal Engineering lab	PCC	0	0	2	2	1
8	ME3802	Computer Aided Engineering Lab	PCC	0	0	3	3	1.5
9	ME3803	Instrumentation and Control system lab	PCC	0	0	2	2	1
10	ME3902	Theme based project		0	0	2	0	1
Total								**22.5**

Category: Professional Core Course **Subject Code:** ME3201

Power Plant Engineering

Internals: 40 Marks
Externals: 60 Marks

L - T - P - C
3 - 0 - 0 - 3

Course Outcomes:

1. After completing this course, the students will get a good understanding of various practical power cycles and heat pump cycles.
2. They will be able to analyze energy conversion in various thermal devices such as combustors, air coolers, nozzles, diffusers, steam turbines
3. They will be able to understand phenomena occurring in high speed compressible flows
4. They will understand the design and performance criteria for steam turbines
5. They will understand the concepts of nuclear and hydro electric power plants

UNIT-1:

Introduction to solid, liquid and gaseous fuels– Stoichiometry, exhaust gas analysis- First law analysis of combustion reactions- Heat calculations using enthalpy tables- Adiabatic flame temperature- Chemical equilibrium and equilibrium composition calculations use free energy.

UNIT-2:

Vapor power cycles Rankine cycle with superheat, Concept of Mean Temperature of Heat addition, Methods to improve cycle performance, reheat and regeneration, exergy analysis. Super-critical and ultra super-critical Rankine cycle- Gas power cycles, Air standard Brayton cycle, essential components – parameters of performance – actual cycle – effect of reheat regeneration and intercooling- Combined gas and vapor power cycles

UNIT-3:

Basics of compressible flow. Stagnation properties, Isentropic flow of a perfect gas through a nozzle, choked flow, subsonic and supersonic flows- normal shocks- use of ideal gas tables for isentropic flow and normal shock flow- Flow of steam and refrigerant through nozzle, super saturation- compressible flow in diffusers, efficiency of nozzle and diffuser.

UNIT-4:

Boilers and Classification based on Working principles & Pressures of operation -L.P & H.P. Boilers – Mountings and Accessories, equivalent evaporation, efficiency and heat balance and draught. Analysis of steam turbines, velocity and pressure compounding of steam turbines, Mechanical details – principle of operation, , degree of reaction –velocity diagram

UNIT-5:

Basics of nuclear energy conversion, Layout and subsystems of nuclear power plants, Boiling Water Reactor, Pressurized Water Reactor, CANDU Pressurized Heavy Water Reactor, Fast Breeder Reactors. Hydroelectric power plants, classification, typical layout and components, principles of wind, tidal, solar PV and solar thermal, geo thermal, biogas and fuel cell power systems.

Text Books:

1. Borgnakke, Claus., Sonntag, Richard E Fundamentals of Thermodynamics. United Kingdom: Wiley, 2020.
2. Jones, J. B. and Duggan, R. E., 1996, Engineering Thermodynamics, Prentice-Hall of India
3. Moran, M. J. and Shapiro, H. N., 1999, Fundamentals of Engineering Thermodynamics, John Wiley and Sons.
4. Nag, P.K, 1995, Engineering Thermodynamics, Tata McGraw-Hill Publishing Co. Ltd
5. Nag, P.K. (2002). Power Plant Engineering: Tata Mcgraw-Hill Publishing Company Limited.
6. Arora and Domkundwar, "A course in Power Plant Engineering" Dhanpat Rai & Sons.
7. M. M. EI- Wakil, "Power plant technology," Tata McGraw - Hill.

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Category: Professional Core Course **Subject Code:** ME3202

Design of Transmission Elements

Internals: 40 Marks
Externals: 60 Marks

L	T	P	C
3	0	0	3

COURSE OUTCOMES:

1. Apply the knowledge of stress analysis, theories of failure, manufacturing and material science, and ergonomics principles in design of machine elements.
2. Analyze the stress and strain on mechanical components under different loadings; and understand, identify and quantify failure modes for mechanical parts.
3. Design various transmission elements such as Shafts, keys, couplings, Gears, Belt, Chain drives, Bearings, flywheels and clutches.
4. Able to make proper assumptions, perform correct analysis and finally decide the size of machine elements while giving due consideration to material, manufacturing method and cost of the element
5. Approach design problem successfully, and able to take decisions when no unique solution exists.

Unit-I Design of shaft, keys and coupling

Design of keys, shafts – solid, hollow shafts and splined shafts under torsion and bending loads.Design of couplings – Muff and Split Couplings, Flange, Flexible and Marine type of couplings.

Unit-II Design of gear drive

Gears: Introduction of gear drives, different types of gears, Materials used for gears. Standards for gears and specifications.

Spur Gear Design: Lewis equation, Beam strength of gear tooth and static design. Wear load and design for Wear. Dynamic loads on gear tooth. Design of Helical, Bevel and Worm gears, concepts of Design for manufacturability.

Unit-III Design of belt drive and chain drive

Design of belt drive systems, selection of belts and design of pulleys.Design of chain drives: Power rating of roller chains. Strength of roller chains

Unit-IV Bearing design

Bearings: Introduction. Materials used for Bearings. Classification of bearings and mounting of bearings.

Design of sliding contact bearings: Properties and types of Lubricants, Design of Hydrostatic and Hydrodynamic sliding contact bearings.

Design of Rolling Contact Bearings: Different types of rolling element bearings and their constructional details, static load carrying capacity. Dynamic load carrying capacity.Load-life relationship, selection of bearing life. Design for cyclic loads and speeds. Design of Ball and Roller bearings.

Unit-V Design of flywheel and clutch
Design of solid and rimmed type flywheel,
Requirement of clutch, Principle of clutch, Design of friction clutch- cone, centrifugal, single disc and multi disc clutch.

Text books:

1. V.B. Bhandari, Machine Design, Tata McGraw Hill Publication, 1991.
2. J.E. Shigley, C.R. Mischne, Mechanical Engineering Design, Tata McGraw Hill Publications, 2003.
3. Robert L. Norton, Machine Design: An Integrated Approach,2/e Pearson Education, 2000

REFERENCES:

1. Robert C. Juvinall, Fundamentals of Machine Component Design, John Wiley & Sons, 2005
2. M.F. Spotts, Design of Machine Elements, Prentice Hall of India, 1964.
3. P. Kannaiah, Machine Design, 2nd Edition, Scitech Publications. 2012,

Web References:

https://archive.nptel.ac.in/courses/112/106/112106137/

Category: **Professional Core Course** Subject Code**: ME3202**

Computer Aided Engineering

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OUTCOMES:

1. Understand the role of CAD in mechanical component and system design by creating geometric models and engineering drawings.
2. Use CAD software collaboratively when designing in a team.
3. Able to select type of modeling technique for given part
4. Use motion and interference checking to ensure that parts will not interfere throughout their complete range of motion.
5. Communicate effectively the geometry and intent of design features.

UNIT I

Computer Graphics:

Introduction to CAE, CAD. Role of CAD in Mechanical Engineering, Design process, software tools for CAD, geometric modelling. Transformations in Geometric Modeling: Translation, Scaling, Reflection, Rotation in 2D and 3D. Homogeneous representation of transformation, Concatenation of transformations. Projections: Projective geometry, transformation matrices for Perspective, Axonometric projections, Orthographic and Oblique projections. Implementation of the transformations and projection formulations using computer codes.

UNIT II

Geometric Modelling of curves

Types of mathematical representation of curves, wire frame models wire frame entities parametric representation of synthetic curves hermite cubic splines Bezier curves B-splines rational curves. Implementation of the all the curve models using computer codes in an interactive manner

UNIT III

Geometric Modelling of surfaces

Mathematical representation surfaces, Surface model, Surface entities surface representation, parametric representation of analytic surfaces, plane surface, rules surface, surface of revolution, Tabulated Cylinder. Parametric Representation Of Synthetic Surfaces: Hermite Bicubic surface, Bezier surface, B- Spline surface, COONs surface, Blending surface Sculptured surface, Implementation of the all the surface models using computer codes.

UNIT IV

Geometric Modelling of solids

Solid Representation, Boundary Representation (B-rep), Constructive Solid Geometry (CSG). CAD/CAM Exchange: Evaluation of data — exchange format, IGES data representations and structure, STEP Architecture, implementation, CAD Applications

UNIT V

Basic Concepts of FEA:

Introduction, One Dimensional Problems: Basis steps, Discretization, Element equations, Linear and quadratic shape functions, Assembly, Local and global stiffness matrix and its properties, boundary conditions, penalty approach, multipoint constraints, Applications to solid mechanics, heat and fluid mechanics problems,

Text Book:

1. Saxena, A., & Sahay, B. (2005). Computer Aided Engineering Design: Springer Netherlands.
2. Zeid, Ibrahim. CAD/CAM theory and practice. Spain: McGraw-Hill, , 2009
3. Adams, J. Alan., Rogers, David F.., Adams, James Alan. Mathematical Elements for Computer Graphics. Germany: McGraw-Hill, 1990.

References:

4. Zeid, Ibrahim. Mastering CAD/CAM. United Kingdom: McGraw-Hill Higher Education, 2005.
5. Reddy, J. N, Junuthula Narasimha. An introduction to the finite element method. India: McGraw-Hill, 1993.

Web References:

1. https://archive.nptel.ac.in/courses/112/102/112102101/
2. https://archive.nptel.ac.in/courses/112/102/112102101/
3. https://nptel.ac.in/courses/112102102

Category: **Professional Core Course** Subject Code**:** ME3202

Instrumentation and Control Systems

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OUTCOMES:

1. elucidate the construction and working of various industrial devices used to measure displacement, pressure, sound, flow, temperature, level, vibration.
2. ability to analyze, formulate and select suitable sensor for the given industrial applications.
3. able to describe the type of System, dynamics of physical systems, to represent system by transfer function
4. demonstrate the working and application of different type of actuators and control valves
5. able to apply techniques for controlling devices automatically.

UNIT I

Transducer Variables And Measurement Signals

Three stages of generalized measurement system– static characteristics of instruments- factors considered in selection of instruments – commonly used terms, error analysis and classification – sources of error – frequency response – displacement transducers – potentiometer, strain gauge – orientation of strain gauge, LVDT – variable reluctance transducers, proximity sensors, capacitance transducers, tacho generator; smart sensors, integrated sensors,, torque measurements,

UNIT II

Vibration And Temperature

Elementary accelerometer and vibrometer – seismic instrument for acceleration – velocity measurement, piezo electric accelerometer, temperature measurement-liquid in glass thermometer,pressure thermometer, resistance temperature detector, themcouples and thermopiles, thermistor, total radiation pyrometer, optical pyrometer – temperature measuring problem in flowing fluid.

UNIT III

Pressure And Flow Measurement

Manometer, elastic transducer, elastic diaphragm transducer – pressure cell, bulk modulus pressure gauge – Mc Leod gauge – thermal conductivity gauge, calibration of pressure gauge, flow measurement – turbine type meter, hotwire anemometer, magnetic flow meter; liquid level sensors, light sensors, selection of sensors.

UNIT IV

Control systems

Basic elements of open/closed loop, design of block diagram; control method – P, PI, PID, when to choose what, tuning of controllers; Pneumatic, hydraulic, electric systems, Basic elements involved with PLC systems.

UNIT V:

System models

Models for physical systems in terms of simple building block, Transfer function – block diagram simplification techniques, and System response-System parameters, Concepts in Stability of systems. Frequency response- Construction of Bode plot,

Text Books:

1. Bolton, William. Instrumentation and Control Systems. Netherlands: Elsevier Science, 2004.
2. Marangoni, Roy D.., Lienhard, John H.., Beckwith, Thomas G.. Mechanical Measurements. United Kingdom: Addison-Wesley, 1993.
3. Doebelin, Ernest O.. Measurement Systems: Application and Design. Japan: McGraw-Hill, 1966.

References:

4. Nise, Norman S.. Control Systems Engineering. United Kingdom: Wiley, 2020.
5. Considine, Douglas M.., McMillan, Gregory K.. Process/Industrial Instruments and Controls Handbook, 5th Edition. United Kingdom: McGraw-Hill Education, 1999.

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1. https://nptel.ac.in/courses/108105064
2. https://nptel.ac.in/courses/103103037
3. https://onlinecourses.nptel.ac.in/noc22_ag04/preview

Category: **HSM Course** Subject Code**: BM3201**

MANAGERIAL ECONOMICS & FINANCIAL ANALYSIS

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Course Outcome: After the successful completion of this course, the learner will be able to know:

1. The dynamic game of demand and supply, and how the trinity of Economics i.e. Demand, Supply and Scarcity make the things move around the globe.
2. Principles of Microeconomics applied to industries.
3. Concept of forecasting and applying forecasting techniques to address the challenges and opportunities in the organization they work.
4. Cost and Production analysis, Break-Even analysis, Opportunity Cost, how to optimize organizational resources and how to minimize cost and maximize production, revenue and profit
5. Different pricing structure and discount mechanism suitable for business firms.
6. Market structure and how to exploit market structure for optimizing the benefits of organization.
7. Capital requirements and sources of capital.

UNIT I:

Introduction to Managerial Economics

Definition, Nature and Scope of Managerial Economics-Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting

UNIT II:

Theory of Production and Cost Analysis

Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs. Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale.000

Cost Analysis: Cost concepts, Opportunity cost. Fixed vs. Variable costs, Explicit costs Vs. Implicit costs. Out of pocket costs vs. Imputed costs. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems)- Managerial Significance and limitations of BEA

UNIT III:

Markets & Pricing Policies

Market structures: Types of competition, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Objectives and Policies of Pricing- Methods of Pricing: Cost Plus Pricing. Marginal Cost Pricing, Sealed Bid Pricing, Going Rate Pricing, Limit Pricing, Market Skimming Pricing, Penetration Pricing

UNIT IV:

Introduction to Financial Accounting

Introduction to Financial Accounting: Double entry Book Keeping, Journal, Ledger, Trail Balance and Final Accounts (Trading account, Profit and Loss Account and Balance sheet with simlvbple adjustments).

UNIT V:

Capital and Capital Budgeting

Capital and Capital Budgeting: Capital and its significance. Types of Capital. Estimation of Fixed and Working capital requirements. Methods and sources of raising finance. Nature and scope of capital budgeting, features of capital budgeting proposals. Methods of Capital Budgeting: Payback Method. Accounting Rate of Return (ARR) and Net Present Value Method, Internal Rate of Return (IRR).

Reference Books:

1. Aryasri. Managerial Economics And Financial Analysis: McGraw-Hill Education (India) Pvt Limited,2009.
2. Maheshwari, Yogesh. Managerial Economics. India: PHI Learning, 2012
3. Ahuja, Amit. Managerial Economics (Analysis of Managerial Decision Making), 9th Edition. India: S Chand Limited, 2017.
4. Raghunatha Reddy &Narasimhachary: Managerial Economics & Financial Analysis, Scitech, 2009.
5. V.Rajasekarn&R.Lalitha. Financial Accounting, Pearson Education. New Delhi. 2010
6. Suma Damodaran, Managerial Economics, Oxford University Press. 2009.

Web References:

1. https://www.digimat.in/nptel/courses/video/110105075/L01.html

Category: Professional Core Course **Subject Code: ME3801**

Thermal Engineering laboratory

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0 - 0 - 2 - 1**

List of experiments:

1. To study Vapour Compression Refrigeration cycle with the help of refrigeration circuit under variable load conditions
2. To determine the Coefficient of Performance, Refrigeration capacity & Compressor work of Vapour Compression Refrigeration cycle with the help of refrigeration circuit under variable load conditions
3. To study Vapour Absorption Refrigeration cycle
4. To determine the Coefficient of Performance, Refrigeration capacity & Compressor work of Vapour Absorption Refrigeration cycle
5. To compare heat transfer for different heating elements in a cross flow heat exchanger
6. To study fundamental principles and various controls used in room air conditioning
7. To study different psychometric processes and estimating the change of state of air using air conditioner and illustrating them on psychometric diagram
8. Study on the characteristics of flame stability and methods to improve stability limits
9. Determination of flame speed based on the cone method
10. Determination of the relationship between flame speed and air/fuel ratio
11. flame separation demonstration
12. To study and determine performance parameters for Petrol Engine
13. To study and determine performance parameters for Diesel Engine

Category: **Professional Core Course** Subject Code**: ME3802**

Computer Aided Engineering laboratory

Internals: 50 Marks **L - T - P - C**
Externals: 50 Marks **0 - 0 - 3 - 1.5**

LIST OF EXPERIMENTS:

One dimensional analysis:

1. Structural analysis of cantilever beam under different types of loading
2. Structural analysis of simply supported beam under different types of loading

Two dimensional analysis:

3. Static structural analysis of Plate with hole
4. Transient structural analysis of rack and pinion

Three dimensional analysis:

5. Static structural analysis of Cotter Joint
6. Rigid dynamic analysis of Oldham coupling
7. Thermal analysis of Piston

Non linear analysis:

8. Non-linear analysis of sheet bending
9. Non-linear buckling analysis of column

Category: **Professional Core Course** Subject Code**: ME3803**

Instrumentation and Control Systems Laboratory

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **0 - 0 - 2 - 1**

Sl.No	**Name of the Experiment**
1.	Calibration of Pressure Gauge using Dead Weight Tester (DWT)
2.	Measurement of displacement using Full bridge Strain Gauge circuit
3.	Measurement of displacement using Linear Variable Differential Transformer (LVDT)
4.	Motor speed measurement using Magnetic Pick Up Sensor, Photo Reflector Sensor, Photo Interruptive Sensor and Hall Effect Sensor
5.	Measurement of Torque Generated by AC (Induction motor) using Force Transducer.
6.	Weight measurement using Load cell
7.	Static Torque Measurement by using load cell.
8.	Measurement of Pressure using Transducer
9.	Strain measurement using strain gauges and cantilever assembly
10	Industrial PLC trainer
11	Motion control of Rotary inverted pendulum

FOURTH YEAR (E4) – SEMESTER – I

Sl. No.	Course Code	Course Title	Course Category	L	T	P	Total Contact Hours	Credits
1	ME4101	Automation in Manufacturing	PCC	3	0	0	3	3
2	ME4102	Refrigeration and Air Conditioning	PCC	3	0	0	3	3
3	ME415X	Professional Elective – 3	PEC	3	0	0	3	3
4	ME415X	Professional Elective – 4	PEC	3	0	0	3	3
5	ME415X	Professional Elective – 5	PEC	3	0	0	3	3
6	ME4701	Automation in Manufacturing Lab	PCC	0	0	2	2	1
7	ME4000	Comprehensive Viva		0	0	0	0	1
8	ME4XXX	Internship		0	0	0	0	1
9	ME4901	Project – I						4
Total								**22**

Category: **Professional Core Course** Subject Code**: ME4101**

Automation in Manufacturing

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Course Objectives:

1. To understand the importance of automation in the of field machine tool based manufacturing
2. To get the knowledge of various elements of manufacturing automation – CAD/CAM, sensors, pneumatics, hydraulics and CNC.
3. To understand the basics of product design and the role of manufacturing automation

Course Outcomes:

Upon completion of this course, the students will get a comprehensive picture of computer based automation of manufacturing operations

UNIT I:

Introduction: Why automation, Current trends, CAD, CAM, CIM; Rigid automation: Part handling, Machine tools.

UNIT II:

Flexible automation: Computer control of Machine Tools and Machining Centers, NC and NC part programming, CNC-Adaptive Control, Automated Material handling, Assembly, Flexible fixturing.

UNIT III:

Computer Aided Manufacturing: CNC technology, PLC, Micro-controllers, CNC- Adaptive Control

UNIT IV:

Low cost automation: Mechanical & Electro mechanical Systems, Pneumatics and Hydraulics, Illustrative Examples and case studies

UNIT V:

Introduction to Modeling and Simulation: Product design, process route modeling, Optimization techniques, Case studies & industrial applications.

Text Books:

1. Jayaprakash, G, Groover, Mikell P.. Automation, Production Systems, and Computer-integrated Manufacturing. United Kingdom: Pearson Education Limited, 2015.
2. SeropeKalpakjian and Steven R. Schmid, Manufacturing –Engineering and Technology, 7th edition,Pearson
3. Koren, Yoram. Computer Control of Manufacturing Systems. Singapore: McGraw-Hill, 1983.
4. Zeid, Ibrahim. CAD/CAM Theory and Practice. Singapore: McGraw-Hill,

Web References:

1. https://onlinecourses.nptel.ac.in/noc22_me123/preview
2. https://nptel.ac.in/courses/112104288

Category: **Professional Core Course** Subject Code**: ME4102**

Refrigeration and Air Conditioning

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT I :

REFRIGERATION SYSTEM

Introduction to Refrigeration system : Necessity and applications – Unit of refrigeration and C.O.P. Mechanical Refrigeration – Types of Ideal cycles of refrigeration. Air Refrigeration: Bell Coleman cycle and Brayton Cycle, Open and Dense air systems – Actual air refrigeration system problems Refrigeration needs of Air crafts.

UNIT II :

VAPOUR COMPRESSION AND ABSORPTION REFRIGERATION

Vapour compression refrigeration – working principle and essential components of the plant simple Vapour compression refrigeration cycle – COP – Representation of cycle on T-S and p h charts effect of sub cooling and super heating – cycle analysis – Actual cycle Influence of various parameters on system performance – Use of p-h charts – numerical Problems

Vapor Absorption System – Calculation of max COP – description and working of NH3 water system and Li Br –water (Two shell & Four shell) System. Principle of operation Three Fluid absorption system, silent features

UNIT III:

SYSTEM COMPONENTS

System Components : Compressors – General classification – comparison – Advantages and Disadvantages. Condensers classification Working Principles Evaporators classification Working Principles Expansion devices Types Working Principles Refrigerants – Desirable properties – classification refrigerants used – Nomenclature – Ozone Depletion Global Warming

UNIT IV :

AIR CONDITIONING

Introduction to Air Conditioning Review of fundamental properties of psychometric – use of sychometric charts – psychometric processes – Grand and Room Sensible Heat Factors – by pass factor – requirements of comfort air conditioning –factors governing optimum effective temperature, recommended design conditions and ventilation standards. Concept of ESHF and ADP Requirements of human comfort and concept of effective temperature- Comfort chart –Comfort Air conditioning – Requirements of Industrial air conditioning, Air conditioning Load Calculations.

UNIT V:

AIR CONDITIONING SYSTEMS AND HEAT PUMP

Air Conditioning systems - Classification of equipment, cooling, heating humidification and dehumidification, filters, grills and registers, fans and blowers. Heat Pump – Heat sources – different heat pump circuits, air conditioning applications

TEXT BOOKS:

1. Arora, C.P. (2010). Refrigeration and Air Conditioning: Tata McGraw-Hill.
2. Ballany P.L., Refrigeration and Air Conditioning, Khanna Publications,2009

REFERENCE BOOKS:

1. Arora, S. C, Domkundwar, S.. A Course in Refrigeration & Air-conditioning: Environmental Engineering. India: Dhanpat Rai, 1980.
2. ASHRAE Handbook: Heating, ventilating, and air-conditioning applications. United States: American Society of Heating, Refrigerating and Air Conditioning Engineers, 2003.
3. Ananthanarayanan. (2005). Basic Refrigeration and Air Conditioning: McGraw-Hill Education.

WEB REFERENCES:

1. https://archive.nptel.ac.in/courses/112/105/112105129/
2. https://archive.nptel.ac.in/courses/112/107/112107208/
3. https://www.youtube.com/watch?v=F2iCHWAibsg

Category: **Professional Core Course** Subject Code**: ME4701**

Automation in Manufacturing Lab

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

List of Experiments

1. Simulation of manufacturing Processes.
2. Machining of Components using CNC Lathe.
3. Machining of Components using CNC Mill.
4. Welding of materials by MIG and TIG
5. Palletization of objects using pick and place Robot.
6. Manufacturing of simple components using 3D Printer.

Program Elective Courses (PECs)

Sl. No.	Course Code	Course Title	Course Category
1	ME2211	Composite materials	PEC-I
2	ME2212	Nanotechnology	PEC-I
3	ME2213	Biomaterials	PEC-I
4	ME3121	Robotics	PEC-II
5	ME3122	Nontraditional Manufacturing Processes	PEC-II
6	ME3123	Green Engineering	PEC-II
7	ME3231	Automotive systems	PEC-III
8	ME3232	Production, Planning and Control	PEC-III
9	ME3233	Computational Fluid Dynamics	PEC-III
10	ME3234	Internet of things	PEC-III
11	ME3235	Mechanical Vibrations	PEC-III
12	ME4141	AI and ML for Mechanical Systems	PEC-IV
13	ME4142	Biomechanical Engineering	PEC-IV
14	ME4143	Turbomachinery	PEC-IV
15	ME4144	Additive Manufacturing Technology	PEC-IV
16	ME4145	Systems Engineering	PEC-IV
17	ME4151	Laser applications in Manufacturing systems	PEC-V
18	ME4152	Solar Technology	PEC-V
19	ME4153	Non-Destructive Technology	PEC-V
20	ME4154	Finite Element Methods	PEC-V
21	ME4155	Industrial Engineering	PEC-V

Category: **Professional Elective Course** Subject Code**: ME2211**

Composite Materials

Internals: 40 Marks **L - T - P – C**

Externals: 60 Marks **3 - 0 - 0 – 3**

Course Objectives:

1. Understanding the fundamentals of composites and types of composites
2. Understanding the properties of fiber and matrix materials used in commercial composites, as well as some common manufacturing techniques.
3. Learn some common manufacturing processes and mechanical behavior of composite materials, especially of fiber composites.
4. Study the advances in composites and its applications

UNIT – I

Introduction to composites: Fundamentals of composites - need for composites – Enhancement of properties - classification of composites – Matrix-Polymer matrix composites (PMC), Metal matrix composites (MMC), Ceramic matrix composites (CMC, Fiber reinforced composites. Applications of various types of composites.

UNIT II

Polymer matrix composites: Polymer matrix resins – Thermosetting resins, thermoplastic resins – Reinforcement fibres – Woven fabrics – Non woven random mats – various types of fibres. PMC processes - Hand layup processes – Spray up processes – Compression moulding – Reinforced reaction injection moulding - Resin transfer moulding – Pultrusion – Filament winding – Injection moulding. Fibre reinforced plastics (FRP).

UNIT III

Metal matrix composites: Characteristics of MMC, Various types of Metal matrix composites Alloy vs. MMC, Advantages of MMC, Limitations of MMC, Metal Matrix. Effect of reinforcement - Volume fraction – Rule of mixtures. Processing of MMC – Powder metallurgy process - diffusion bonding – stir casting – squeeze casting.

UNIT IV

Ceramic matrix composites: Engineering ceramic materials – properties – advantages – limitations – Monolithic ceramics - Need for CMC – Ceramic matrix - Various types of Ceramic Matrix composites- oxide ceramics – non oxide ceramics – aluminium oxide – silicon nitride – reinforcements – particles- fibres- whiskers. Sintering - Hot pressing – Cold isostatic pressing (CIPing) – Hot isostatic pressing (HIPing).

UNIT V

Advances in composites: Carbon / carbon composites – Advantages of carbon matrix – limitations of carbon matrix Carbon fibre – chemical vapour deposition of carbon on carbonfibre perform. Sol gel technique. Composites for aerospace applications.

Text Books:

1. Matthews, F., and R.D. Rawlings. Composite Materials: Engineering and Science. Springer Netherlands, 1994.
2. Chawla K.K., Composite materials, Springer – Verlag, 1987

References:

1. Agarwal, B.D., L.J. Broutman, and K. Chandrashekhara. Analysis and Performance of Fiber Composites. Wiley, 2017.
2. Agarwal, B. D., and L. J. Broutman. Analysis and performance of fiber composites Second edition. John Wiley & Sons, 1990.
3. Chawla, Krishan K.. Composite Materials: Science and Engineering. United States: Springer New York, 2016.
4. Mouritz, Adrian P. Introduction to Aerospace Materials. United Kingdom: Elsevier Science, 2012.
5. Leonard Hollaway (1994), Handbook of Polymer Composite for Engineers, ISBN: 1-85573- 1290, Woodhead Publishing Ltd.

Web Resources:

1. https://nptel.ac.in/courses/112104168
2. https://archive.nptel.ac.in/courses/112/104/112104229/
3. https://nptel.ac.in/courses/101104010

Category: **Professional Elective Course** Subject Code**: ME2212**

Nanotechnology

Internals: 40 Marks **L - T - P – C**

Externals: 60 Marks **3 - 0 - 0 – 3**

Course Outcomes: Upon successful completion of this course, students will be able to

- Identify 0D,1D,2D and 3D nanomaterials.
- Gain knowledge the optical and mechanical properties
- Interpret the magnetic and electrical properties.
- Illustrate the use of nanomaterials for different applications

Unit I

Introduction of nanomaterials and nanotechnologies, influence of nanoover micro/macro, Comparison of nanotechnology with micromanufacturing, Features of nanostructures, Background of nanostructures,Techniques of synthesis of nanomaterials, Tools of the nanoscience,Applications of nanomaterials and technologies.

Unit II

Bonding and structure of the nanomaterials, Predicting the Type ofBonding in a Substance crystal structure, One dimensional, Twodimensional and Three dimensional nanostructured materials, Metallicnanoparticles, Surfaces of Materials, Nanoparticle Size and Properties

Unit III

Mechanical properties of materials, theories relevant to mechanicalproperties, techniques to study mechanical properties of nanomaterials,adhesion and friction, thermal properties of nanomaterials.

Unit IV

Electrical properties, Conductivity and Resistivity, Classification ofMaterials based on Conductivity, magnetic properties, electronic propertiesof materials, classification of magnetic phenomena.

Unit V

Nano thin films, nanocomposites, new application of nanoparticles inmanufacturing of bearings, cutting tools, cutting fluids, medical science,soil science, membrane-based application, polymer based application

Text Books:

1. Wilson, M., K. Kannangara, G. Smith, M. Simmons, and B. Raguse. *Nanotechnology: Basic Science and Emerging Technologies.* CRC Press, 2005.
2. Poole, C.P., C.P. Poole, F.J. Owens, F.J.A. OWENS, and F.J. Owens. *Introduction to Nanotechnology.* Wiley, 2008.

3. Ratner, D. *Nanotechnology: A Gentle Introduction to the Next Big Idea.* Pearson Education, 2003.

References:

1. Shanmugam, S. *Nanotechnology.* MJP Publisher, 2019.
2. Sengupta, A., and C.K. Sarkar. *Introduction to Nano: Basics to Nanoscience and Nanotechnology.* Springer Berlin Heidelberg, 2015.

Web Resources:

1. https://nptel.ac.in/courses/118102003
2. https://nptel.ac.in/courses/113106093
3. https://archive.nptel.ac.in/courses/118/107/118107015/

Category: **Professional Elective Course** Subject Code**: ME2213**

Biomaterials

Internals: 40 Marks **L - T - P – C**

Externals: 60 Marks **3 - 0 - 0 – 3**

Course Objectives:

At the end of the course the student will be able to:

CO1: Understand the structure and properties of biomaterials

CO2: Classify implant biomaterials

CO3: Evaluate biocompatibility of implants

CO4: Identify appropriate biomaterials for specific medical applications

Unit-1

Overview of biomaterials: Historical developments, impact of biomaterials, interfacial phenomena, tissue responses to implants. Structure and properties of biomaterials: Crystal structure of solids, phase changes, imperfections in solids, non-crystalline solids, surface properties of solids, mechanical properties, surface improvements.

Unit-2

Types of biomaterials: Metallic implant materials, ceramic implant materials, polymeric implant materials composites as biomaterials.

Unit-3

Characterization of materials: Electric properties, optical properties, X-ray absorption, acoustic and ultrasonic properties.

Unit-4

Bio implantation materials: Materials in ophthalmology, orthopedic implants, dental materials and cardiovascular implant materials.

Unit-5

Tissue response to implants: Normal wound healing processes, body response to implants, blood compatibility, structure - property relationship of tissues.

Text Books:

1. Dahman, Yaser. Biomaterials Science and Technology: Fundamentals and Developments. United States: CRC Press, 2019.
2. Biomaterials Science: An Introduction to Materials inMedicine. Netherlands: Elsevier Science, 2020.

References:

1. JoonPark, R.S. Lakes, Biomaterials an introduction; 3rd Edition, Springer, 2007
2. Sujatha V Bhat, Biomaterials; 2nd Edition, NarosaPublishing House, 2006.

Web Resources:

1. https://nptel.ac.in/courses/113104009
2. https://onlinecourses.nptel.ac.in/noc19_mm24/preview
3. https://onlinecourses.nptel.ac.in/noc20_bt12/preview

Category: Professional Elective Course **Subject Code: ME3121**

Robotics

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT - I

Introduction: Automation and Robotics, CAD/CAM And Robotics -An over view of Robotics – presentand future applications - classification by coordinate system and control systems. Components of the Industrial Robotics: End effectors-types, Mechanical grippers, and other typesof grippers, comparison of Electric, Hydraulic and pneumatic types of locomotion devices.

UNIT - II

Motion Analysis: Homogeneous transformations as applicable to rotation and translation — Problems.
Manipulator Kinematics: Specifications of matrices, D-H notation Joint coordinates and worldcoordinates – Forward and inverse kinematics — problems.

UNIT - III

Manipulator jacobians: Differential transformation and manipulators, Jacobians — problems.
Dynamics: Lagrange — Euler and Newton-Euler formulations — Problems.

UNIT - IV

Trajectory Planning: Path planning and avoidance of obstacles, Slew motion, joint interpolated motion — straight line motion.
Programming Languages: problems. Robot programming, languages and software packages

UNIT - V

Robot actuators and Feed-back components: Actuators: Pneumatic, Hydraulic actuators, electric and stepper motors. Feedback components, position sensors – potentiometers, resolvers, encoders – Velocity sensors.
Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading –processing – spot and continuous arc welding & spray painting – Assembly and Inspection

Text Books:

1. Nagarajan Ramachandran, Introduction to Industrial Robotics. India: Pearson India, 2016.
2. Mikell P. Groover, Industrial Robotics: Technology, Programming, and Applications. India: McGraw-Hill, 2012.
3. Nagel, Roger N.., Groover, Mikell P.., Weiss, M. Industrial Robotics: Technology, Programming, and Applications. Taiwan: McGraw-Hill, 1986.

Reference Books:

1. K S Fu, Ralph Gonzalez, C S G Lee. Robotics: Control Sensing. Vis. McGraw-Hill, 1987.
2. Kuttan, Appuu. Robotics. India: I.K. International Publishing House Pvt. Limited, 2013.

3. Slotine, J.-J. E.., Slotine, Jean-Jacques., Asada, H. Robot analysis and control. United Kingdom: Wiley, 1986.
4. Mark W. Spong and M. Vidya Sagar I John Wiley & Sons, Robot Dynamics and Control. India: Wiley India Pvt. Limited, 2008.
5. Robotics and Control. India: Tata McGraw-Hill, 2003.
6. Nagrath, M. Robotics and Control. Tata McGraw-Hill, 2003.

Web Resources:

1. https://nptel.ac.in/courses/112105249
2. https://nptel.ac.in/courses/107106090
3. https://archive.nptel.ac.in/courses/112/105/112105249/

Category: Professional Elective Course **Subject Code: ME3122**

Non-Traditional Manufacturing Process

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OBJECTIVES:

1. Understand the need and importance of non-traditional machining methods and process selection.
2. Gain the knowledge to remove material by thermal evaporation, mechanical energy process.
3. Apply the knowledge to remove material by chemical and electro chemical methods.
4. Analyze various material removal applications by unconventional machining process.

COURSE OUTCOMES:

1. Students will be able to categorize different material removal, joining processes as per the requirements of material being used to manufacture end product.
2. Students will be able to select material processing technique with the aim of cost reduction, reducing material wastage & machining time.
3. Students will be able to identify the process parameters affecting the product quality in various advanced machining of metals/ non-metals, ceramics and composites.
4. Students will be able to combine & develop novel hybrid techniques from the state of art techniques available.

UNIT— I

Need for Modern Manufacturing Methods: Non-traditional machining methods and rapid prototyping methods - their relevance for precision and lean manufacturing. Classification of non-traditional processes - their selection for processing of different materials and the range of applications.

Introduction to rapid prototyping - Classification of rapid prototyping methods - stereolithography, fused deposition methods - materials, principle of prototyping and various applications.

UNIT—II

Abrasive jet, Water jet and abrasive water jet machining: Basic mechanics of material removal, descriptive of equipment, process variables, applications and limitations.

Ultrasonic machining – Elements of the process, mechanics of material removal, process parameters, applications and limitations.

UNIT – III

Electro – Chemical Processes: Fundamentals of electro chemical machining, electrochemical grinding, metal removal rate in ECM, Tooling, process variables, applications, economic aspects of ECM.

Chemical Machining: Fundamentals of chemical machining- Principle of material removal- maskants – etchants- process variables, advantages and applications.

UNIT—IV

Thermal Metal Removal Processes: Basic principle of spark erosion (EDM), Wire cut EDM, and Electric Discharge Grinding processes - Mechanics of machining, process parameters,

selection of tool electrode and dielectric fluids, choice of parameters for improved surface finish and machining accuracy - Applications of different processes and their limitations.
Plasma Machining: Principle of material removal, description of process and equipment, process variables, scope of applications and the process limitations.

UNIT-V

Electron Beam Machining: Generation and control of electron beam for machining, theory of electron beam machining, comparison of thermal and non-thermal processes - process mechanics, parameters, applications and limitations.
Laser Beam Machining: Process description, Mechanism of material removal, process parameters, capabilities and limitations, features of machining, applications and limitations.

Text Book

1. Jain, P.V.K. Advanced Machining Processes. Allied Publishers Pvt Limited, 2009.
2. Pandey, P.C., and H.S. Shan. Modern Machining Processes. McGraw-Hill, 1980.
3. “Modern Machining Processes” P.C Pandey & H.S. Shan, McGraw Hill Education.
4. Benedict, G.F. Nontraditional Manufacturing Processes. CRC Press LLC, 2019.

References

1. Adithan, M. Unconventional Machining Processes. Atlantic Publishers & Distributors (P) Limited, 2009.
2. C. Elanchezhian,, B. Vijaya Ramnath and M Vijayan, Unconventional Machining Processes, Anuradha Publications, 2005.
3. Singh, M.K. Unconventional Manufacturing Process. New Age International (P) Limited, 2008.
4. Manufacturing Technology, P.N. Rao,TMH
5. Singh K K, Unconventional Manufacturing Processes, Dhanpat Rai & Company, New Delhi.

Web Resources:

1. https://archive.nptel.ac.in/courses/112/105/112105212/
2. https://www.udemy.com/course/non-conventional-machining-processes/
3. https://onlinecourses.nptel.ac.in/noc22_me119/preview

Category: Professional Elective Course **Subject Code: ME3123**

Green Engineering

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 – 3**

UNIT - I:
Introduction
Energy sources and their availability- commercial and non commercial energy sources. Need of Renewable Energy Sources (RES), classification of RES, Role and potential of RES in India. Solar Radiation: Structure of the sun, Solar constant, environmental impact of solar radiation, Radiation at the earth surfaces, solar radiation Geometry, extraterrestrial and terrestrial solar radiation, Spectral Distribution of Extraterrestrial Radiation, solar radiation on tilted surfaces and Empirical equations for predicting the availability of solar radiation at any given location. Solar energy - Thermal applications.

UNIT - II:
Solar Collectors
Principle of solar energy conversion into heat, classification of solar collectors, Flat plate collectors, basic energy balance equation, collector efficiency, thermal analysis of flat plate collector. Concentrating collectors and its advantages and disadvantages. Performance analysis of concentrating collectors, selection of absorber coating materials

UNIT - III:
Solar Energy Storage and Applications
Solar Energy Storage: Different storage methods- sensible, latent heat and stratified storage, solar ponds. Solar Energy Applications: Solar water, space heating /cooling, solar thermal electric conversion, direct solar electric power generation- solar photovoltaic, solar distillation, Solar Pumping, Solar furnace, Solar cocking and solar green house.

UNIT - IV:
Wind Energy, Biomass Energy Conversion Systems and Geothermal Thermal Energy
Wind Energy: Working principle of wind energy conversion, Wind patterns, Components of wind energy conversion system (WECS), Types of Wind machines – horizontal axis and vertical axis, Betz coefficient. Biomass Energy Conversion Systems: Biomass Energy: Fuel classification – Pyrolysis – Different digesters and sizing. Geothermal Thermal Energy: Classification – Dry rock and acquifer – Energy analysis

UNIT - V:
Ocean Thermal Energy, Tidal Power System and Wave Energy
Ocean Thermal Energy: Method s of Ocean Thermal Electric power generation-Open cycle systems, closed cycle systems Tidal Power System: Working principle, components of Tidal Power plant, single basin and double basin tidal energy system advantages and limitations . Wave Energy: Wave energy conversion Devices-wave energy conversion by floats, high

level reservoir wave machine and dolphin type wave power machine. Advantages and disadvantages

UNIT - VI:

Direct Energy Conversion, MHDPower Generation and Fuel Cell

Direct Energy Conversion (DEC): Need for DEC, limitations, principles of DEC. Thermoelectric Power – See-beck, Peltier, Joule -Thomson effects, Thermo-electric Power generators MHD Power Generation: Principles, dissociation and ionization, Hall effect, magnetic flux, MHD accelerator, MHD engine, power generation systems, electron gas dynamic conversion. Fuel Cell: working principle, classification – Efficiency – VI characteristics.

Text Books

1. SP Sukhatme, Solar Energy: Principles of thermal collection and storage, Tata McGraw Hill
2. Tiwari, G.N., and M.K. Ghosal. Renewable Energy Resources: Basic Principles and Applications. Alpha Science International, 2005.

References

1. Allen, D.T., and D.R. Shonnard. Green Engineering: Environmentally Conscious Design of Chemical Processes. Pearson Education, 2001.
2. Jena, O.P., A.R. Tripathy, and Z. Polkowski. Green Engineering and Technology: Innovations, Design, and Architectural Implementation. CRC Press, 2021.

Web Resources:

1. https://archive.nptel.ac.in/courses/127/105/127105018/
2. https://nptel.ac.in/courses/127105018
3. https://archive.nptel.ac.in/courses/105/102/105102195/

Category: **Professional Elective Course** Subject Code**: ME3231**

Automotive systems

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT I

Introduction: Components of a Four Wheeler Automobile – Chassis and Body – Power Unit –Power Transmission – Rear Wheel Drive, Front Wheel Drive, Four Wheel Drive – Types of Automobile Engines, Engine Construction, Turbo Charging and Super Charging – Oil Filters, Oil Pumps – Crank Case Ventilation.

UNIT II

Emissions from Automobiles – Pollution Standards National and International – Pollution Control–Techniques – Multipoint Fuel Injection for SI Engines- Common Rail Diesel Injection, Emissions from Alternative Energy Sources– Hydrogen, Biomass, Alcohols, LPG, CNG - Their Merits And Demerits. Electrical System: Charging Circuit, Generator, Current – Voltage Regulator – Starting System, Bendix Drive, Mechanism of Solenoid Switch, Lighting Systems, Horn, Wiper, Fuel Gauge – Oil Pressure Gauge, Engine Température Indicator.

UNIT III

Transmission System: Clutches- Principle- Types: Cone Clutch, Single Plate Clutch, Multi Plate Clutch, Magnetic and Centrifugal Clutches, Fluid Fly Wheel – Gear Box- Types: Sliding Mesh, Constant Mesh, Synchromesh, Epi-Cyclic, Over Drive, Torque Converter. Propeller Shaft – Hotch – Kiss Drive, Torque Tube Drive, Universal Joint, Differential, Rear Axles.

UNIT IV

Steering System: Steering Geometry – Camber, Castor, King Pin Rake, Combined Angle Toe-In, Center Point Steering. Types Of Steering Mechanism – Ackerman Steering Mechanism, Davis Steering Mechanism, Steering Gears – Types, Steering Linkages.

UNIT V

Suspension System: Objects of Suspension Systems – Rigid Axle Suspension System, Torsion Bar, Shock Absorber, Independent Suspension System. Braking System: Mechanical Brake System, Hydraulic Brake System, Pneumatic and Vacuum Brake Systems.

Text Books:

1. Maurer, M., and H. Winner. Automotive Systems Engineering. Springer Berlin Heidelberg, 2013.

2. Heisler, H. Advanced Vehicle Technology. Butterworth-Heinemann, 2002.

Reference Books:

1. Awari, G.K., V.S. Kumbhar, and R.B. Tirpude. Automotive Systems: Principles and Practice. CRC Press, 2021.
2. Sivakumar, P., B.V. Kumar, and R.S.S. Devi. Software Engineering for Automotive Systems: Principles and Applications. CRC Press, 2022.
3. R.K.Rajput,Laxmi, Automobile Engineering, Pub, 1st edition, 2011

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc23_de01/preview
2. https://www.udemy.com/topic/automobile-engineering/

Category: **Professional Elective Course** Subject Code**: ME3232**

Production Planning & Control

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 – 3**

Unit I PPC performance

PPC – Requirements, Benefits, Factors influencing PPC performance, 3 types of decisions – 3 Phases of PPC – Aggregate and Disaggregate Planning – Master Production Schedule (MPS) – Techniques & Hour Glass Principle – Bill of Material (BOM) structuring

Unit II MRP

Material Requirements Planning (MRP) System – Inputs, Outputs, Benefits, Technical issues – MRP system nervousness – Manufacturing Resources Planning (MRP II) – Resource Planning - Final assembly scheduling

Unit III Capacity management

Capacity Planning using overall factors (CPOF) – Capacity Bills – Resource Profiles – Capacity requirements planning (CRP) – I/O Control - Shop floor control – Basic concepts, Gantt Chart, Priority sequencing rules and Finite Loading – Inventory models.

Unit IV Shop floor control

Shop floor control – Just in time (JIT) – Key elements, techniques – JIT & PPC – Pull & Push Systems – Kanban system – Types, number of kanban calculations, Design, advantages and disadvantages

Unit V ERP System

ERP systems – Components, Modules, Implementation, advantages and disadvantages - Technical aspects of SAP - Supply Chain Management (SCM) – Components, stages, Decision phases – Supply chain macro processes in a firm

Text Books

1. Vollmann, T.E., Berry, W.L., Whybark, D.C., and Jacobs, F.R., Manufacturing Planning and Control for Supply Chain Management (5th ed.), Irwin. 2005
2. Kiran, D.R. Production Planning and Control: A Comprehensive Approach. Elsevier Science, 2019.

Reference Books

1. Mukhopadhyay, S.K. Production Planning and Control: Text and Cases. PHI Learning, 2015.
2. Sipper, D., Bulfin, R.L., Production Planning, Control, and Integration, McGraw Hill. 2007.

Web Resources:

1. https://nptel.ac.in/courses/112107143
2. https://www.npcindia.gov.in/NPC/Homes1/e-learning/course?search=PRODUCTION+PLANNING+AND+CONTROL
3. http://www.digimat.in/nptel/courses/video/110107141/L34.html

Category: **Professional Elective Course** Subject Code**: ME3233**

Computational Fluid Dynamics

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OUTCOMES:

- Illustrate governing equations for various applications.
- Identify, formulate and solve the fluid dynamics problems using Boundary conditions.
- Apply fluid dynamics principles in Internal flows as well as External Flows
- Able to analyze different flow characteristics of laminar and turbulent flows
- Analyze various problems involving fluid properties and shear forces resulting from Newtonian fluids
- Evaluate fluid systems using the integral form of the continuity, momentum, and energy equation

UNIT I

Introduction: Methods to solve a physical problem , numerical methods , brief comparison between FDM, FEM & FVM, applied numerical methods. Solution of a system of simultaneous linear algebraic equations, Iterative schemes of matrix inversion, direct methods for matrix inversion, direct methods for baned matrices. Finite difference applications in heat conduction and convention, heat conduction, steady heat conduction in a rectangular geometry, transient heat conduction, finite difference application in convective heat transfer.

UNIT II

Finite differences: Discretization, consistency, stability, and fundamentals of fluid flow modeling. Introduction, elementary finite difference quotients, implementation aspects of finite-difference equations, consistency, explicit and implicit methods.

UNIT III

Errors and stability analysis: introduction, first order wave equation, stability of hyperbolic and elliptic equations, fundamentals of fluid flow modeling, conservative property, the upwind scheme. Review of equations governing fluid flow and heat transfer: Introduction, Conservation of mass Newton's second law of motion, expanded forms of Navier-stokes equations, conservation of energy principle, special forms of the Navier stokes equations.

UNIT IV

Steady flow: Dimensions form of momentum and energy equations, navier stokes equation, and conservative body force fields, stream function, vorticity formulation, boundary, layer theory, buoyancy, driven convection and stability.

UNIT V

Simple cfd techniques: Viscous flows conservation form space marching, relovation techniques, viscous flows, conservation from space marching relovation techniques, artificial viscosity, the alternating direction implicit techniques, pressure correction technique, computer graphic techniques used in CFD. Quasi one dimensional flow through a nozzle, turbulence models, standard and high reynolds number models and their applications.

Text Books:

1. T. J. Chung, Computational Fluid Dynamics, Cambridge University Press. 2010
2. John D.Anderson Jr, Computational Fluid Dynamics, McGraw Hill Book Company
3. Pradip Niyogi, Chakrabarty, S.K., Laha, M.K. "Introduction to Computational FluidDynamics", Pearson Education, 2005.

Reference Books:

1. Anil W. Date, Introduction to Computational Fluid Dynamics, Cambridge UniversityPress, 2005.
2. Patankar, S.V.Numerical Heat Transfer and Fluid Flow, Hemisphere PublishingCorporation, 2004.
3. Muralidhar, K., and Sundararajan, T., Computational Fluid Flow and Heat Transfer,Narosa Publishing House, New Delhi, 1995.
4. S.V.Patankar, Numerical Heat Transfer and Fluid Flow, McGraw-Hill.

Web Resources:

1. https://nptel.ac.in/courses/112105045
2. https://onlinecourses.nptel.ac.in/noc21_me126/preview
3. https://onlinecourses.nptel.ac.in/noc23_ch10/preview
4. https://www.coursera.org/lecture/digital-thread-implementation/computational-fluid-dynamics-HXjWG

Category: Professional Elective Course **Subject Code: ME3234**

Internet of things

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT-I
Fundamentals of IoT: Introduction, Definitions & Characteristics of IoT, IoT Architectures, Physical & Logical Design of IoT, Enabling Technologies in IoT, History of IoT, About Things in IoT, The Identifiers in IoT, About the Internet in IoT, IoT frameworks, IoT and M2M.

UNIT-II
Sensors Networks: Definition, Types of Sensors, Types of Actuators, Examples and Working,IoT Development Boards: Arduino IDE and Board Types, Raspberri PI Development Kit, RFID

Principles and components, Wireless Sensor Networks: History and Context, The node, Connecting nodes, Networking Nodes, WSN and IoT.

UNIT-III
Wireless Technologies for IoT: WPAN Technologies for IoT: IEEE 802.15.4, Zigbee, HART, NFC, Z-Wave, BLE, Bacnet, Modbus.

IP Based Protocols for IoT IPv6, 6LowPAN, RPL, REST, AMPQ, CoAP, MQTT.

Edge connectivity and protocols

Unit- IV Cloud Platforms for IOT
Virtualization concepts and Cloud Architecture, Cloud computing, benefits, Cloud services -- SaaS, PaaS, IaaS, Cloud providers & offerings, Study of IOT Cloud platforms, Interfacing ESP8266 with Web services

Unit-V: IoT Applications
IoT applications for Home Automation, Cities, Environment, Energy,Retail, Logistics, Agriculture, Industry, health and Lifestyle

Text Books:

1. Bahga, A. and Madisetti, V.,Internet of Things - A Hands-on Approach, Universities Press, 2015
2. Hajjaj, S S H. and Gsangaya, K. R., The Internet of Mechanical Things: The IoT Framework for Mechanical Engineers,CRC Press, 2022.
3. Raj, P. and Raman, A. C., The Internet of Things: Enabling Technologies, Platforms, and Use Cases, Auerbach Publications/CRC Press, 2017.

Reference Books:

1. Adrian McEwen, A. and Cassimally, H., Designing the Internet of Things, John Wiley and Sons, 2013.
2. Veneri, G., Capasso, A., Hands-On Industrial Internet of Things: Create a powerful Industrial IoT infrastructure using Industry 4.0, Packt Publishing, 2018.

3. Hersent, O, Boswarthick, D., Elloumi, O., The Internet of Things: Key Applications and Protocols, 2012.

Web Resources:

1. https://nptel.ac.in/courses/106105166
2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
3. https://www.classcentral.com/course/swayam-introduction-to-internet-of-things-10093

Category: Professional Elective Course **Subject Code: ME3235**

Mechanical Vibrations

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

Unit I
Introduction: Causes and effects of vibration, Classification of vibrating system, Discrete and continuous systems, degrees of freedom, Identification of variables and Parameters, Linear and nonlinear systems, linearization of nonlinear systems, Physical models, Schematic models and Mathematical models.

Unit II
SDF systems: Formulation of equation of motion: Newton -Euler method, De Alembert's method, Energy method, Undamped Free vibration response and Damped Free vibration response, Case studies on formulation and response calculation**.**

Unit III
Forced vibration response: Response to harmonic excitations, solution of differential equation of motion, Vector approach, Complex frequency response, Magnification factor Resonance, Rotating/reciprocating unbalances, Force Transmissibility, Motion Transmissibility, Vehicular suspension, Vibration measuring instruments, Case studies on forced vibration.

Unit IV
Two degree of freedom systems: Introduction, Formulation of equation of motion: Equilibrium method, Lagrangian method, Case studies on formulation of equations of motion.
Free vibration response, Eigen values and Eigen vectors, Normal modes and mode superposition, Coordinate coupling, decoupling of equations of motion, Natural coordinates, Response to initial conditions, free vibration response case studies, Forced vibration response, undamped vibration absorbers, Case studies on undamped vibration absorbers.

Unit V
Multi degree of freedom systems: Introduction , Formulation of equations of motion, Free vibration response, Natural modes and mode shapes, Orthogonally of model vectors, normalization of model vectors, Decoupling of modes, model analysis, mode superposition technique, Free vibration response through model analysis, Forced vibration analysis through model analysis, Model damping, Rayleigh's damping, Introduction to experimental model analysis.
Continuous systems: Introduction to continuous systems, Exact and approximate solutions, free vibrations of strings, bars and beams.

Text Books:
1. L. Meirovich, Elements of Vibration analysis, Tata Mc-Grawhill, 2nd Ed., 2007.

2. Singiresu S Rao, Mechanical Vibrations, , Pearson education,4th Ed.2011.
3. W.T., Thompson, Theory of Vibration, , CBS Publishers.
4. Clarence W. de Silva, Vibration: Fundamentals and Practice, CRC Press LLC, 2000.

Reference Books:

1. Den Hartog, J.P. Mechanical Vibrations. Dover Publications, 2013.
2. Rao, J.S. Introductory Course on Theory and Practice of Mechanical Vibrations. New Age International, 1999.
3. Venkatachalam, R.,Mechanical Vibrations. PHI Learning, 2014.

Web Resources:

1. https://nptel.ac.in/courses/112107212
2. https://archive.nptel.ac.in/courses/112/103/112103112/
3. https://in.coursera.org/learn/introduction-basic-vibrations

Category: Professional Elective Course **Subject Code: ME4141**

AI and ML for Mechanical Systems

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OUTCOMES:

At the end of the course, the student will be able to

5. Understand the core concepts of Mechanical Systems in the context of Industry 4.0
6. Apply AI, ML and Deep Learning concepts on Various Mechanical Systems
7. Apply the statistical and optimization techniques on Mechanical
8. Evaluate the Mechanical System performance using simulation and experimental analysis

UNIT— I

Introduction to Mechanical Systems evolution in the context of Industry 4.0, Key issues: Adaptability, Intelligence, Autonomy, Safety, Sustainability, Interoperability, Flexibility of Mechanical Systems.

UNIT—II

Introduction of Statistics; Descriptive statistics: Central tendency measures, Dispersion measures, data distributions, centre limit theorem, sampling, sampling methods; Inferential Statistics: Hypothesis testing, confidence level, degree of freedom, P-value, Chi-square test, ANOVA, Correlation V's Regression, Uses of Correlation and regression.

UNIT – III

Artificial Intelligence: Brief review of AI history, Problem formulation: Graph structure, Graph implementation, state space representation, search graph and search tree, Search Algorithms: random search, Depth-first, breadth-first search and uniform-cost search. Heuristic: Best first search, A* and AO* algorithm, generalization of search problems. Ontology; Fuzzy; Meta- heuristics.

UNIT – IV

Machine Learning: Overview of supervised and unsupervised learning; Supervised Learning: Linear Regression, Non-linear Regression Model evaluation methods, Logistic Regression, Neural Networks; Unsupervised Learning: K-means clustering, C-means Clustering. Convolutional Neural Networks (CNN), Pooling, Padding Operations, Interpretability in CNNs, Limitations in CNN. Cases with respect to different mechanical systems.

UNIT – V

Introduction to Raspberry PI; Installation of Raspbian OS on Raspberry PI; Controlling LED using Raspberry PI; Integrating IR Sensor with Raspberry PI; Controlling LED with IR Sensor; Integrating Temperature and amp; Humidity Sensor with Raspberry PI read Current Environment Values, Collecting the sensor data using Raspberry PI; Matlab toolboxes -

Simulink, Mechanical Systems implementation: From features to software components, Mapping software components to ECUs.

Text Books:

1. Kumar, K., D. Zindani, and J.P. Davim. Artificial Intelligence in Mechanical and Industrial Engineering. CRC Press, 2021.
2. Johri, P., J.K. Verma, and S. Paul. Applications of Machine Learning. Springer Nature Singapore, 2020.

Reference Books:

1. Rajkumar, Dionisio De Niz, and Mark Klein, Cyber-Physical Systems, Wesley Professional.
2. Rajeev Alur, Principles of Cyber-Physical Systems, MIT Press, 2015.
3. Robert Levine et al., A Comprehensive guide to AI and Expert Systems, McGraw Hill Inc, 1986.
4. E. A. Lee and S. A. Seshia, Introduction to Embedded Systems: A Cyber-Physical Systems Approach, 2011.
5. C. Cassandras, S. Lafortune, “Introduction to Discrete Event Systems”, Springer 2007.
6. Constance Heitmeyer and Dino Mandrioli, “Formal methods for real-time computing”, Wiley publisher, 1996.
7. Montgomery Douglas, 2017. Design of Experiments, John Wiley and Sons, Inc.

Web Resources:

1. https://archive.nptel.ac.in/courses/112/103/112103280/
2. https://onlinecourses.nptel.ac.in/noc19_cs82/preview
3. https://nptel.ac.in/courses/106106202

Category: Professional Elective Course **Subject Code: ME4142**

Biomechanical Engineering

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

Course OUTCOMES:
At the end of the course, the student should be able to:
• Understand the principles of mechanics
• Outline the principles of biofluid dynamics.
• Explain the fundamentals of bio-solid mechanics.
• Apply the knowledge of joint mechanics.
• Give Examples of computational mathematical modelling applied in biomechanics.

Unit I
Introduction to Biomechanics
Fundamentals, Key anatomical concepts, muscle actions, mechanical methods of muscle analysis, sport, medicine and rehabilitation applications

UNIT II Biofluid Mechanics
Intrinsic fluid properties – Density, Viscosity, Compressibility and Surface Tension, Viscometers –Capillary, Coaxial cylinder and cone and plate, Rheological properties of blood, Pressure-flowrelationship for Non-Newtonian Fluids, Fluid mechanics in straight tube – Steady Laminar flow,Turbulent flow, Flow development, Viscous and Turbulent Sheer Stress, Effect of pulsatility,Boundary Layer Separation, Structure of blood vessels, Material properties and modeling of Bloodvessels, Heart –Cardiac muscle characterisation, Native heart valves – Mechanical properties andvalve dynamics, Prosthetic heart valve fluid dynamics.

UNIT III Biosolid Mechanics
Constitutive equation of viscoelasticity – Maxwell &Voight models, anisotropy, Hard Tissues –Structure, blood circulation, elasticity and strength, viscoelastic properties, functional adaptation,
Soft Tissues – Structure, functions, material properties and modeling of Soft Tissues – Cartilage,Tendons and Ligaments Skeletal Muscle – Muscle action, Hill's models, mathematical modeling,Bone fracture mechanics, Implants for bone fractures.

UNIT IV Biomechanics Of Joints
Skeletal joints, forces and stresses in human joints, Analysis of rigid bodies in equilibrium, Freebody diagrams, Structure of joints, Types of joints, Biomechanical analysis of elbow, shoulder,spinal column, hip, knee and ankle, Lubrication of synovial joints, Gait analysis, Motion analysisusing video.
UNIT V Modeling and Ergonomics
Introduction to finite element analysis, finite element analysis of lumbar spine; ergonomics –

Musculoskeletal disorders, ergonomic principles contributing to good workplace design, design ofA computer work station, whole body vibrations, hand transmitted vibrations.

Text Books:

1. Y.C. Fung, "Bio-Mechanics- Mechanical Properties of Tissues", Springer-Verlag, 1998.
2. Subrata Pal, "Textbook of Biomechanics", Viva Books Private Limited, 2009.

Reference Books:

1. Krishna B. Chandran, Ajit P. Yoganathan and Stanley E. Rittgers, Biofluid Mechanics: TheHuman Circulation", Taylor and Francis, 2007.
2. Sheraz S. Malik and Shahbaz S. Malik, Orthopaedic Biomechanics Made Easy, Cambridge University Press, 2015.
3. Jay D. Humphrey, Sherry De Lange, An Introduction to Biomechanics: Solids and FluidsAnalysis and Design", Springer Science Business Media, 2004.
4. Shrawan Kumar, Biomechanics in Ergonomics, Second Edition, CRC Press 2007.
5. Neil J. Mansfeild, Human Response to Vibration, CRC Press, 2005.
6. Carl J. Payton, Biomechanical Evaluation of movement in sports and Exercise, 2008.

Web Resources:

1. https://nptel.ac.in/courses/102101068
2. https://www.classcentral.com/tag/biomedical-engineering
3. https://in.coursera.org/courses?query=biomedical%20engineering

Category: **Professional Elective Course** Subject Code**: ME4143**

Turbomachinery

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

COURSE OUTCOMES:

- To design and analyse the performance of Turbo machines for engineering applications
- To understand the energy transfer process in Turbo machines and governing equations of various forms.
- To understand the structural and functional aspects of major components of Turbo machines.
- To design various Turbo machines for power plant and aircraft applications
- Understand the design principles of the turbo machines
- Analyze the turbo machines to improve and optimize their performance

UNIT-I:

FUNDAMENTALS OF TURBO MACHINES: Classifications, Applications, Thermodynamic analysis, isentropic flow, Energy transfer, Efficiencies, Static and Stagnation conditions, Continuity equations, Euler's flow through variable cross sectional areas, unsteady flow in turbo machines

UNIT -II:

STEAM NOZZLES: Convergent and Convergent-Divergent nozzles, Energy Balance, Effect of back pressure of analysis, Designs of nozzles.

Steam Turbines: Impulse turbines, Compounding, Work done and Velocity triangle, Efficiencies, Constant reactions, Blading, Design of blade passages, Angle and height, Secondary flow. Leakage losses, Thermodynamic analysis of steam turbines.

UNIT-III:

GAS DYNAMICS: Fundamental thermodynamic concepts, isentropic conditions, mach numbers and area, Velocity relations, Dynamic Pressure, Normal shock relation for perfect gas. Super sonic flow, oblique shock waves. Normal shock recoveries, Detached shocks, Aerofoil theory.

Centrifugal compressor: Types, Velocity triangles and efficiencies, Blade passage design, Diffuserand pressure recovery. Slip factor, Stanitz and Stodolas formula's, Effect of inlet mach numbers, Pre whirl, Performance

UNIT-IV:

AXIAL FLOW COMPRESSORS: Flow Analysis, Work and velocity triangles, Efficiencies, Thermodynamic analysis. Stage pressure rise, Degree of reaction, Stage Loading, General design, Effect of velocity, Incidence, Performance

Cascade Analysis: Geometrical and terminology. Blade force, Efficiencies, Losses, Free end force, Vortex Blades.

UNIT-V:

AXIAL FLOW GAS TURBINES: Work done. Velocity triangle and efficiencies, Thermodynamic flow analysis, Degree of reaction, Zweifels relation, Design cascade analysis, Soderberg, Hawthrone, Ainley, Correlations, Secondary flow, Free vortex blade, Blade angles for variable degree of reaction. Actuator disc, Theory, Stress in blades, Blade assembling, Material and cooling of blades, Performances, Matching of compressors and turbines, off design performance..

Textbooks

1. Yahya, S.M. Turbines Compressors and Fans. McGraw-Hill Education (India) Pvt Limited, 2010.
2. Lal, Jagdish. Hydraulic Machines. Metropolitan Book Company, 1959.
3. Som, S.K., and G. Biswas. Introduction to Fluid Mechanics and Fluid Machines. Tata McGraw-Hill, 2008
4. M.M. Das, Fluid Mechanics & Turbo Machines, PHI, 2010.
5. R.K. Bansal, Fluid Mechanics & Machinery, Luxmi Publications.

References:

6. C. Ratnam, A.V. Kothapalli, Fluid Mechanics & Machinery, I.K. International Publishing House Ltd, 2010.
7. C.S.P. Ojha, R. Berndtsson, P.N. Chandramouli, Fluid Mechanics & Machinery, Oxford University Press.
8. Gupta, Fluid Mechanics and Hydraulic Machines, Pearson Publication.
9. A.T. Sayers, Hydraulic and Compressible Flow Turbomachines.
10. R.K. Bansal, Fluid Mechanics and Hydraulic Machines.

Web Resources:

https://onlinecourses.nptel.ac.in/noc21_me127/preview
https://nptel.ac.in/courses/101101058

Category: **Professional Elective Course** Subject Code**: ME4144**

Additive Manufacturing Technology

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

Course Outcomes: On successful completion of the course, the learner will be able to
CO1: Develop CAD models for 3D printing and Import and Export CAD data.
CO2: Select a specific material for the given application.
CO3: Select a 3D printing process for an application.
CO4: Apply in product manufacturing using 3D Printing or Additive Manufacturing (AM).

Unit I
Introduction: Introduction to 3D Printing, Overview of additivemanufacturing techniques, Additive v/s Conventional Manufacturingprocesses, Applications.
CAD for Additive Manufacturing: CAD Data formats, Slicing, Datatranslation, Data loss, STL format
Unit II
3D Printing: Process, Equipment, Process parameter, Process Selectionfor various applications.Application Domains: Aerospace, Electronics, Health Care, Defence, Automotive, Construction, Food Processing, Machine Tools
Unit III
Materials: Polymers, Metals, Non-Metals, Process parameter, ProcessSelection for various applications. Various forms of raw material and their desired properties, SupportMaterials
Unit IV
Core issues in 3D Printing:
Design and process parameters, Governing Bonding Mechanism,Common faults and troubleshooting
Unit V
Post Processing: Requirement and TechniquesSupport Removal, Sanding, Acetone treatment, polishing, Inspection andtesting, Defects and their causes
Text Books:

1. Soloman, S. 3d Printing & Design. Khanna Publishing House, 2020.
2. Srivatsan, T.S., and T.S. Sudarshan. Additive Manufacturing: Innovations, Advances, and Applications. CRC Press, 2015.
3. Gibson, I., D. Rosen, and B. Stucker. Additive Manufacturing Technologies: 3d Printing, Rapid Prototyping, and Direct Digital Manufacturing. Springer New York, 2014.
4. Kumar, L.J., P.M. Pandey, and D.I. Wimpenny. 3d Printing and Additive Manufacturing Technologies. Springer Nature Singapore, 2018.

Reference Books:

1. Yang, L., K. Hsu, B. Baughman, D. Godfrey, F. Medina, M. Menon, and S. Wiener. Additive Manufacturing of Metals: The Technology, Materials, Design and Production. Springer International Publishing, 2017.

2. Yan, C., Y. Shi, L. Zhaoqing, S. Wen, and Q. Wei. Selective Laser Sintering Additive Manufacturing Technology. Elsevier Science, 2020.
3. Singh, S., C. Prakash, and S. Ramakrishna. Innovative Processes and Materials in Additive Manufacturing. Elsevier Science, 2022.
4. Gebhardt, A. Understanding Additive Manufacturing: Rapid Prototyping, Rapid Tooling, Rapid Manufacturing. Hanser Publications, 2012.
5. Amitava Ghosh, Rapid Prototyping, McGraw hill Publishers.

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc22_me122/preview
2. https://archive.nptel.ac.in/courses/112/107/112107078/
3. https://nptel.ac.in/courses/112103202
4. https://nptel.ac.in/courses/112103306
5. https://in.coursera.org/learn/advanced-manufacturing-process-analysis

Category: **Professional Elective Course** Subject Code**: ME4145**

Systems Engineering

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Course outcomes: Students will be able to

1. develop system solutions that optimally fulfill customer objectives with available resources.
2. solve open-ended problems, utilizing creativity, problem formulation, generation of need statements, requirements analysis, alternative solutions generation and examination, concurrent engineering design,
3. enforce various realistic aspects such as safety, reliability, manufacturability, operations, aesthetics, ethics, and sustainability.

Unit I Introduction

Definitions of Systems Engineering, Systems Engineering Knowledge, Life cycles, Life-cycle phases, logical steps of systems engineering, Frame works for systems engineering.

Unit II Systems Engineering Processes

Formulation of issues with a case study, Value system design, Functional analysis, Business Process Reengineering, Quality function deployment, System synthesis, Approaches for generation of alternatives

Unit III- Analysis of Alternatives- I

Cross-impact analysis, Structural modeling tools, System Dynamics models with case studies, Economic models: present value analysis – NPV, Benefits and costs over time, ROI, IRR; Work and Cost breakdown structure,

Unit IV- Analysis of Alternatives- II

Reliability, Availability, Maintainability, and Supportability models; Stochastic networks and Markov models, Queuing network optimization, Time series and Regression models, Evaluation of large scale models

Unit V- Decision Assessment

Decision assessment types, Five types of decision assessment efforts, Utility theory, Group decision making and Voting approaches, Social welfare function; Systems Engineering methods for Systems Engineering Management,

TEXT BOOK:

1. Andrew P. Sage, James E. Armstrong Jr. "Introduction to Systems Engineering", John Wiley and Sons, Inc,2000.
2. Systems Engineering Second ed. – Kossiakoff, A., Sweet, W.N., Seymour, S.J., and Biemer S.M., John Wiley Sons Inc., New Jersey, 2011
3. Blanchard, B. S., and Fabrycky, W. J. (2006). Systems Engineering and Analysis (5th Ed). Pearson Prentice Hall: Upper Saddle River, NJ. [ISBN 978-0-13-221735-4]

Web Resources:

1. **https://onlinecourses.nptel.ac.in/noc21_mg39/preview**

Category: **Professional Elective Course** Subject Code**: ME4151**

Laser Applications in Manufacturing

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Unit I

Main industrial lasers: He, Ne, CO2,Excimer, Nd:YAG, Diode, Fiber and Ultra, short pulse lasers and their output beam characteristics; laser beam delivery systems.

Overview of Laser Industrial and Scientific Applications: Metrological applications, Holography, Laser Isotope Separation, Laser fusion.

Unit II

Laser processing fundamentals: Laser beam interaction with metal, semiconductor and insulator, Ultra, short laser pulse interaction, heat flow theory and metallurgical considerations.

Laser Material Processing Applications: Laser Cutting and drilling: Process characteristics, material removal modes, practical performances

Unit III

Laser Welding: Process mechanisms like keyhole and plasma effect, operating characteristics and process variation

Laser Surface modifications: Heat treatment, surface remelting, surface alloying and cladding, surface texturing, LCVD and LPVD Laser rapid manufacturing

Unit IV

Laser metal forming: Mechanisms involved including thermal temperature gradient, buckling, upsetting.

Laser peening: Fundamentals of Laser Shock Processing, Effects of various laser and process parameters, Mechanical effects and microstructure modification during laser shock processing. Theoretical modeling of laser material processing

Unit V

On-line Process monitoring and control: Laser and process parameters, and workpiece characteristics

Economics of Laser Applications in Manufacturing Laser Safety: Laser safety standards and safety procedures

Text Books:

1. Brandt, M. Laser Additive Manufacturing: Materials, Design, Technologies, and Applications. Elsevier Science, 2016.
2. Lawrence, J.R. Advances in Laser Materials Processing: Technology, Research and Applications. Elsevier Science, 2017.
3. Martellucci, S., A.N. Chester, and A.M.V. Scheggi. Laser Applications for Mechanical Industry. Springer Netherlands, 2012.

Reference Books:

1. Joshi, S.N., and U.S. Dixit. Lasers Based Manufacturing: 5th International and 26th All India Manufacturing Technology, Design and Research Conference, Aimtdr 2014. Springer India, 2015.
2. Luxon, J.T., D.E. Parker, and P.D. Plotkowski. Lasers in Manufacturing: An Introduction to the Technology. IFS, 1987.
3. Dixit, U.S., S.N. Joshi, and J.P. Davim. Application of Lasers in Manufacturing: Select Papers from Aimtdr 2016. Springer Nature Singapore, 2018.
4. Steen, W.M., and IFS Ltd. Lasers in Manufacturing: Proceedings of the 4th International Conference, 12-14 May 1987, Birmingham, Uk. IFS Publications, 1987.
5. Engineers, Society of Photo-optical Instrumentation. Laser Applications in Microelectronic and Optoelectronic Manufacturing. SPIE--The International Society for Optical Engineering, 2000.

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc22_me92/preview
2. https://archive.nptel.ac.in/courses/104/104/104104085/
3. https://www.coursera.org/lecture/3d-printing-revolution/selective-laser-melting-rodrigo-gutierrez-kkpwz

Category: **Professional Elective Course** Subject Code: **ME4152**

SOLAR ENERGY TECHNOLOGY

Internals: 40 Marks L - T - P - C
Externals: 60 Marks 3 - 0 - 0 - 3

Course Outcomes:

At the end of the course, the student will be able to:

CO1: Understand the fundamentals of solar energy and its conversion techniques
CO2: Estimate solar energy through radiation principles and solar collectors
CO3 : Design thermal and electrical energy storage systems
CO4 : Design solar thermal and photovoltaic systems
CO5 : Understand solar passive architecture

UNIT - I:

Introduction

Energy sources and their availability- commercial andnon commercial energy sources. Need of Renewable Energy Sources (RES), classification of RES, Role and potential of RES in India.Solar Radiation: Structure of the sun, Solar constant, environmental impact of solar radiation, Radiation at the earth surfaces, solar radiation Geometry, extraterrestrial and terrestrial solar radiation, Spectral Distribution of Extraterrestrial Radiation, solar radiation on tilted surfaces and Empirical equations for predicting the availability of solar radiation at any given location. Solar energy - Thermal applications.

UNIT - II:

Solar Collectors

Principle of solar energy conversion into heat, classification of solar collectors, Flat plate collectors, basic energy balance equation, collector efficiency, thermal analysis of flat plate collector. Concentrating collectors and its advantages and disadvantages. Performance analysis of concentrating collectors, selection of absorber coating materials

UNIT - III:

Solar Energy Storage and Applications

Solar Energy Storage: Different storage methods- sensible, latent heat and stratified storage, solar ponds. Solar Energy Applications: Solar water, space heating /cooling, solar thermal electric conversion, direct solar electric power generation- solar photovoltaic, solar distillation, Solar Pumping, Solar furnace, Solar cocking and solar green house.

UNIT-IV

Semiconductor – properties - energy levels - basic equations of semiconductor devices physics. Solar cells - p-n junction: homo and hetro junctions – metal semiconductor interface - dark and illumination characteristics. Solar cell array system analysis and performance prediction- Shadow analysis: reliability - solar cell array design concepts - PV system design - design process and optimization - detailed array design

UNIT-V

SOLAR PASSIVE ARCHITECTURE: Thermal comfort - heat transmission in buildingsbioclimatic classification – passive heating concepts: direct heat gain - indirect heat gain - isolated gain and sunspaces - passive cooling concepts: evaporative cooling - radiative cooling - application of wind, water and earth for cooling; shading - paints and cavity walls for cooling – roof radiation traps - earth air-tunnel. – energy efficient landscape design – thermal comfort – concept of solar temperature and its significance - calculation of instantaneous heat gain through building envelope.

Text Books:

1. Sukhatme, K., and S.P. Sukhatme. Solar Energy: Principles of Thermal Collection and Storage. Tata McGraw-Hill, 1996
2. Duffie, J.A., and W.A. Beckman. Solar Engineering of Thermal Processes. Wiley, 2013.
3. Prakash, G., and H.P. Garg. Solar Energy: Fundamentals and Applications. McGraw-Hill Education (India) Pvt Limited, 2000.
4. Tiwari, G.N., and M.K. Ghosal. Renewable Energy Resources: Basic Principles and Applications. Alpha Science International, 2005.

References

1. Allen, D.T., and D.R. Shonnard. Green Engineering: Environmentally Conscious Design of Chemical Processes. Pearson Education, 2001.
2. Jena, O.P., A.R. Tripathy, and Z. Polkowski. Green Engineering and Technology: Innovations, Design, and Architectural Implementation. CRC Press, 2021.
3. SOLANKI, C.S. Solar Photovoltaics: Fundamentals, Technologies and Applications. PHI Learning, 2015.

Web Resources:

4. https://archive.nptel.ac.in/courses/127/105/127105018/
5. https://nptel.ac.in/courses/127105018
6. https://archive.nptel.ac.in/courses/105/102/105102195/

Category: **Professional Elective Course** Subject Code**: ME4153**

Non-Destructive Testing

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

Unit I Introduction to NDET and Surface NDT Techniques
Introduction to non-destructive testing and evaluation, visual examination, liquid penetrant testing and magnetic particle testing. Advantages and limitations of each of these techniques.

Unit II Radiographic Testing
Radiography principle, electromagnetic radiation sources, X-ray films, exposure, penetrameter, radiographic imaging, inspection standards and techniques, neutron radiography. Radiography applications, limitations and safety.

Unit III Eddy Current Testing and Ultrasonic Testing
Eddy current principle, depth of penetration, eddy current response, eddy current instrumentation, probe configuration, applications and limitations. Properties of sound beam, ultrasonic transducers, inspection methods, flaw characterization technique, immersion testing.

Unit IV Special/Emerging Techniques
Leak testing, Acoustic Emission testing, Holography, Thermography, Magnetic Resonance Imaging, Magnetic Barkhausen Effect. In-situ metallography.

Unit V Defects in materials / products and Selection of NDET Methods
Study of defects in castings, weldments, forgings, rolled products etc. and defects arising during service. Selection of NDET methods to evaluate them. Standards and codes.

Text Books

1. Baldevraj, Jayakumar T., Thavasimuthu M.,Practical Non-Destructive Testing, 3rd edition, Narosa Publishers, 2008.
2. Boogaard, J., and G.M. van Dijk. Non-Destructive Testing. Elsevier Science, 2012.
3. Prakash, R. Non-Destructive Testing Techniques. New Age Science, 2009.

Reference Books

1. American Society for Metals, Non-Destructive Evaluation and Quality Control: Metals Hand Book: 1992, Vol. 17, 9th Ed, Metals Park, OH.
2. Paul E Mix, Introduction to nondestructive testing: a training guide, Wiley, 2nd edition New Jersey, 2005.
3. Ravi Prakash, Nondestructive Testing Techniques, New Age International Publishers, 1st rev. edition, 2010.

Web Resources:

1. https://archive.nptel.ac.in/courses/113/106/113106070/
2. https://elearn.nptel.ac.in/shop/nptel/theory-and-practice-of-non-destructive-testing/
3. https://www.edx.org/course/fundamentals-of-non-destructive-testing

Category: **Professional Elective Course** Subject Code**: ME4154**

FINITE ELEMENT METHODS

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Course Outcomes:

At the end of the course, the student will be able to,

- Apply finite element method to solve problems in solid mechanics, fluid mechanics and heat transfer.
- Formulate and solve problems in one dimensional structures including trusses, beams and frames. Formulate FE characteristic equations for two dimensional elements and analyze plain stress, plain strain, axi-symmetric and plate bending problems.
- Implement and solve the finite element formulations using MATLAB.

UNIT – I:

Introduction to Finite Element Methods for solving field problems, Methods of Engineering Analysis, Functional Approximation Methods: Rayleigh- Ritz Method, Weighted Residual Methods, Applications of FEM, Advantages and Disadvantages of FEM, Stress and Equilibrium, Strain – Displacement relations, Stress – strain relations for 2D and 3D Problems. Basic Steps of FEM, Characteristics of Finite Element, Principle of Minimum Potential Energy, Convergence Requirements.

UNIT – II:

One Dimensional Problems: Formulation of Stiffness Matrix for a Bar Element by the Principle of Minimum Potential Energy, Properties of Stiffness Matrix, Characteristics of Shape Functions, Quadratic shape functions. Problems on uniform and stepped bars for different loading conditions.

Analysis of Trusses: Derivation of Stiffness Matrix for Trusses, Stress and strain Calculations, Calculation of reaction forces and displacements.

UNIT – III:

Analysis of Beams: Derivation of Stiffness matrix for two noded, two degrees of freedom per node beam element, Load Vector, Deflection, Stresses, Shear force and Bending moment, Problems on uniform and stepped beams for different types of loads applied on beams.

UNIT – IV:

Finite element – formulation of 2D Problems: Derivation of Element stiffness matrix for two dimensional CST Element, Derivation of shape functions for CST Element, Elasticity Equations, constitutive matrix formulation, Formulation of Gradient matrix. Two dimensional Isoparametric Elements and Numerical integration.

Finite element – formulation of 3D problems: Derivation of Element stiffness matrix for

Tetrahedron Element, Properties of Shape functions for 3D Tetrahedral Element, Stress-Strain Analysis for 3D Element, Strain Displacement for Relationship Formulation.

UNIT – V:

Steady state heat transfer analysis: One Dimensional Finite Element analysis of fin and composite slabs. Two dimensional steady state heat transfer problems: Derivation of Thermal Stiffness matrix for 2D heat transfer problems-CST, Derivation of thermal force vector for 2D heat transfer problems.

Dynamic Analysis: Formulation of mass matrices for uniform bar and beam Elements using lumped and consistent mass methods, Evaluation of Eigen values and Eigen vectors for a stepped bar and beam Problems.

Text Books:

1. Chandrupatla, T.R., and A.D. Belegundu. Introduction to Finite Elements in Engineering: International Edition. Pearson Education, 2014.
2. Rao, S.S. The Finite Element Method in Engineering. Elsevier Science, 2011.
3. SESHU, P. Textbook of Finite Element Analysis. PHI Learning, 2003.

Reference Books:

1. ALAVALA, C.R. Finite Element Methods: Basic Concepts and Applications. PHI Learning, 2008.
2. Zienkiewicz, O.C., R.L. Taylor, and J.Z. Zhu. The Finite Element Method: Its Basis and Fundamentals. Elsevier Science, 2005.
3. S.Md.Jamaluddin, Introduction to Finite element analysis Anuradha Publications, print-2012
4. Reddy, J.N. An Introduction to the Finite Element Method. McGraw-Hill, 2006.
5. Hutton, D.V. Fundamentals of Finite Element Analysis. McGraw-Hill, 2004.
6. Krishnamoorthy, C.S. Finite Element Analysis Theory and Programming. Tata McGraw Hill Education Private Limited, 2011.
7. Liu, G.R., and S.S. Quek. Finite Element Method: A Practical Course. Elsevier Science, 2003.

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc22_me43/preview
2. https://nptel.ac.in/courses/112104116
3. https://nptel.ac.in/courses/112106135
4. https://nptel.ac.in/courses/112104193
5. https://in.coursera.org/courses?query=finite%20element
6. https://www.udemy.com/topic/finite-element-analysis/

Category: **Professional Elective Course** Subject Code**: ME4155**

Industrial Engineering

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Course outcome: On successful completion of the course learners will be able to

1: To recall and identify the relevance of management concepts.

2: To apply management techniques for meeting current and future management challenges faced by the organization

3: To compare the management theories and models critically to solve real-life problems in an organization.

4: To apply principles of management in order to execute the role as a manager in an organization

UNIT-I

Management concept: administration, management,functions of management, organization, types of organizations, principles of management, scientific management, Fayol's and Taylor's contributions to management

UNIT-II

Sales forecasting: need, classification, sales forecasting methods: moving average, exponential smoothing,linear regression, measures of forecast accuracy, marketing: definition, principles, marketing functions, marketing management, marketing research

Plant location: influencing factors, Weber's theory,choice of city, suburban and country locations, plant layout: definition, objectives, types – product, process and fixed position layouts,plant maintenance: importance, types – preventive maintenance, predictive and breakdown maintenance, introduction to total productive maintenance (TPM).

UNIT-III

Work study: basic procedure, method study – definition, objectives and procedure, work measurement: objectives, techniques of work measurement, time study, work sampling and analytical sampling, predetermined motion time systems (PMTs), determination of standard time

UNIT-IV

Personnel management: functions of personnel management, methods of job evaluation,Methods of merit rating, incentive plans – piece rate system, Taylor's differential piece rate system,Halsey 50-50 plan, rowan plan and bedaux system

UNIT-V

Quality control: introduction to inspection and quality control, variables and attributes, types of sampling plans, acceptance sampling for attributes – description, advantages and disadvantages of sampling, OC curve for single and double sampling plans, design of sampling plans, total quality management: introduction, six sigma concepts, tools for continuous quality improvement

Text Books:

1. Essentials of Management, by Harold Kooritz& Heinz Weihrich Tata McGraw
2. Production and Operations Management-K. Aswathapa, K. Shridhara Bhat, Himalayan Publishing House

References:

1. Organizational Behavior, Stephen Robbins Pearson Education, New Delhi
2. New era Management, Daft, 11th Edition, Cengage Learning
3. Principles of Marketing, Kotlar Philip and Armstrong Gary, Pearson publication

Web Resources:

1. **https://onlinecourses.nptel.ac.in/noc22_me04/preview**

Open Elective Courses (OECs)

Sl. No.	Course Code	Course Title	Course Category
1	ME3301	Smart materials	OEC-I
2	ME3302	Finite element analysis	OEC-I
3	ME3303	Advanced Fluid Mechanics	OEC-I
4	ME4401	3D Printing Technology	OEC-II
5	ME4402	Alternative sources of Energy	OEC II
6	ME4403	Computer aided design	OEC II
7	ME4411	Nano Science and Technology	OEC III
8	ME4412	Welding Technology	OEC III
9	ME4413	Industrial Robotics	OEC III

Category: **Open Elective Course** Subject Code:**ME3301**

Smart materials

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Unit I: Overview of Smart Materials

Introduction to Smart Materials, Principles of Piezoelectricty, Perovskyte Piezoceramic Materials, Single Crystals vs Polycrystalline Systems, Piezoelectric Polymers, Principles of Magnetostriction, Rare earth Magnetostrictive materials, Giant Magnetostriction and Magneto-resistance Effect, Introduction to Electro-active Materials, Electronic Materials, Electro-active Polymers, Ionic Polymer Matrix Composite (IPMC), Shape Memory Effect, Shape Memory Alloys, Shape Memory Polymers, Electro-rheological Fluids, Magneto Rhelological Fluids

Unit II: High-Band Width, Low Strain Smart Sensors

Piezoelctric Strain Sensors, In-plane and Out-of Plane Sensing, Shear Sensing, Accelerometers, Effect of Electrode Pattern, Active Fibre Sensing, Magnetostrictive Sensing, Villari Effect, Matteuci Effect and Nagoka-Honda Effect, Magnetic Delay Line Sensing, Application of Smart Sensors for Structural Health Monitoring (SHM), System Identification using Smart Sensors

Unit III: Smart Actuators

Modelling Piezoelectric Actuators, Amplified Piezo Actuation – Internal and External Amplifications, Magnetostrictive Actuation, Joule Effect, Wiedemann Effect, Magnetovolume Effect, Magnetostrictive Mini Actuators, IPMC and Polymeric Actuators, Shape Memory Actuators, Active Vibration Control, Active Shape Control, Passive Vibration Control, Hybrid Vibration Control

Unit IV: Smart Composites

Review of Composite Materials, Micro and Macro-mechanics, Modelling Laminated Composites based on Classical Laminated Plate Theory, Effect of Shear Deformation, Dynamics of Smart Composite Beam, Governing Equation of Motion, Finite Element Modelling of Smart Composite Beams

Unit V: Advances in Smart Structures & Materials

Self-Sensing Piezoelectric Transducers, Energy Harvesting Materials, Autophagous Materials, Self Healing Polymers, Intelligent System Design, Emergent System Design

Text Books:

1. Brian Culshaw, Smart Structures and Materials, Artech House, 2000
2. Gauenzi, P., Smart Structures, Wiley, 2009
3. Cady, W. G., Piezoelectricity, Dover Publication

Web Resources:

1. https://nptel.ac.in/courses/112104173

2. https://onlinecourses.nptel.ac.in/noc22_me17/preview

Category: **Open Elective Course** Subject Code:**ME3302**

Finite element analysis

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Course Outcomes:

At the end of the course, the student will be able to,

- Apply finite element method to solve problems in solid mechanics, fluid mechanics and heat transfer.
- Formulate and solve problems in one dimensional structures including trusses, beams and frames. Formulate FE characteristic equations for two dimensional elements and analyze plain stress, plain strain, axi-symmetric and plate bending problems.
- Implement and solve the finite element formulations using MATLAB.

UNIT – I:

Introduction to Finite Element Methods for solving field problems, Methods of Engineering Analysis, Functional Approximation Methods: Rayleigh- Ritz Method, Weighted Residual Methods, Applications of FEM, Advantages and Disadvantages of FEM, Stress and Equilibrium, Strain – Displacement relations, Stress – strain relations for 2D and 3D Problems. Basic Steps of FEM, Characteristics of Finite Element, Principle of Minimum Potential Energy, Convergence Requirements.

UNIT – II:

One Dimensional Problems: Formulation of Stiffness Matrix for a Bar Element by the Principle of Minimum Potential Energy, Properties of Stiffness Matrix, Characteristics of Shape Functions, Quadratic shape functions. Problems on uniform and stepped bars for different loading conditions.

Analysis of Trusses: Derivation of Stiffness Matrix for Trusses, Stress and strain Calculations, Calculation of reaction forces and displacements.

UNIT – III:

Analysis of Beams: Derivation of Stiffness matrix for two noded, two degrees of freedom per node beam element, Load Vector, Deflection, Stresses, Shear force and Bending moment, Problems on uniform and stepped beams for different types of loads applied on beams.

UNIT – IV:

Finite element – formulation of 2D Problems: Derivation of Element stiffness matrix for two dimensional CST Element, Derivation of shape functions for CST Element, Elasticity Equations, constitutive matrix formulation, Formulation of Gradient matrix. Two dimensional Isoparametric Elements and Numerical integration.

Finite element – formulation of 3D problems: Derivation of Element stiffness matrix for

Tetrahedron Element, Properties of Shape functions for 3D Tetrahedral Element, Stress-Strain Analysis for 3D Element, Strain Displacement for Relationship Formulation.

UNIT – V:

Steady state heat transfer analysis: One Dimensional Finite Element analysis of fin and composite slabs. Two-dimensional steady state heat transfer problems: Derivation of Thermal Stiffness matrix for 2D heat transfer problems-CST, Derivation of thermal force vector for 2D heat transfer problems.

Dynamic Analysis: Formulation of mass matrices for uniform bar and beam Elements using lumped and consistent mass methods, Evaluation of Eigen values and Eigen vectors for a stepped bar and beam Problems.

Text Books:

1. Belegundu, Ashok D.., Chandrupatla, Tirupathi R.. Introduction to Finite Elements in Engineering. United Kingdom: Prentice Hall, 2011.
2. Rao, S. S.. The Finite Element Method in Engineering. Germany: Pergamon Press, 1989..
3. Seshu, P.. Textbook Of Finite Element Analysis. India: Phi Learning, 2004.

Reference Books:

1. Alavala, ChennakesavaR.. Finite Element Methods: Basic Concepts And Applications. India: Phi Learning, 2008.
2. Zienkiewicz, O. C.., Taylor, Robert Leroy. The Finite Element Method. China: World Books Publishing Corporation, 2005.
3. S. Md. Jalaludeen, "Introduction of Finite Element Analysis", Anuradha publications, 2st Edition,2012
4. Hutton. Fundamentals of Finite Element Analysis. McGraw-Hill Education (India) Pvt Limited, 2005.
5. Bathe, Klaus-Jürgen. Finite Element Procedures. United Kingdom: Prentice Hall, 1996.
6. Krishnamoorthy, C. S.. Finite Element Analysis: Theory AndProgramming. India: Tata Mcg-Hill, 1995.

Web References:

1. http://nptel.ac.in/courses/112104116/
2. http://nptel.ac.in/courses/112104116/
3. http://nptel.ac.in/courses/112104116/ui/TableofContents.html

Category: **Open Elective Course** Subject Code**: ME3103**

Advanced Fluid Mechanics

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT – I: Inviscid Flow of Incompressible Fluids:

Lagrangian and Eulerain Descriptions of fluid motion- Path lines, Stream lines, Streak lines, stream tubes – velocity of a fluid particle, types of flows – Stream and Velocity potential functions. Basic Laws of fluid Flow: Potential flow, Condition for irrotationality, circulation & vorticity Accelerations in Carte systems normal and tangential accelerations, Euler's, Bernouli equations Dimensional Analysis & Similarity

UNIT – II: Viscous Flow:

Equation of Fluid flow-Continuity & Momentum equation. Derivation of Navier-Stoke's Equations for viscous compressible flow – Exact solutions to certain simple cases : Plain Poisoulle flow – Coutte flow with and without pressure gradient – Hagen Poisoulle flow.

UNIT III: Boundary Layer Concepts :

External Flow-Prandtl's contribution to real fluid flows –Blasius solution-Prandtl's boundary layer theory – Boundary layer thickness for flow over a flat plate – Approximate solutions – Creeping motion (Stokes) – Oseen's approximation – Von-Karman momentum integral equation for laminar boundary layer — Expressions for local and mean drag coefficients for different velocity profiles for flow over flat plates-Flow over Blunt objects-Boundary layer development-Drag calculation.

UNIT IV: Internal Flow:

Boundary layer development-Hydrodynamic entry length-Smooth and rough boundaries – Equations for Velocity Distribution and frictional Resistance in smooth rough Pipes – Roughness of Commercial Pipes – Moody's diagram. Introduction to Turbulent Flow: Fundamental concept of turbulence – Time Averaged Equations – Boundary Layer Equations – Prandtl Mixing Length Model – Universal Velocity Distribution Law: Van Driest Model – Approximate solutions for drag coefficients – More Refined Turbulence Models – kepsilon model.

UNIT V: Compressible Fluid Flow:

Thermodynamic basics – Sonic Velocity – Mach Number – Generalized and simple 1D compressible flows – Development of Equations – Acoustic Velocity Derivation of Equation for Mach Number – Area – Pressure Velocity Relationship

Compressible Fluid Flow – II: Nozzles, Diffusers – Isothermal Flow in Long Ducts – Fanno and Reyleigh Lines, Property Relations — Normal Compressible Shock, Oblique Shock: Expansion and Compressible Shocks – Supersonic Wave Drag

Text book:

1. Frank M. White, Fluid Mechanics, Mc Graw Hill, 8th Edition
2. Potter, Fluid Mechanics, Cengage Learning
3. Finnemore & Franzini, Fluid Mechanics with Engineering Applications, McGraw Hill

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc20_me54/preview

Category: **Open Elective Course** Subject Code**: ME4144**

3D Printing Technology

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

Course Outcomes: On successful completion of the course, the learner will be able to
CO1: Develop CAD models for 3D printing and Import and Export CAD data.
CO2: Select a specific material for the given application.
CO3: Select a 3D printing process for an application.
CO4: Apply in product manufacturing using 3D Printing or Additive Manufacturing (AM).

Unit I
Introduction: Introduction to 3D Printing, Overview of additivemanufacturing techniques, Additive v/s Conventional Manufacturingprocesses, Applications.
CAD for Additive Manufacturing: CAD Data formats, Slicing, Datatranslation, Data loss, STL format
Unit II
3D Printing: Process, Equipment, Process parameter, Process Selectionfor various applications.Application Domains: Aerospace, Electronics, Health Care, Defence, Automotive, Construction, Food Processing, Machine Tools
Unit III
Materials: Polymers, Metals, Non-Metals, Process parameter, ProcessSelection for various applications. Various forms of raw material and their desired properties, SupportMaterials
Unit IV
Core issues in 3D Printing:
Design and process parameters, Governing Bonding Mechanism,Common faults and troubleshooting
Unit V
Post Processing: Requirement and TechniquesSupport Removal, Sanding, Acetone treatment, polishing, Inspection andtesting, Defects and their causes
Text Books:
5. Soloman, S. 3d Printing & Design. Khanna Publishing House, 2020.
6. Srivatsan, T.S., and T.S. Sudarshan. Additive Manufacturing: Innovations, Advances, and Applications. CRC Press, 2015.
7. Gibson, I., D. Rosen, and B. Stucker. Additive Manufacturing Technologies: 3d Printing, Rapid Prototyping, and Direct Digital Manufacturing. Springer New York, 2014.
8. Kumar, L.J., P.M. Pandey, and D.I. Wimpenny. 3d Printing and Additive Manufacturing Technologies. Springer Nature Singapore, 2018.

Reference Books:

6. Yang, L., K. Hsu, B. Baughman, D. Godfrey, F. Medina, M. Menon, and S. Wiener. Additive Manufacturing of Metals: The Technology, Materials, Design and Production. Springer International Publishing, 2017.
7. Yan, C., Y. Shi, L. Zhaoqing, S. Wen, and Q. Wei. Selective Laser Sintering Additive Manufacturing Technology. Elsevier Science, 2020.
8. Singh, S., C. Prakash, and S. Ramakrishna. Innovative Processes and Materials in Additive Manufacturing. Elsevier Science, 2022.
9. Gebhardt, A. Understanding Additive Manufacturing: Rapid Prototyping, Rapid Tooling, Rapid Manufacturing. Hanser Publications, 2012.
10. Amitava Ghosh, Rapid Prototyping, McGraw hill Publishers.

Web Resources:

6. https://onlinecourses.nptel.ac.in/noc22_me122/preview
7. https://archive.nptel.ac.in/courses/112/107/112107078/
8. https://nptel.ac.in/courses/112103202
9. https://nptel.ac.in/courses/112103306
10. https://in.coursera.org/learn/advanced-manufacturing-process-analysis

Category: **Open Elective Course** Subject Code**: ME4402**

Alternative Sources of Energy

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT - I:

Introduction

Energy sources and their availability- commercial and non commercial energy sources. Need of Renewable Energy Sources (RES), classification of RES, Role and potential of RES in India. Solar Radiation: Structure of the sun, Solar constant, environmental impact of solar radiation, Radiation at the earth surfaces, solar radiation Geometry, extraterrestrial and terrestrial solar radiation, Spectral Distribution of Extraterrestrial Radiation, solar radiation on tilted surfaces and Empirical equations for predicting the availability of solar radiation at any given location. Solar energy - Thermal applications.

UNIT - II:

Solar Collectors

Principle of solar energy conversion into heat, classification of solar collectors, Flat plate collectors, basic energy balance equation, collector efficiency, thermal analysis of flat plate collector. Concentrating collectors and its advantages and disadvantages. Performance analysis of concentrating collectors, selection of absorber coating materials

UNIT - III:

Solar Energy Storage and Applications

Solar Energy Storage: Different storage methods- sensible, latent heat and stratified storage, solar ponds. Solar Energy Applications: Solar water, space heating /cooling, solar thermal electric conversion, direct solar electric power generation- solar photovoltaic, solar distillation, Solar Pumping, Solar furnace, Solar cocking and solar green house.

UNIT - IV:

Wind Energy, Biomass Energy Conversion Systems and Geothermal Thermal Energy

Wind Energy: Working principle of wind energy conversion, Wind patterns, Components of wind energy conversion system (WECS), Types of Wind machines – horizontal axis and vertical axis, Betz coefficient. Biomass Energy Conversion Systems: Biomass Energy: Fuel classification – Pyrolysis – Different digesters and sizing. Geothermal Thermal Energy: Classification – Dry rock and acquifer – Energy analysis

UNIT - V:

Ocean Thermal Energy, Tidal Power System and Wave Energy

Ocean Thermal Energy: Method s of Ocean Thermal Electric power generation-Open cycle systems, closed cycle systems Tidal Power System: Working principle, components of Tidal Power plant, single basin and double basin tidal energy system advantages and limitations . Wave Energy: Wave energy conversion Devices-wave energy conversion by floats, high level reservoir wave machine and dolphin type wave power machine. Advantages and disadvantages

UNIT - VI:

Direct Energy Conversion, MHDPower Generation and Fuel Cell

Direct Energy Conversion (DEC): Need for DEC, limitations, principles of DEC. Thermoelectric Power – See-beck, Peltier, Joule -Thomson effects, Thermo-electric Power generators MHD Power Generation: Principles, dissociation and ionization, Hall effect, magnetic flux, MHD accelerator, MHD engine, power generation systems, electron gas dynamic conversion. Fuel Cell: working principle, classification – Efficiency – VI characteristics.

Text Books

1. Sukhatme, K., and S.P. Sukhatme. Solar Energy: Principles of Thermal Collection and Storage. Tata McGraw-Hill, 1996.
2. Tiwari, G. N, Ghosal, M. K.. Fundamentals of Renewable Energy Sources. India: Narosa Publishing House, 2007.
3. Maczulak, Anne Elizabeth. Renewable Energy: Sources and Methods. United States: Facts On File, 2010.

Reference Books:

1. Biram, T. Renewable Energy. Independence Educational Publishers, 2019.
2. Sayigh, A.A.M. Comprehensive Renewable Energy. Elsevier, 2012
3. Jelley, Nicholas Alfred., Jelley, Nick. Renewable Energy: A Very Short Introduction. United Kingdom: Oxford University Press, 2020.
4. Kataki, Rupam., Patra, S. C., Kusre, B. C. Renewable Energy and Energy Management. India: International Book Distributing Company, 2007.
5. Boyle, G., and Open University. Renewable Energy: Power for a Sustainable Future. Oxford University Press, 1996.
6. Smith, Trevor. Renewable Energy Resources. United States: Weigl Publishers, 2004.
7. Quaschning, Volker. UnderstandingRenewable Energy Systems. Iran: Earthscan, 2005.

Web Resources:

7. https://archive.nptel.ac.in/courses/127/105/127105018/
8. https://nptel.ac.in/courses/127105018
9. https://archive.nptel.ac.in/courses/105/102/105102195/

Category: **Open Elective Course** Subject Code**: ME4403**

Computer Aided Design

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT I

Computer Graphics:

Introduction, CAD Tools, Types of system, CAD system evaluation criteria, brief treatment of input and output devices. Graphics standard, functional areas of CAD, Modeling and viewing, software documentation, graphic primitives, Transformation in graphics, coordinate systems, view port, 2D and 3D transformation, Visual realism, hidden line and surface removal,

.

UNIT II

Geometric Modelling of curves

Types of mathematical representation of curves, wire frame models wire frame entities parametric representation of synthetic curves hermite cubic splines Bezier curves B-splines rational curves

UNIT III

Geometric Modelling of surfaces

Mathematical representation surfaces, Surface model, Surface entities surface representation, parametric representation of analytic surfaces, plane surface, rules surface, surface of revolution, Tabulated Cylinder. Parametric Representation Of Synthetic Surfaces: Hermite Bicubic surface, Bezier surface, B- Spline surface, COONs surface, Blending surface Sculptured surface, Surface manipulation — Displaying, Segmentation, Trimming, Intersection, Transformations (both 2D and 3D).

UNIT IV

Geometric Modelling of solids

Solid Representation, Boundary, Representation (13-rep), Constructive Solid Geometry (CSG). CAD/CAM Exchange: Evaluation of data — exchange format, IGES data representations and structure, STEP Architecture, implementation, ACIS & DXF. Design Applications: Mechanical tolerances, Mass property calculations, Mechanical Assembly.

UNIT V

Basic Concepts of FEA:

Introduction, One Dimensional Problems: Basis steps, Discretization, Element equations, Linear and quadratic shape functions, Assembly, Local and global stiffness matrix and its properties, boundary conditions, penalty approach, multipoint constraints, Applications to solid mechanics, heat and fluid mechanics problems,

Text Books

1. Saxena, Anupam., Sahay, Birendra. Computer Aided Engineering Design. Germany: Springer, 2005.
2. Zeid, Ibrahim. CAD/CAM Theory and Practice. Singapore: McGraw-Hill.
3. Rogers, David F.., Adams, James Alan., Adams, J. Alan. Mathematical Elements for Computer Graphics. Germany: McGraw-Hill, 1976.

Reference Books:

4. Zeid, Ibrahim. Mastering CAD/CAM. United Kingdom: McGraw-Hill Higher Education, 2005.
5. Reddy, J. N, Junuthula Narasimha. An introduction to the finite element method. India: McGraw-Hill, 1993.

Web Resources:

1. https://nptel.ac.in/courses/112102101
2. https://nptel.ac.in/courses/112104031
3. https://in.coursera.org/courses?query=cad
4. https://www.careers360.com/courses-certifications/coursera-computer-aided-design-courses-brp-org

Category: **Open Elective Course** Subject Code**: ME4412**

Nano Science and Technology

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

Unit-I:

Introduction to Nano: Importance, Definition and scope, Nano size, challenges, applications. Electrons, Other Materials, Nano magnetism as a case study; Fundamental terms (Physics & Chemistry) in nano-science and technology; Feynman's perspective; Scaling laws pertaining to mechanics, optics, electromagnetism; Importance of Quantum mechanics, statistical mechanics and chemical kinetics in nano-science and technology;

Unit-II:

Classification of nano materials: Scientific basis for top-down and bottom up approaches to synthesize Nanomaterials; How to characterize Nanomaterials?

Unit-III:

Tools for Nanoscience and Technology: Tools for measuring properties of Nanostructures, Tools to Make Nanostructures. Nano scale Bio-structures, Modelling

Unit-IV:

Nano-Biotechnology: Bio-molecules, Biosensors; Nanomaterials in drug delivery,Working in clean room environments, Safety and related aspects of Nanomaterials.

Unit – V:

Carbon Nanomaterials and Applications: Carbon Nano structures and types of Carbon Nano tubes, growth mechanisms of carbon nanotubes. Carbon clusters and Fullerenes, Lithium & Hydrogen adsorption & storages, Fuel cell applications and energy storage, Chemical Sensors applications of CNTs

TEXT BOOKS

1. B.S.Murthy,P.Shankar, Baldev Raj,B.B.Rath and James Murday, Textbook of Nanoscience and Nanotechnology , University Press-IIM Series in Metallurgy and Materials Science.

2. T.Pradeep, A Textbook of Nanoscience and Nanotechnology –, Tata McGraw Hill edition.

3. Sengupta, A., and C.K. Sarkar. Introduction to Nano: Basics to Nanoscience and Nanotechnology. Springer Berlin Heidelberg, 2015.

4. Poole, C.P., C.P. Poole, F.J. Owens, F.J.A. OWENS, and F.J. Owens. Introduction to Nanotechnology. Wiley, 2008.

Web Resources:

4. https://nptel.ac.in/courses/118102003
5. https://nptel.ac.in/courses/113106093
6. https://archive.nptel.ac.in/courses/118/107/118107015/

Category: **Open Elective Course** Subject Code**: ME4412**

Welding Technology

Internals: 40 Marks **L - T - P - C**

Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT-I

Introduction: Welding as compared with other fabrication processes, Importance and application of welding, classification of welding processes, Health & safety measures in welding. Welding Power Sources: Physics of welding Arc, Basic characteristics of power sources for various arc welding processes, Transformer, rectifier and generators.

Physics of Welding Arc: Welding arc, arc initiation, voltage distribution along the arc, arc characteristics, arc efficiency, heat generation at cathode and anode, Effect of shielding gas on arc, isotherms of arcs and arc blow. Metal Transfer: Mechanism and types of metal transfer in various arc welding processes.

UNIT-II

Welding Processes: Manual Metal Arc Welding (MMAW), TIG, MIG, Plasma Arc, Submerged Arc Welding, Electrogas and Electroslag, Flux Cored Arc Welding, Resistance welding, Friction welding, Brazing, Soldering and Braze welding processes, Laser beam welding, Electron beam welding, Ultrasonic welding, Explosive welding, Friction Stir Welding, Underwater welding & Microwave welding.

UNIT-III

Heat Flow Welding: Calculation of peak temperature; Width of Heat Affected Zone (HAZ); cooling rate and solidification rates; weld thermal cycles; residual stresses and their measurement; weld distortion and its prevention.

UNIT-IV

Repair & Maintenance Welding: Hardfacing, Cladding, Surfacing, Metallizing processes and Reclamation welding. Weldability: Effects of alloying elements on weld ability, welding of plain carbon steel, Cast Iron and aluminium. Micro & Macro structures in welding.

UNIT-V

Weld Design: Types of welds & joints, Joint Design, Welding Symbols, weld defects, Inspection/testing of welds.

Text Book:

1. Richard L. Little, Welding and Welding Technology, McGraw Hill Education. 1973
2. Edwars R. Bohnart, Welding Principals and Practices, McGraw Hill Education.

3. Little, Richard L. Welding and Welding Technology. United States: McGraw-Hill, 1973.
4. Sacks, Raymond., Bohnart, Edward. Welding: Principles & Practices. United Kingdom: McGraw-Hill Companies,Incorporated, 2004.

Reference Books:

1. R. S. Parmar, Welding Engineering and Technology, Khanna Publishsers.
2. Welding Handbooks (Vol. I & II).
3. Parmar, R. S.. Welding Engineering and Technology. India: Khanna Publishers, 2010.
4. Society, American Welding, and A.L. Phillips. Welding Handbook-1 . American Welding Society, 1968.
5. Society, American Welding. Welding Handbook-2. American Welding Society, 1969

   Society, American Welding. Welding Handbook-3. American Welding Society, 1969

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc22_me121/preview
2. https://www.asme.org/learning-development/find-course/practical-welding-technology-(2)/online--apr-17-27th--2023

Category: Open Elective Course **Subject Code: ME4413**

Industrial Robotics

Internals: 40 Marks **L - T - P - C**
Externals: 60 Marks **3 - 0 - 0 - 3**

UNIT - I

Introduction: Automation and Robotics, CAD/CAM And Robotics -An over view of Robotics – presentand future applications - classification by coordinate system and control systems. Components of the Industrial Robotics: End effectors-types, Mechanical grippers, and other typesof grippers, comparison of Electric, Hydraulic and pneumatic types of locomotion devices.

UNIT - II

Motion Analysis: Homogeneous transformations as applicable to rotation and translation — Problems.
Manipulator Kinematics: Specifications of matrices, D-H notation Joint coordinates and worldcoordinates – Forward and inverse kinematics — problems.

UNIT - III

Manipulator jacobians: Differential transformation and manipulators, Jacobians — problems.
Dynamics: Lagrange — Euler and Newton-Euler formulations — Problems.

UNIT - IV

Trajectory Planning: Path planning and avoidance of obstacles, Slew motion, joint interpolated motion — straight line motion.
Programming Languages: problems. Robot programming, languages and software packages

UNIT - V

Robot actuators and Feed-back components: Actuators: Pneumatic, Hydraulic actuators, electric and stepper motors. Feedback components, position sensors – potentiometers, resolvers, encoders – Velocity sensors.
Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading –processing – spot and continuous arc welding & spray painting – Assembly and Inspection

Text Books:

1. Groover. Industrial Robotics. McGraw-Hill Education (India) Pvt Limited, 1986.
2. Keramas, James G. Robot Technology Fundamentals. United States: Delmar Publishers, 1999.

Reference Books:

1. K S Fu, Ralph Gonzalez, C S G Lee. Robotics: Control Sensing. Vis. McGraw-Hill, 1987.
2. Klafter, Richard David., Chmielewski, Thomas A.., Negin, Michael. Robotic engineering : an integrated approach. United Kingdom: Prentice Hall, 1989.
3. Slotine, J.-J. E.., Slotine, Jean-Jacques., Asada, H.. Robot analysis and control. United Kingdom: Wiley, 1986.
4. Spong, M.W., and M. Vidyasagar. Robot Dynamics and Control. Wiley India Pvt. Limited, 2008.
5. Nagrath, M. Robotics and Control. Tata McGraw-Hill, 2003.

Web Resources:

1. https://nptel.ac.in/courses/112105249
2. https://nptel.ac.in/courses/107106090
3. https://nptel.ac.in/courses/112101098
4. https://nptel.ac.in/courses/112107289
5. https://nptel.ac.in/courses/112101099
6. https://in.coursera.org/specializations/robotics